Dell CloudLink 8.3.x

Administration Guide

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DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Revision History

Table 1. Revisions

Date	Document Revision	Description of Changes
June 2026	1.0	Initial release

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About Dell CloudLink

About Dell CloudLink for Enterprise and Microsoft Azure and Azure Stack

Cloud computing offers significant benefits for deployment flexibility, infrastructure scalability, and cost-effective use of IT resources. You can take advantage of these benefits by deploying enterprise workloads in the cloud. However, because cloud computing is based on a shared, multi-tenant compute, network, and storage architecture, traditional security controls are not sufficient. Data owners must secure sensitive data that is saved in the cloud to address privacy and regulatory compliance requirements, and satisfy requirements that are related to data that might remain in the cloud after it is no longer used.

Dell CloudLink secures sensitive information within machines across both public and private clouds. It provides encryption for the boot volume and additional data volumes with prestartup authorization for cloud-hosted machines. CloudLink provides this encryption by using the following native OS encryption features:

- Microsoft BitLocker for Windows
- dm-crypt for Linux

BitLocker and dm-crypt are proven high-performance volume encryption solutions that are widely implemented for physical machines. However, customers have not been able to use these solutions in the cloud, where you cannot use the native OS encryption features alone to encrypt the boot volume. CloudLink solves this problem.

CloudLink's VM encryption functionality enables you to use native OS encryption features to encrypt a machine's boot and data volumes in a multi tenant cloud environment. This encryption enables you to protect the integrity of the machine itself against unauthorized modifications.

CloudLink encrypts the machine boot and data volumes with unique keys that enterprise security administrators control. Neither cloud administrators nor other tenants in the cloud have access to the keys. By securing machines, you can define the security policy that must be met before passing the prestartup authorization, including verifying the integrity of the machine's boot chain. This offers protection against tampering.

CloudLink ensures that only trusted and verified machines can run and access sensitive data that is stored in the cloud. As part of the CloudLink solution, CloudLink Center defines the key release policy, performs prestartup authorization, and monitors all CloudLink Agents, events, and logs.

About Dell CloudLink for PowerFlex

Enterprises have many reasons for encrypting their data—addressing regulatory compliance, protecting against theft of customer data, and sensitive intellectual property.

CloudLink offers significant benefits for environments that use Dell PowerFlex resources. PowerFlex is a software-defined solution that enables you to transform Direct Attached Storage (DAS) on existing hardware into shared block storage. It offers considerable scalability and extreme performance with flexible and elastic storage capacity and nodes.

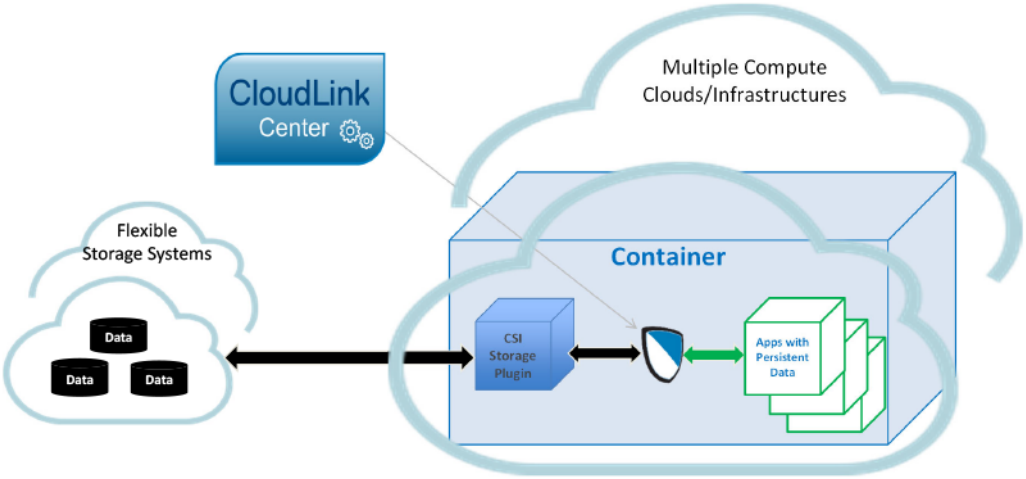
CloudLink provides software-based Data at Rest Encryption (DARE) for PowerFlex storage node that is transparent to the features and operation of the PowerFlex solution. It uses dm-crypt, a native Linux encryption package, to secure storage node devices. A proven high-performance volume encryption solution, dm-crypt is widely implemented for Linux machines.

CloudLink encrypts the storage node devices with unique keys that are controlled by enterprise security administrators. CloudLink Center provides centralized, policy-based management for these keys, enabling single-screen security monitoring and management across one or more PowerFlex deployments.

About Dell CloudLink for Containers

CloudLink supports data encryption in a Kubernetes containerized environment. CloudLink encryption for containers enables you to encrypt shared volumes in a Kubernetes cluster. This functionality leverages Kubernetes 1.14 to 1.21 Container Storage Interface (CSI), which is customizable to the user environment, and features a quick, easy setup with the UI or REST-API.

Encryption of Containers Agents sits between the Application and the CSI Storage Plugin encrypting the application data before it is sent to storage-thus providing both Data at Rest and Data in Motion. One CloudLink Center instance can support multiple Kubernetes clusters. Each Kubernetes cluster node can have multiple Container agents running on it, which includes one Encryption for Containers agent for each driver.



About Dell CloudLink Administration Guide

This guide contains instructions for managing day-to-day operations and administering Dell CloudLink.

Intended audience for the CloudLink Administration Guide

This guide is intended for CloudLink Center administrators who use the CloudLink Center administration interface to manage the security of machines that are registered to CloudLink Center.

This guide is also intended for IT administrators who are responsible for the deployment and maintenance of machines in the CloudLink Center environment, but not necessarily for the security of data on those machines. The reader of this Administration Guide is expected to have prior working knowledge of VMs and data encryption.

Deprecated support

The KMIP server and Machine Encryption functionalities are deprecated in CloudLink version 8.3.

The KMIP server and Machine Encryption functionalities will be disabled in a later release. Dell Technologies recommends customers using these functionalities to plan accordingly and transition to alternative supported solutions.

For guidance, impact assessment, or additional information, contact your Account Team.

Get started with Dell CloudLink

Before you can use CloudLink Center, you must deploy CloudLink into your enterprise infrastructure or into the public cloud. For deployment information, see the *Dell CloudLink Deployment Guide*.

Use a web browser to access the CloudLink Center management interface. For more information, see Access CloudLink Center.

NOTE: CloudLink Center uses a self-signed certificate by default. You can import a certificate issued by a certification authority.

CloudLink Center server address

CloudLink Center server address is used frequently. For example, you provide the address in the URL used to access the CloudLink Center Graphical User Interface (GUI), and in commands used to download installation files.

The CloudLink Center address can be configured by using IPv4, IPv6, and hostname addresses. By default, IPv4 is used but you can change it to IPv6 or hostname address using the Intial Configuration wizard. If the Domain Name System (DNS) has an entry for CloudLink Center, it is recommended that you specify the CloudLink Center server address as a hostname in a Fully Qualified Domain Name (FQDN) format, such as clc.example.com. For more information, see DNS servers in CloudLink Center. If you want to use an IP address, use a static one.

NOTE: In a CloudLink Center cluster, the cluster node servers and CloudLink Agents use this server address for communication. Before creating the cluster, specify the server address in the format you prefer for each server. You can use a mix of FQDNs and IP addresses in a cluster, but you cannot change the format after creating a cluster.

For more information about prerequisites and requirements for server addresses in clusters, see the *Dell CloudLink Deployment Guide*.

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CloudLink 8.3.x license deployment or migration offerings

If you are upgrading to CloudLink 8.3.x from an earlier version, the license of the earlier version is retained and is displayed on the CloudLink interface. However, for 8.3.x and later, you do not require explicit licenses to access CloudLink.

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Log in to CloudLink Center

Access CloudLink Center

Most management tasks are performed from CloudLink Center. Access CloudLink Center from an HTTPS session using a web browser by enabling the JavaScript.

Steps

1. In the web browser, type the CloudLink Center URL in the following format:

   ```
   https://clc_address
   ```

   Where,

   clc_address is the CloudLink Center address.

   The *clc_address* must be in either the FQDN, IPv4, or IPv6 format. For more information, see CloudLink Center server address.

2. On the CloudLink Center home page, do one of the following:
   - Type a username and password.

     For information about the first-time login to CloudLink Center or about the username or password, see the *Dell CloudLink Deployment Guide*.

   - Click **Log in with my Windows credentials**.

     This option is available only for domain users if CloudLink Center has been added to the Microsoft Windows domain. For more information, see Assign Microsoft Windows User Account for CloudLink Center User Roles.

CloudLink Center server address

This topic provides information about the CloudLink Center server address.

You use the CloudLink Center server address frequently. For example, you provide the address in the URL used to access the CloudLink Center user interface (UI), and the commands that are used to download installation files.

The CloudLink Center address can be configured as an IPv4 address, IPv6 address, or hostname. The address is set to IPv4 by default, but it can be changed to an IPv6 address or hostname in the Initial Configuration wizard. If the Domain Name System (DNS) has an entry for CloudLink Center, Dell Technologies recommends that you specify the CloudLink Center server address as a hostname in fully qualified domain name (FQDN) format, such as clc.example.com. For more information, see DNS servers in CloudLink Center. If you choose to use an IP address, use a static one.

NOTE: In a CloudLink Center cluster, cluster node servers and CloudLink Agents use this server address for communication. Before creating the cluster, you must specify the server address in the format you prefer for each server. You can use FQDNs and IP addresses in a cluster, but you cannot change the format after creating a cluster.

For more information about prerequisites for server addresses in clusters, see the *Dell CloudLink Deployment Guide*.

Change maximum session timeout

For security, CloudLink Center automatically ends a session that has been active for a specified duration. You can set this maximum session timeout.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Login Options**.
3. Click **Change Max Session Timeout**.
4. In the **Change Max Session Timeout** dialog box, enter the max session timeout in minutes.
5. Click **Change**.

Change the automatic logout interval

Use this procedure to change the automatic logout interval in CloudLink Center.

About this task

If no activity has occurred for a specified period, the web application automatically logs off a user. You can configure this timeout from zero to 60 minutes, where zero implies that no automatic logout occurs.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Login Options**.
3. Click **Change UI Idle Timeout**.
4. In the **Change UI Idle Timeout** dialog box, enter the UI idle timeout in minutes. The default timeout is 15 minutes.
5. Click **Change**.

Change the number of login attempts before lockout

Use this procedure to change the number of login attempts before lockout.

About this task

You can specify the number of times that a user can provide an incorrect password before CloudLink Center locks the user out. For more information about unlocking users, see Unlock CloudLink accounts.

NOTE: By default, you are allowed five consecutive failed login attempts before your account is automatically locked for 15 minutes. After this, you are allowed three consecutive failed login attempts for every 15 minutes until you can log in successfully.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Login Options**.
3. Click **Change Login Attempts**.
4. In the **Change Login Attempts** dialog box, enter the number of attempts after which the CloudLink Center session must be automatically locked for a user.
5. Click **Change**.

Enable or disable special characters in password

This topic provides information about enabling or disabling the use of special characters in your password.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Login Options**.
3. Click **Require Special Characters In Password**.
   The **Require Special Characters In Password** dialog box is displayed.
4. From the **Require Special Characters** list, select the required value, and then click **Change**.

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CloudLink Center Graphical User Interface (GUI)

CloudLink Center provides an easy-to-use interface with many features to help you manage registered machines, and to configure and monitor the environment.

Navigate through CloudLink Center user interface

This topic provides information about how to navigate through the CloudLink Center user interface.

The following screenshot shows CloudLink Center and identifies the primary navigation features.

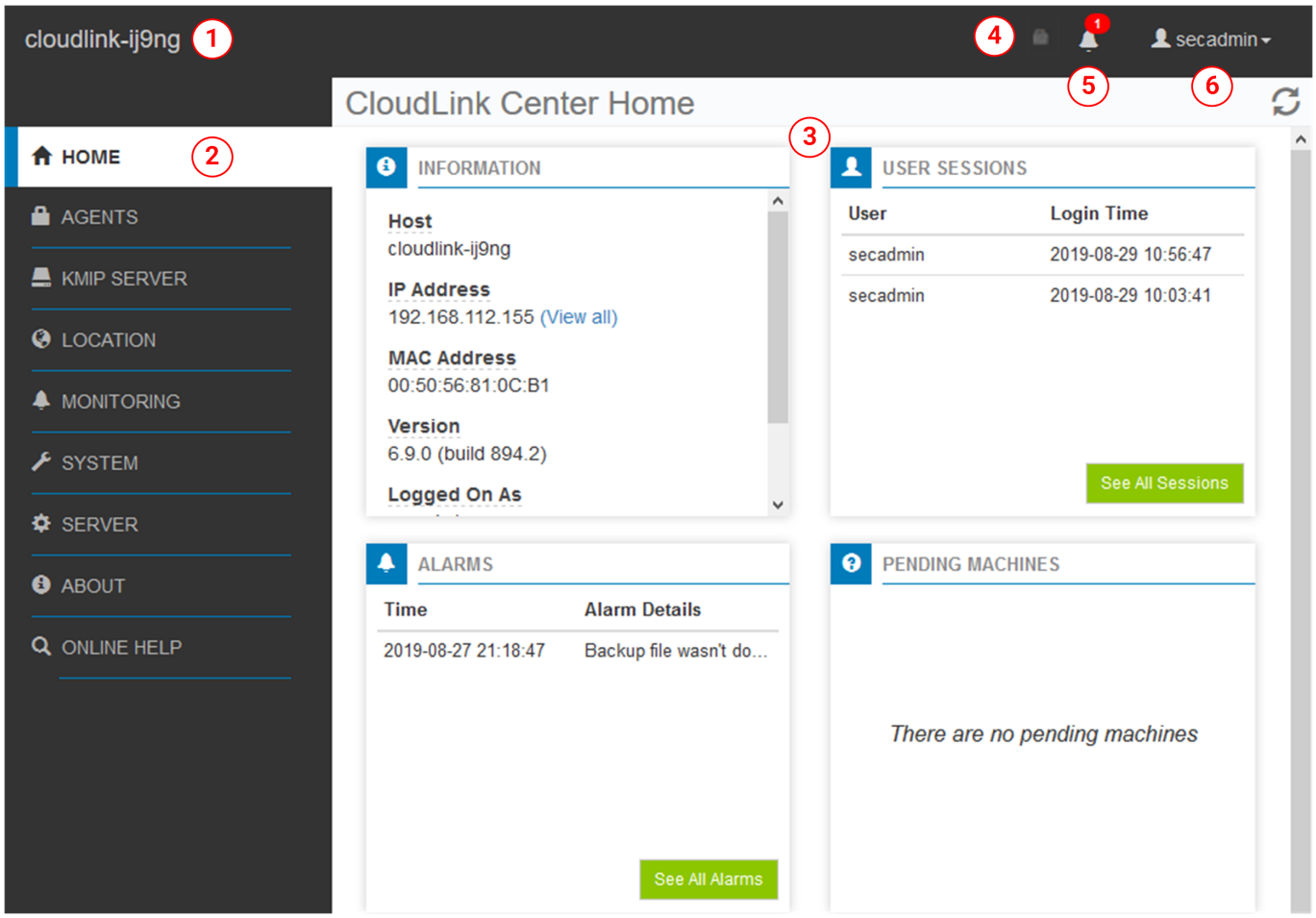


Figure 1. CloudLink Center Home

- **1—CloudLink Center server**CloudLink Center

  Identifies the CloudLink Center server that you are logged in to.
- **2—Contents panel**

  Lists the pages that you can select in the **Edit** panel.
- **3—Edit panel**

Shows the page that is selected in the **Contents** panel.

- **4—Manual lock**

  Indicates that CloudLink Vault is locked.

  The Manual lock icon disappears when the CloudLink Center Vault is unlocked. For more information, see Manage CloudLink Vault.

- **5—Alarms notification**

  Displays a badge with the number of current alarms, if any.

  An alarm represents a state or condition that you should be aware of. Click the Alarms notification icon to view alarms. For more information, see View CloudLink Center alarms.

- **6—User Name menu**

  Displays your username.

  You can log out of CloudLink Center from this menu.

  NOTE: Every CloudLink Center user is assigned a role that determines the permissions of the user. If you notice that few CloudLink Center features are not available to you, contact your secadmin user.

Dell CloudLink Center home page

The CloudLink Center **Home** page provides a dashboard view of CloudLink Center.

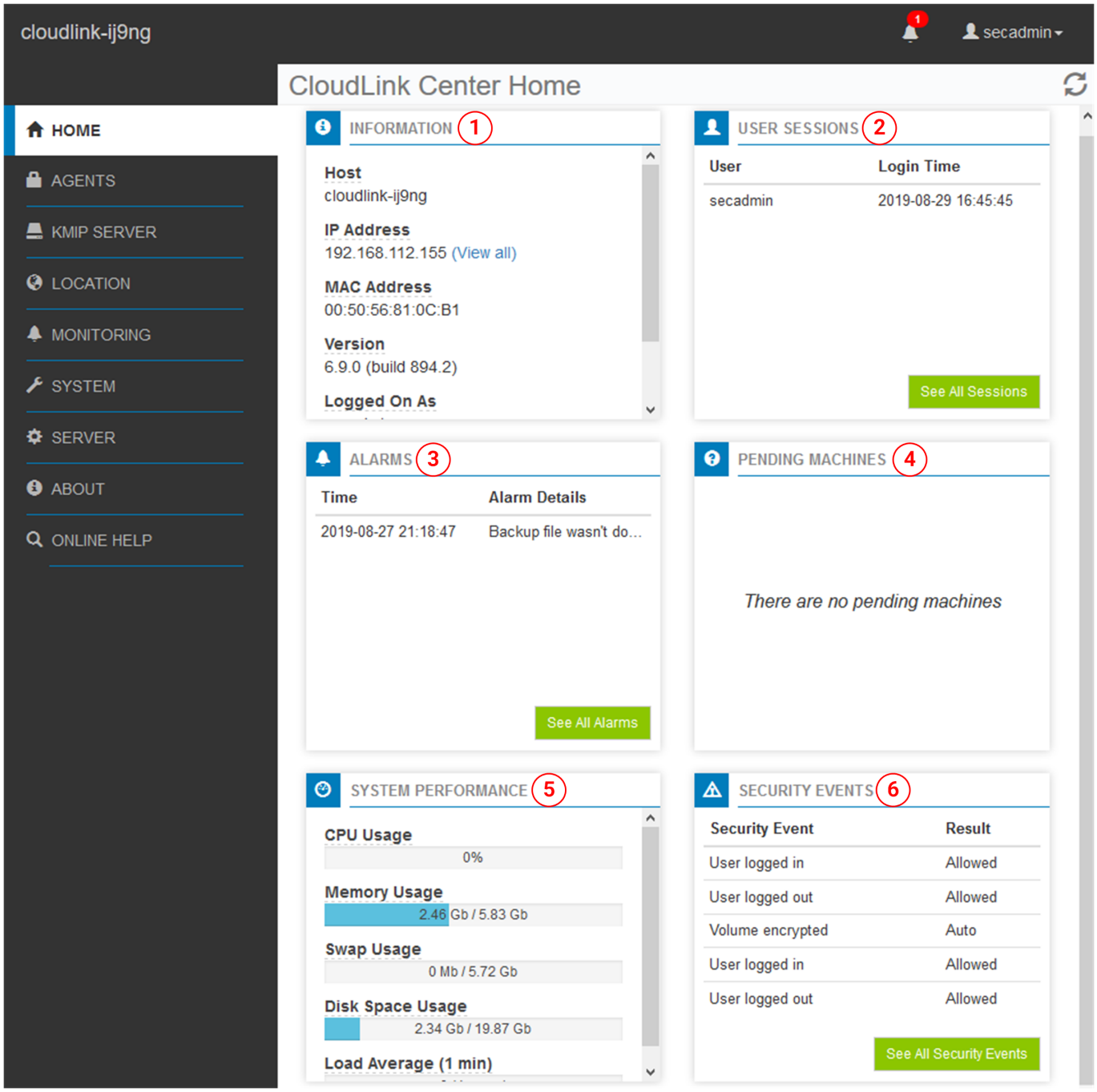


Figure 2. CloudLink Center Home page

The **Home** page includes the following panels:

- **1—Information**

  Provides connection information about the server on which CloudLink Center is running.
- **2—User Sessions**

  Lists the users who are logged in to CloudLink Center, including the time of the user's login and last action. For more information, see View user sessions in CloudLink Center.
- **3—Alarms**

Lists the alarms that are active. An alarm represents a state or condition of which you must be aware. For more information, see View and manage alarms.

- **4—Pending Machines**

  Lists the machines that are waiting for you to accept or reject startup. For more information, see Accept or reject pending machines in CloudLink Center.

- **5—System Performance**

  Provides information about the server that is hosting CloudLink Center, including CPU and memory usage, available disk space, and so on.

- **6—Security Events**

  Provides information about user logins, failed vault unlock attempts, machine registration, changes to the CloudLink Center Vault, secure user actions, and encryption key activities. For more information, see Monitor CloudLink Center.

6

Common tasks you can perform in CloudLink Center

This chapter provides information on the common tasks you can perform in CloudLink Center.

Save tabulated data to a CSV file

Use this procedure to save data to a CSV file.

About this task

All the CloudLink Center features that have data available in tables, enable you to export information to a CSV file. For example, on the **Machines** page, you can export the list of machines that are registered with CloudLink Center, and then include the list in a distribution report.

Steps

1. Log in to CloudLink Center.
2. Go to a page in the CloudLink Center menu that supports the file export feature.
3. Click  .

   The data is exported to a standard comma-separated value (CSV) file that is downloaded to your Downloads folder.

Filter table data

About this task

For tables with many rows, it might be difficult to quickly find a particular row or subset of rows. For example, if hundreds of machines are registered or pending registration with CloudLink Center, the Machines table spans many pages.

For convenience, use the Filter icon to quickly search for table rows that match criteria that you specify.

To help you define a search criteria:

- Type text in the filter box to see all table rows that contain the text. For example, type **`Connect`** in the **Status** box to see all table rows where the status is Connected or Disconnected. Both these words contain the criteria Connect.
- Type an exclamation mark as the first symbol in a filter box to see all table rows that do not match the criteria. For example, type **`!102`** in the IP address box to see all table rows that do not contain 102 in the IP address.
- Type text that is enclosed in double or single quotation marks to see all table rows with an exact match for the criteria. For example, type **`"Connected"`** in the Status box to see only table rows where the status is Connected.

  Similarly, type **`!"vmware"`** in the Platform box to see all table rows where the platform is not VMware.
- Search criteria is case-insensitive. For example, **`vmware`** and **`VMware`** are considered to be the same text.

Steps

1. Log in to CloudLink Center.
2. Go to a page in the CloudLink Center menu that supports the filter feature.
3. Click  .

   A set of search boxes appears above the table header. Type your search criteria in any of the search boxes.

Refresh CloudLink Center page data

About this task

Generally, CloudLink Center pages display live information in real time. However, to ensure that you are viewing the most current information:

Steps

1. Log in to CloudLink Center.
2. Click  icon.

View CloudLink Center alarms

An alarm represents a state or condition. When one or more alarm conditions exist, CloudLink Center displays a badge on the **Alarms** icon on the home page. The badge indicates the number of alarms.

Steps

1. Log in to CloudLink Center.
2. On the **Home** page, click 

.

Next steps

CloudLink Center removes alarms from the **Alarms** page when the condition or state that triggered the alarm is resolved. For example, CloudLink Center triggers an alarm if a backup file is generated but not downloaded. When you download the backup file, the alarm is automatically deleted.

7

Best practices to secure data in CloudLink Center

After configuring and setting up CloudLink Center, it is recommended that you follow certain best practices to secure your data in CloudLink Center.

Following are the best practices to manage and secure the CloudLink Center and encryption keys:

- Rotate keys automatically—It is recommended to rotate keys on a scheduled basis. By setting an automatic rotation interval, encryption keys are updated without further effort . These rotations can be set at one day, seven days, 30 days, and 365 days interval. You can also customize the rotation interval. You can set the key rotation interval while creating a machine group or update it while modifying the machine group properties. For more information about setting key lifetime, see Create a machine group to CloudLink Center.
- Shred keys periodically—It is recommended to shred keys periodically to protect the data. Key shredding is destroying or deleting the encryption keys that are used for securing the data that was previously stored on the respective machine. You can shred a key by shredding a machine from CloudLink Center. By shredding a machine, you are deleting keys in all accessible keystores. For more information about shredding a machine from CloudLink Center, see Shred a machine from CloudLink Center.
- Automatically backup CloudLink Center—To prevent data loss or database corruption, it is important to have a backup of CloudLink Center so that you can deploy a new server and restore CloudLink Center from the backup. CloudLink Center automatically generates a backup file each day at midnight (UTC time). You can choose to change this backup schedule to an hourly basis or how frequently you want to backup CloudLink Center. For more information about changing the schedule for automatic backups, see Change the schedule for automatic backups.

8

Manage secure machines on CloudLink Center

From CloudLink Center, manage the secure machines to which CloudLink Center Agent has been deployed. To help you administer and manage these machines, organize them using machine groups. For all machines in a group, the machine group policies determine the role that administers the machines, the conditions under which machines may start up automatically, the encryption that must be in effect for the machine, and so on.

For individual machines in a group, you can:

- Encrypt and decrypt volumes.
- Encrypt and erase devices.
- Decide whether or not to enable a particular volume or device to be exempt from the policy for the machine group.
- Accept, reject, or remove a machine.

NOTE: CloudLink does not support network drives encryption or decryption, except VMware vSphere Virtual volumes (vVols).

CloudLink self-encrypting drives (SEDs)

This topic provides information about the SEDs in CloudLink Center.

When managing SEDs from CloudLink Center, be aware of the following:

- CloudLink Center can manage encryption keys for self-encrypting drives (SEDs).

  NOTE: You must not encrypt SEDs even when you have a license that supports SED features because such a license supports only managing and releasing SEDs. To identify if a drive is of SED type, look at the SED section in the Machines page. If info about the SED drive is displayed, then it implies that the drive is of SED type. Else, it is not.

- Manage SEDs in CloudLink Center by locking and unlocking when the CloudLink agent is installed on machines with SEDs.
- When CloudLink Center manages SEDs, SED encryption keys are stored in the current keystore for the machine group they are in.
- The functionality for managing SEDs requires a separate SED license.
- When CloudLink Center takes ownership of an SED, it releases SED encryption keys when the physical machine containing the SEDs is powered on or restarted.
- If the SED cannot retrieve the key from CloudLink Center, the SED remains locked.

CloudLink prestartup authorization

Prestartup authorization of CloudLink Center machines applies to only Enterprise and Microsoft Azure and Azure Stack.

Pre-startup authorization enables a machine to start up automatically when:

- The machine has been previously registered with CloudLink Center and can connect to it.
- The boot volume of a machine is encrypted, and the machine meets key release policies for boot volume encryption. For more information, see CloudLink key release policies.

  NOTE: If the boot volume of a machine is not encrypted, but one or more data volumes are encrypted, the machine is allowed to start. CloudLink Center determines whether encryption keys for encrypted data volumes can be released automatically based on key release policies. If key release policies are not met for the data volume, CloudLink Center puts the machine in the pending state.

If a machine does not pass prestartup authorization, CloudLink Center puts the machine in the pending state and you must explicitly accept the machine before startup is allowed to continue.

CloudLink Center registered machines

From CloudLink Center, manage individual machines on which CloudLink Agent has been deployed. For example, you can view a list of machines and their current states, and accept or reject machine startup.

Each machine registered with CloudLink Center is assigned a unique serial number that is stored in the CloudLink Center database. If a machine has more than one volume or device, CloudLink Center management operations can be performed on individual volumes or devices. For example, you can:

- Encrypt or decrypt volumes on an individual basis
- Encrypt or erase devices on an individual basis

CloudLink Center machine startup

This topic provides information about machines starting automatically using CloudLink key release policies.

Enable machines to start up automatically using key release policies set for the machine group. For more information, see CloudLink key release policies.

After starting, if a machine does not meet the requirements of all the key release policies, CloudLink Center puts the machine in the pending state. In this state, an administrator must explicitly accept the startup of the machine. For example, you can set a key release policy so that CloudLink Center does not allow a machine to start up automatically if its IP address does not belong to an approved network for the machine group. For more information about the pending state and manually accepting the startup of a machine, see Accept or reject pending machines in CloudLink.

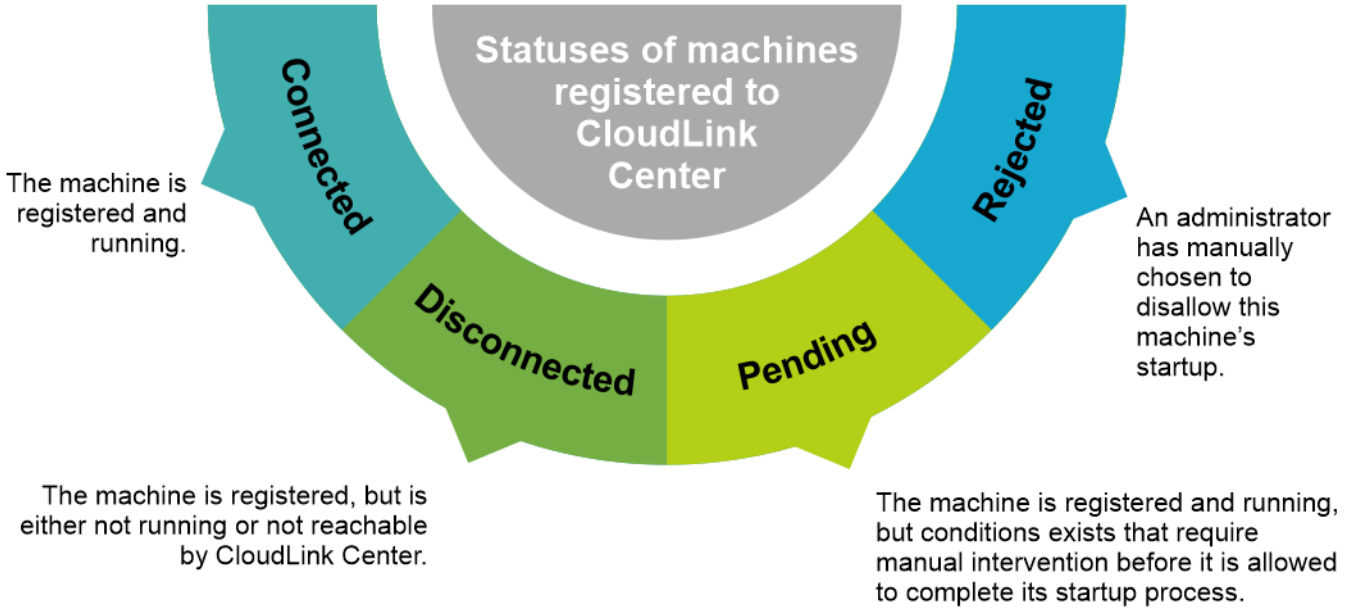
CloudLink Center machine states

This topic provides information about the different statuses of machines in CloudLink Center.

Each registered machine is assigned to a state that you can view in CloudLink Center. For more information, see View registered machines. The state of a machine determines the actions that you can perform, including:

- Encrypting volumes and devices
- Decrypting volumes
- Erasing devices
- Accepting or rejecting startup of a machine
- Removing machines

The preceding actions depend on whether the machine is running Windows or Linux.



Accept or reject pending machines in CloudLink Center

This topic provides information about accepting or rejecting pending machines in CloudLink Center.

When a registered machine with an encrypted boot volume or a device attempts to start up, CloudLink Center may cancel the startup process and put the machine in the pending state. This state implies that key release policies are not met, and CloudLink Center cannot release encryption keys for the boot volume or device of the machine. For more information, see CloudLink key release policies.

You can view the reason for the pending state in the Details column of the Machines panel. As a one-time option, you can then choose to manually:

- Accept the startup of the machine. CloudLink Center releases encryption keys for the boot volume or device, and the machine continues its startup process. CloudLink Center displays the machine in a connected state.
- Reject the startup of the machine. CloudLink Agent disconnects from CloudLink Center and stops requesting encryption keys. CloudLink Center displays the machine in the disconnected state.

For example, by default, CloudLink Center ensures that the IP address of the machine belongs to an approved network for the group of the machine. For more information, see Manage approved networks for machine groups. If the IP address is not in this list, CloudLink Center puts the machine in the pending state. After determining whether you want to allow this machine to start up, you select to accept or reject the startup process.

Accepting or rejecting a machine applies to the current startup process only. The next time a previously accepted or rejected machine starts, it is put in the pending state again if any reason for this state is detected. You must once again choose whether to accept or reject the machine. If no conditions exist for the pending state (for example, the previous issue has been resolved), the machine is in a connected state.

Accept a pending machine

Use this procedure to accept a pending machine.

Prerequisites

You can accept a machine only if the machine is in the pending state.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the machine you want to accept.
4. Click **Accept**.
5. In the **Confirm Accept Machine** dialog box, when prompted to confirm the accept request, click **Accept**.

Reject a pending machine

Use this procedure to reject a pending machine.

Prerequisites

You can reject a machine only if the machine is in the pending state.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the machine you want to reject.
4. Click **Reject**.
5. In the **Confirm Reject Machine** dialog box, when prompted to confirm the reject request, click **Reject**.

View registered machines

You can view the list of machines that are registered in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
   All the machines that are registered in CloudLink Center are displayed.

Move a registered machine to a different machine group

Every machine that is registered with CloudLink Center belongs to a machine group. You can change the machine group to which a machine belongs.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the machine you want to move to a different machine group.
4. Click **Actions** > **Move to Group**.
5. In the **Move Machine to Another Group** dialog box, select the required group from the **Move to** list.
6. Click **Move**.

Scenarios for removing machine from CloudLink Center

This topic provides information about the scenarios for removing a machine from CloudLink Center.

Remove a machine when you no longer want it registered with CloudLink Center. Typically, you remove a machine because:

- You no longer want to use the machine in the CloudLink environment.
  - In this case, decrypt any encrypt volumes and remove CloudLink Agent from the machine. The machine must be in the disconnected state.
  - In this case, erase any encrypted devices and remove CloudLink Agent from the machine. The machine must be in the disconnected state.
- You want to release a machine instance license.
  - In this case, deregister the machine from CloudLink Center. At some time in the future, you may re-register the machine. The machine must be in the disconnected state.

NOTE: The capacity of an encrypted capacity license is increased when a device is decrypted.

Remove a machine from CloudLink Center

Use this procedure to remove a machine from CloudLink Center. If you no longer want to use a machine in the CloudLink environment, you can remove it so that the machine is unregistered from CloudLink Center and CloudLink Agent is automatically uninstalled.

Prerequisites

The machine must be in the disconnected state.

It is recommended that you decrypt the boot volume and any additional data volumes (Windows) or mounted devices (Linux) that you want to use after the machine has been de-registered from CloudLink Center, as they will be inaccessible otherwise. For more information, see CloudLink Center machine volumes.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the checkbox next to the machine you want to remove.
4. Click **Remove**.
5. In the **Confirm Machine Removal** dialog box, when prompted to confirm the removal of the machine request, click **Remove**.

Release a license

Use this procedure to release a license in CloudLink Center.

Prerequisites

The machine must be in the disconnected state.

About this task

If you want to temporarily remove a machine instance, encrypted capacity, or a KMIP client from CloudLink Center to release a license, ensure it is in the disconnected state. Once removed, the machine does not appear in the **Machines** panel. However, if the machine is restarted, it is listed in this panel in the pending state. For more information, see Accept or reject pending machines in CloudLink.

Steps

1. Log in to CloudLink Center.

2. Click **Agents** > **Machines**.
3. Select the checkbox next to the machine that you want to release a license.
4. Click **Remove**.
5. In the **Confirm Machine Removal** dialog box, when prompted to confirm the removal of the machine request, click **Remove**.

Shred encrypted CloudLink machines

Shred encrypted machines to make the machines and their volumes or devices completely inaccessible by destroying their keys in the keystore.

If you attempt to shred a machine and a keystore that contains keys for the machine is not accessible, CloudLink Center does not shred the machine. CloudLink Center deletes keys in all accessible keystores. You can try to shred the machine again after the inaccessible keystore is accessible.

With one exception, shredding a machine prevents that machine, including all backups, from being started again. The exception is when the keys are in the CloudLink Vault keystore and a backup of CloudLink Center exists that might have those keys.

Shred a machine from CloudLink Center

Use this procedure to shred a machine from CloudLink Center.

Prerequisites

- The machine must be in the disconnected state.
- Decrypt the disks of the machines.
- Uninstall CloudLink Agent on the machine.

About this task

If you shred a machine that has older shared keys from cloning or upgrading, the shared keys are not deleted and an error message stating “Some keys have not been deleted” is displayed. The current keys for the machine are deleted. However, if the machine boot drive was encrypted, it no longer boots.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the machine whose keys you want to shred and remove the machine from the CloudLink Center.
4. Click **Actions** > **Shred**.
5. In the **Confirm Machine Shredding** dialog box, enter the name of the machine to confirm it.
6. Click **Shred**.

View event history of a machine

Use this procedure to view the history of a machine.

About this task

You can view only the events for a selected machine. For information about choosing the timeframe in which events occurred and the information that is provided on the Events page, see CloudLink events and corresponding syslog severity numbers.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the checkbox next to the machine to view the event history of that machine.

4. Click **Actions** > **Show Event History**.
   The event history of the selected machine is displayed in the **Events** page.

Refresh mounted devices or devices of a Linux machine

Use this procedure to refresh mounted devices or devices of a Linux machine.

About this task

If you add a new mounted device or a device for a Linux machine, you must refresh the machine, so that the new device is added to the **Machines** panel.

Steps

Enter the following command at the Command-Line Interface (CLI):

```
svm reload [-v ].
```

Next steps

For information about optional `svm` parameters, see Linux commands for CloudLink Agent.

Work With Cloned Machines in CloudLink Center

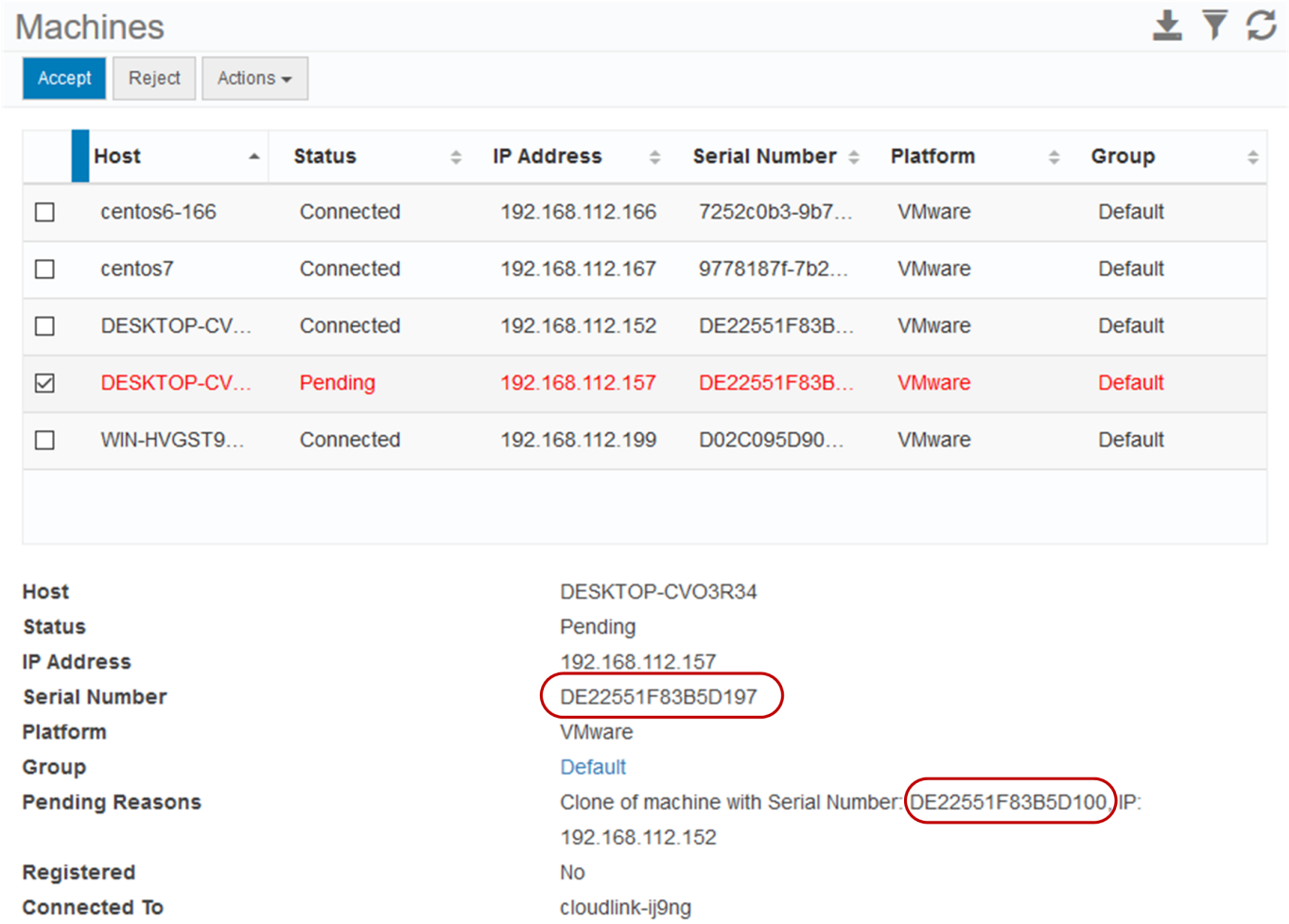
This topic gives information about cloned machines in CloudLink Center.

Each machine that is registered with CloudLink Center is assigned a unique serial number that is stored in the CloudLink Center database. By default, if you clone a registered machine in the connected state, and then start the clone, CloudLink Center puts the clone in the pending state. For more information, see CloudLink Center machine states.

CloudLink Center puts the clone in the pending state because the default key release policy does not enable CloudLink Center to release keys to a clone of a previously registered machine. You can change this key release policy, enabling automatic release of keys to clones of machines belonging to the group. For more information, see CloudLink key release policies.

To indicate the relationship with the machine in the pending state to the machine from which it was cloned, the serial number for the machine clone is the same as the original machine, except for the last two digits.

As shown in the following screenshot, data of the selected machine indicates that the machine is in pending state. The serial number for the pending machine ends with 60 and the serial number for the machine from which it was cloned ends with 00.



To enable the startup process for the cloned machine to continue, you must accept the machine. For more information, see Accept or reject pending machines in CloudLink. After a cloned machine is accepted, it is restarted and assigned a new serial number.

For Enterprise and Microsoft Azure and Azure Stack—If the machine passes the pre-startup authorization, all subsequent startup operations of the machine are automatically enabled. For more information about pre-startup authorization, see the section *CloudLink pre-startup authorization*.

Change encryption keys on Linux machines

This topic provides information about changing encryption keys on Linux machines.

If PowerFlex storage node devices on the original machine were encrypted when the clone was created, the data on both machines is encrypted with the same key. If you want data on each machine encrypted with different keys, you must erase, and then encrypt any encrypted devices on one of the machines. For more information, see CloudLink Center machine devices.

If boot or data volumes on the original machine were encrypted when the clone was created, the data on both machines is encrypted with the same key. If you want data on each machine encrypted with different keys, you must decrypt and then encrypt any encrypted volumes on one of the machines. For more information, see CloudLink Center machine volumes.

Change encryption keys on Windows machines

New keys are generated for the cloned machine when it is accepted.

Change hostnames

The hostname for a cloned machine is the same as the machine from which it was cloned. To help distinguish machines from their clones, you may want to change the hostname for clones. See the documentation provided for the clone's operating system.

Restart the CloudLink Agent service on Linux machines

Use this procedure to restart the CloudLink Agent service on Linux machines.

About this task

For Linux machines, you must restart the CloudLink Agent service if the networking configuration is changed after deploying the CloudLink Agent.

Steps

Enter the following command at the CLI:

```
service svmd restart
```

CloudLink Center Machine Volumes

This topic gives information about machine volumes in CloudLink Center.

Each machine that is registered to CloudLink Center is assigned a unique serial number that is stored in the CloudLink Center database. If a machine has more than one volume, CloudLink Center management operations (such as encrypting or decrypting the boot or data volumes of a machine) can be performed on individual volumes.

Table 2. Boot and Data Encryption Supported on Bare-Metal and VMs by CloudLink

Encryption On	Boot Volume	Data Volume
VM	Yes	Yes
Physical machine	No. CloudLink does not support boot volume encryption on physical machines.	Yes

Before removing a machine that you no longer want under CloudLink Agent control, you should decrypt the volumes if you want to continue using the machine. Otherwise, its volumes remain encrypted and inaccessible.

For information about monitoring encryption or decryption processes on a machine, see Monitor the real-time progress of encryption and decryption processes.

Encrypt a volume

By using CloudLink Center, you can encrypt system partitions or data volumes.

The following Linux system partitions are automatically encrypted or decrypted with the / partition:

- /
- /bin
- /sbin
- /root
- /lib
- /var
- /usr

- /usr/local
- /initrd
- /tmp
- /home
- /opt

WARNING: If at least one system partition is encrypted by CloudLink 6.7 or earlier, you cannot encrypt an individual system partition. You must first decrypt all system partitions, then encrypt / , to encrypt all system partitions at once.

If more than one volume on the machine is unencrypted, select the one that you want to encrypt. You can encrypt only one volume at a time.

For Linux machines, CloudLink encrypts using dm-crypt. The device must be mounted. CloudLink Center first unmounts the mount point. If files are open, an error occurs and the encryption does not occur.

When encryption is completed, CloudLink Center restarts the machine.

Encrypt a data volume on a self-encrypting drive

Use this procedure to encrypt a data volume on a self-encrypting drive.

About this task

You can encrypt a data volume on a self-encrypting drive (SED), as long as SED encryption is not being managed by CloudLink. If SED encryption is managed by CloudLink, then volume encryption is blocked. You do not need an SED license to encrypt data volumes on an SED.

NOTE: CloudLink does not support boot volume encryption on physical machines.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the SED whose data volume you want to encrypt.
4. Click **Actions** > **Encrypt**.
   The **Encrypt** dialog box is displayed.
5. From the **Volumes** list, select the data volume.
6. Click **Encrypt**.

Encrypt devices of a Linux Machine from the CLI

As an alternative to encrypting a mounted device of a Linux machine from CloudLink Center, you can encrypt it from the CLI.

Prerequisites

The device must be mounted.

About this task

The following Linux system partitions are automatically encrypted or decrypted with the / partition:

- /
- /bin
- /sbin
- /root
- /lib
- /var
- /usr
- /usr/local

- /initrd
- /tmp
- /home
- /opt

WARNING: If at least one system partition is encrypted in CloudLink 6.7 or earlier, you cannot encrypt an individual system partition. You must first decrypt all system partitions, then encrypt / , to encrypt all system partitions at once.

You can encrypt the data volume for a Linux machine from the CLI. If more than one volume on the machine is unencrypted, you choose the volume that you want to encrypt. You can encrypt only one volume at a time.

The process used to encrypt the data volume is the same as when encrypting from CloudLink Center. For more information, see Encrypt a volume.

Use one of the following methods to encrypt devices of the Linux machine:

Steps

1. Enter the following command:

   ```
   svm encrypt [mount_point]
   ```

   For example:

   ```
   svm encrypt /MyData/MyMountPoint
   ```

2. You can also force encryption. Enter the following command from the CLI:

   ```
   svm encrypt -f [mount_point]
   ```

   For example:

   ```
   svm encrypt -f /MyData/MyMountPoint
   ```

   NOTE: The `-f` option restarts a Linux machine and encrypts the data partition. The `-f` option is for users who want to encrypt a Linux data partition that is in use.

   For more information about the `svm` parameters, see Linux commands for CloudLink Agent.

Decrypt a volume

This topic provides information about decrypting a volume.

By using CloudLink Center, you can decrypt boot or data volumes in the connected state on Windows or Linux machines.

If more than one volume is encrypted, select the volume that you want to decrypt.

Decrypting a volume means that it no longer complies with the volume encryption policy for the machine group, CloudLink Center prompts you to confirm that you want to decrypt the volume. For information, see CloudLink Center volume encryption policy. If you select to decrypt the volume, CloudLink Center triggers an alarm and, you can opt to enable its noncompliance through an exemption. For more information, see Types of volume encryption policies in CloudLink.

Decrypt a data volume on a self-encrypting drive

Use this procedure to decrypt a data volume on a self-encrypting drive (SED).

Prerequisites

The machine must be in the connected state.

About this task

NOTE: CloudLink does not support boot volume encryption or decryption on physical machines.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the SED whose data volume you want to decrypt.
4. Click **Actions** > **Decrypt**.
   The **Decrypt** dialog box is displayed.
5. From the **Volumes** list, select the data volume.
6. Click **Decrypt**.

Decrypt boot volume of a linux machine from the CLI

As an alternative to decrypting the boot volume of a Linux machine from CloudLink Center, you can decrypt it from the CLI.

If you decrypt a volume and it no longer complies with the volume encryption policy of a group, then CloudLink Center triggers an alarm. You can choose to allow its noncompliance through an exemption. For more information, see Types of volume encryption policies in CloudLink.

The procedure used to decrypt the boot volume depends on the version of CloudLink that was used to encrypt the boot volume. CloudLink version 5.0 and earlier uses eCryptfs for encryption. CloudLink version 6.9 and later uses dm-crypt.

Decrypt a boot volume encrypted using CloudLink version 6.9 and later

Use this procedure to decrypt a boot volume that was encrypted using CloudLink version 6.9 and later.

Steps

1. Back up all data.
2. At the CLI, enter the following command:

   ```
   svm decrypt / [-v ]
   ```

   For more information about this command, see Linux commands for CloudLink Agent.
3. When prompted to restart the machine, click **Yes**.
4. Wait until the machine is displayed under **Agents** > **Machines**. The machine must be in the Connected state.

   NOTE: The time required to complete this operation depends on the size of the boot disk and the environment type.

Decrypt a boot volume encrypted using CloudLink 5.0 or earlier

Use this procedure to decrypt a boot volume that was encrypted using CloudLink 5.0 or earlier version.

Steps

1. Back up all data.
2. Attach a second disk that is as large as the root disk.

   This disk must not contain any data because it is overwritten with contents from the encrypted disk. Do not mount this drive.
3. At the CLI, enter the following command:

   ```
   svm decrypt /[target_disk] [-v ]
   ```

   where, *target_disk* is the drive that was attached in Step 2.

For example:

```
svm decrypt /dev/sdb
```

For more information about this command, see Linux commands for CloudLink Agent.

4. When prompted to restart the machine, click **Yes**.
5. Wait until the machine is displayed under **Agents** > **Machines**. The machine must be in connected state.

   NOTE: The time required to complete this operation depends on the size of the boot drive and the environment type.

6. Detach the second disk when the status of the boot volume is connected and decrypted in CloudLink Center.

Decrypt the mounted volumes of a Linux machine from the CLI

Use this procedure to decrypt the mounted volumes of a Linux machine from the CLI.

As an alternative to decrypting the mounted volumes of a Linux machine from CloudLink Center, you can decrypt them from the CLI. This process decrypts the data to a new volume that is mounted with the same name as the encrypted mount point. In addition to decrypting the data, this process unregisters the mount point from CloudLink Center.

If you decrypt a volume and it no longer complies with the volume encryption policy of a group, then CloudLink Center generates an alarm. You can choose to allow its noncompliance through an exemption. For more information, see Types of volume encryption policies in CloudLink.

The procedure used to decrypt a volume depends on the version of CloudLink that was used to encrypt the volume.

- CloudLink version 5.0 and earlier uses eCryptfs to encrypt volumes.
- CloudLink version 6.9 and later uses dm-crypt.

Decrypt an encrypted volume using CloudLink version 6.9 and later

Use this procedure to decrypt a volume encrypted using CloudLink version 6.9 and later.

Use one of the following methods to encrypt devices of the Linux machine:

- Enter the following command:

  ```
  svm decrypt [mount_point]
  ```

  For example:

  ```
  svm decrypt /MyData/MyMP
  ```

- You can also force encryption. Enter the following command at the CLI:

  ```
  svm decrypt -f [mount_point]
  ```

  For example:

  ```
  svm decrypt -f /MyData/MyMountPoint
  ```

NOTE: The `-f` option restarts a Linux machine and decrypts the data partition. The `-f` option is for users who want to decrypt a Linux data partition that is in use.

For more information about `svm` parameters, see Linux commands for CloudLink Agent.

Decrypt a volume encrypted using CloudLink version 5.0

Use this procedure to decrypt a volume encrypted using CloudLink version 5.0.

Steps

1. Add a drive that has a similar or more disk space as the encrypted drive.

For example:

```
/dev/sdc
```

This disk must not contain any data as it is overwritten with contents from the encrypted disk.

2. Mount the new disk.

   For example, mount it to the following directory:

   ```
   /MyData/AdditionalMP
   ```

3. Decrypt the encrypted data disk by entering the following command:

   ```
   svm decrypt [encrypted_mount_point] [additional_mp]
   ```

   For example:

   ```
   svm decrypt /MyData/MyMP /MyData/AdditionalMP
   ```

   All the decrypted data is now on an additional disk `/dev/sdc`, which is mounted to `/MyData/AdditionalMP`.

   If you want to keep your decrypted data mounted to the original location (for example, `/MyData/MyMP`), complete Steps 4 to 7.

   If you want to keep your decrypted data on the new mount point (for example, `/MyData/AdditionalMP`), go to Step 8.

4. Copy your data from the new mount point to the original mount point:

   ```
   mv /MyData/AdditionalMP /MyData/MyMP
   ```

5. Unmount the new disk:

   ```
   umount /MyData/AdditionalMP
   ```

6. Edit the `/etc/fstab` file.

   For example, you must update any lines that refer to the original mount point:

   ```
   /dev/sdc1 /MyData/MyMP ext4 defaults 1 2
   ```

7. Mount the original mount point `/MyData/MyMP`.
8. Reload the configuration into CloudLink by entering the following command:

   ```
   svm reload
   ```

Unlock a moved volume

Use this procedure to unlock a moved volume.

About this task

A key release policy determines the conditions under which CloudLink Center enables keys to be released (if any) when it detects that a volume is associated with a different machine than the one recorded in the CloudLink Center database. For more information, see CloudLink key release policies.

The policy may require that you manually unlock the volume. For information about the procedure for moving a volume and manually unlocking the volume, see Move an encrypted drive to another machine.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.

3. Select the checkbox next to the machine whose moved data volume you want to accept.
4. Click **Actions** > **Pending Volumes/Disks**.
   The **Pending Volumes/Disks** dialog box is displayed.
5. From the **Pending Volume/Disk** list, select the required data volume.
6. Click **Accept**.

Machine devices

This topic provides information about encrypting empty device that can be added to the PowerFlex storage node.

CloudLink encrypts empty devices so that the encrypted device can be added to the PowerFlex storage node. An empty device does not contain partitions or a file system. If the device is already added to PowerFlex, the device should be removed before you encrypt it because the encrypted device no longer contains the original data. An encrypted device can be erased so it can be used for another purpose, but this process destroys all data.

Before removing a machine from CloudLink Center, you must erase its devices to continue using them. Else, the devices of the machine remain encrypted and inaccessible.

Encrypt a PowerFlex storage node

Use this procedure to encrypt a PowerFlex storage node.

Prerequisites

The storage node must not have a file system.

About this task

By using the CloudLink Center, you can software-encrypt a PowerFlex storage node when the machine is in the Connected state. Encrypting a device is a quick operation because typically the storage node contains no data.

If more than one storage node on the machine is unencrypted, select the one you want to encrypt. You can only encrypt one device at a time. The software-encrypted device will have a new mapper. For example, `/dev/sdc` will be mapped to `/dev/mapper/ svm_sdc`. Only the new encrypted mapper (`dev/mapper/svm_sdc`) can be used as a PowerFlex storage node device.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the PowerFlex storage node you want to encrypt.
4. Click **Actions** > **Encrypt**.
   The **Encrypt** dialog box is displayed.
5. From the **Volumes/Devices** list, select the required PowerFlex device.
6. Click **Encrypt**.

   NOTE: The **Boot and Manual Data** and **Boot and All Data** encryption policies are not supported on the storage node drives installed on a PowerFlex device.

Erase a PowerFlex storage node device

Use this task to erase a PowerFlex storage node device.

Prerequisites

The machine must be in the connected state.

About this task

If more than one device is encrypted, choose the one that you want to erase.

NOTE: Erase a storage node device before you remove it.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the checkbox next to the machine whose device data you want to erase.
4. Click **Actions** > **Erase**.
5. In the **Erase** dialog box, perform the following:
   a. From the **Devices** list, select the required PowerFlex storage node device.
   b. In the **Confirm Device Name** box, enter the PowerFlex storage node device name to confirm it, and then click **Erase**.

Erase a PowerFlex storage node

Use this procedure to erase a PowerFlex storage node using the PowerFlex user interface.

Steps

1. Use the PowerFlex user interface to remove the device from the PowerFlex Data Server.

   The device type is changed to an encrypted raw device in CloudLink Center.

2. Log in to CloudLink Center.
3. Click **Agents** > **Machines**.
4. Select the storage node that you want to erase.
5. Click **Actions** > **Erase**.
6. In the **Erase** dialog box, confirm the device path and name.

   NOTE: All data on the storage node is erased.

7. The device is changed to an unencrypted raw device.

Unlock a moved device

Use this procedure to unlock a moved device.

About this task

A key release policy determines the conditions under which CloudLink Center enables keys to be released (if any) when it detects that a device is associated with a different machine than the one recorded in the CloudLink Center database. For more information, see CloudLink key release policies.

The policy may require that you manually unlock the device. For information about the procedure for moving a device and manually unlocking the device, see Move an encrypted drive to another machine.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the machine whose moved disk you want to accept.
4. Click **Actions** > **Pending Volumes/Disks**.
   The **Pending Volumes/Disks** dialog box is displayed.
5. From the **Pending Volume/Disk** list, select the required device.
6. Click **Accept**.

Manage a Self-encrypting Drive from CloudLink Center

Use this procedure to manage a Self-Encrypting Drive (SED). Manage SED option is only available if an SED license is uploaded and an SED is detected in the physical machine managed by CloudLink Center.

About this task

When CloudLink Center manages an SED, the SED is locked and the encryption key must be released by CloudLink Center to unlock the SED. When a machine with SEDs is powered on, or when an SED is removed from a machine, the SED must be unlocked by CloudLink Center.

NOTE: The Manage SED option does not change any data on an SED. It only takes ownership of the encryption key.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the SED you want to manage.
4. Click **Actions** > **Manage SED**.
   The **Encrypt** dialog box is displayed.
5. From the **Volumes** list, select the required data volume.
6. Click **Encrypt**.

Manage a self-encrypting drive from the CLI

Use this procedure to manage a SED drive from the CLI.

About this task

As an alternative to managing an SED from CloudLink Center, you can manage it from the CLI.

Steps

Enter the following command at the CLI:

```
svm manage [device_name]
```

For example:

```
svm manage /dev/sdb
```

Encrypt the devices of a machine from the CLI

As an alternative to encrypting the device of a machine from CloudLink Center, you can encrypt it from the CLI.

Prerequisites

The device must not have a file system.

About this task

The process used to encrypt the data device is the same as when encrypting from CloudLink Center. For more information, see Encrypt a PowerFlex storage node device

Steps

Enter the following command at the CLI:

```
svm encrypt [device_name]
```

For example:

```
svm encrypt /dev/sdb
```

For more information about the `svm` parameters, see Linux commands for CloudLink Agent.

Release management of a self-encrypting drive

You can release management of an SED.

About this task

The Release SED option is available only if an SED license is uploaded and an SED is detected in the physical machine managed by CloudLink Center.

When CloudLink releases an SED, the SED is unlocked and CloudLink Center releases the encryption key.

NOTE: The Release SED option does not change any data on an SED. It only releases ownership of the encryption key.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the checkbox next to the machine whose SED you want to release.
4. Click **Actions** > **Release SED**.
5. From the **SEDs** list, select the required SED.
6. Click **Release**.

Release management of a self-encrypting drive using the CLI

You can release management of an SED from the CLI.

Steps

Run the following command:

```
svm release [device name]
```

For example:

```
svm release /dev/sdb
```

Monitor the real-time progress of encryption and decryption processes

This topic provides information about monitoring progress of the encryption and decryption process on a Windows or a Linux machine.

Windows machines

Use this procedure to monitor the progress of encryption or decryption processes on a Windows machine.

About this task

You can also monitor the progress in the Machines panel. For more information, see View registered machines.

Steps

1. Click the **CloudLink Agent** icon in the Windows taskbar.
2. Click the **Volumes Status** option to display the percentage of encryption or decryption that is complete.

Linux machines

After initiating an encryption process on a Linux machine, the current encryption progress is displayed in the machine console. You can also monitor progress in the **Machines** panel. For more information, see View registered machines.

The progress of decryption processes is not available for Linux machines.

View volume encryption policy compliance

This topic provides information about the different states of volume encryption policies.

For each machine volume, CloudLink Center indicates its state relative to the volume encryption policy of the machine group.

For a selected machine, you can view the policy states for each volume on the Machines panel. For more information, see View registered machines.

Volume policy states include:

- **OK**—The volume complies with the volume encryption policy for the machine group. Any policy exemptions have been allowed.
- **Violated**—The volume does not comply with the volume encryption policy for the machine group.

Exempt volumes from encryption in CloudLink Center

This topic provides information about exempting volumes from encryption.

The volume encryption policy for a machine group applies to all volumes for all machines belonging to the group. You may want to exempt a particular volume from the policy. For example, you might have assigned the Boot and All Data volume encryption policy to a machine group. However, you want one volume on an individual machine in that group to be decrypted. When you manually decrypt this volume, it no longer complies with the volume encryption policy for the machine group.

To indicate that a volume does not comply with the machine group's volume encryption policy, CloudLink Center:

- Triggers an alarm
- Displays an exempt link next to the volume name on the Machines panel

You can specify that you want to allow the non-compliance by allowing an exemption to the volume.

Only decryption of a volume that does not comply with the existing volume encryption policy generates an alarm. Encryption of a volume will not make a volume non-compliant.

Exempt a volume from the volume encryption policy

Use this procedure to exempt a volume from the volume encryption policy.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the machine whose volume you want to exempt.
4. Click **Actions** > **Exempt**.

Change the CloudLink Center IP address

This topic provides information about changing the CloudLink Center IP address.

If you want to change the static IP of CloudLink Center and you deployed CloudLink Agent using the static IP, you must change the CloudLink Center IP address.

You must update each CloudLink Agent with the new CloudLink Center IP address, which is used by CloudLink Agent to communicate with CloudLink Center.

Change the CloudLink Center IP address on a Windows machine

Use this procedure to change the CloudLink Center IP address on a Windows machine.

Steps

Enter the following command at the PowerShell CLI:

```
svm  /S clc_address
```

For more information about this command, see Linux commands for CloudLink Agent.

Change the CloudLink Center IP address on a Linux machine

Use this procedure to change the CloudLink Center IP address on a Linux machine.

Steps

Enter the following command at the CLI of a Linux machine:

```
svm -S clc_address [-v ]
```

For more information about this command, see Linux commands for CloudLink Agent.

Move a Machine to a different CloudLink Center

Use this procedure to move a machine to a different CloudLink Center.

About this task

For Enterprise and Microsoft Azure and Azure Stack—You may want to move a machine under the management of a different CloudLink Center. You must decrypt the volumes before moving the machine. Encrypted volumes moved to another machine are not accessible.

For PowerFlex—You may want to move a machine under the management of a different CloudLink Center. You must erase the devices before moving the machine. Moving encrypted devices to a new CloudLink Center that is not part of the CloudLink Center cluster is not supported.

However, you can add a new CloudLink Center to the cluster, and then remove the old CloudLink Center. For more information, see Add a new CloudLink Center to an existing cluster and remove the old CloudLink Center.

Steps

1. Depending on whether it is a volume or a device, do the following :
   - Decrypt any encrypted volumes on the machine
   - Erase any encrypted devices on the machine
2. Uninstall the CloudLink Agent.
3. Deploy CloudLink Agent to the machine from the new CloudLink Center.

   For information about deploying CloudLink Agent, see the *Dell CloudLink Deployment Guide* .

Add a new CloudLink Center to an existing cluster and remove the old CloudLink Center

Use this procedure to add a new CloudLink Center to an existing cluster and remove the old CloudLink Center.

Steps

1. Create a backup of the original CloudLink Center cluster. See CloudLink Center backup for more information.
2. Add a new CloudLink Center to the original CloudLink Center cluster. For more information, see Associate a server to a CloudLink Center cluster for more information.
3. Wait for the initial synchronization to complete.
4. Wait at least 10 minutes for the machines to discover the new CloudLink Center.
5. Power off the original CloudLink Center cluster.
6. Wait for all machines to connect to the new CloudLink Center cluster.
7. Remove the original CloudLink Center from the cluster. For more information, see Remove a CloudLink Center cluster server.

Unlock out-of-band data disks with an ISO image file

You can unlock data disks attached to machines that are disconnected from CloudLink Center using an `ISO` image file.

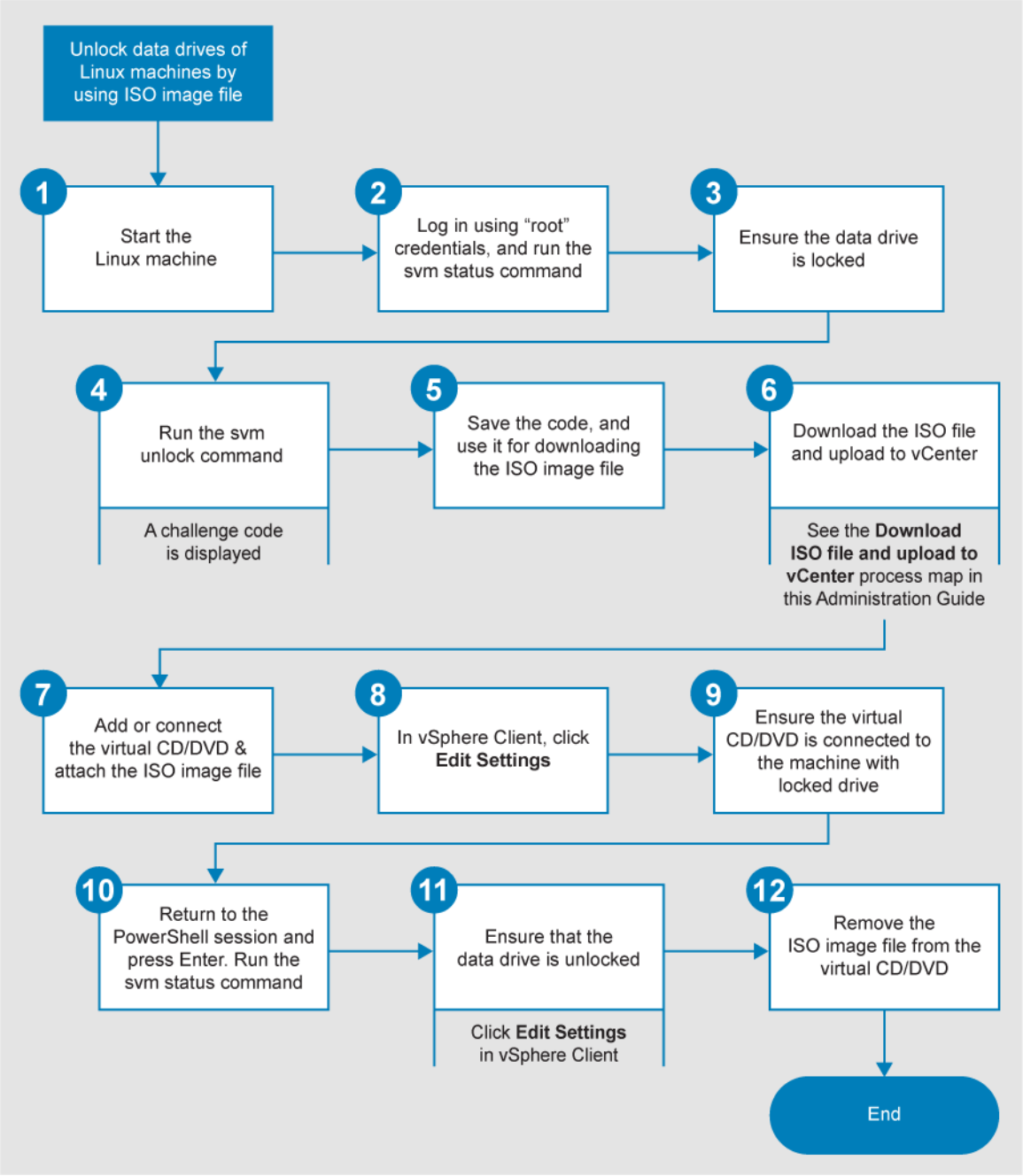
CAUTION: Because you can possibly lose data, it is recommended that you submit a support ticket at Dell Technologies Online Support so that a CloudLink support team member can assist you.

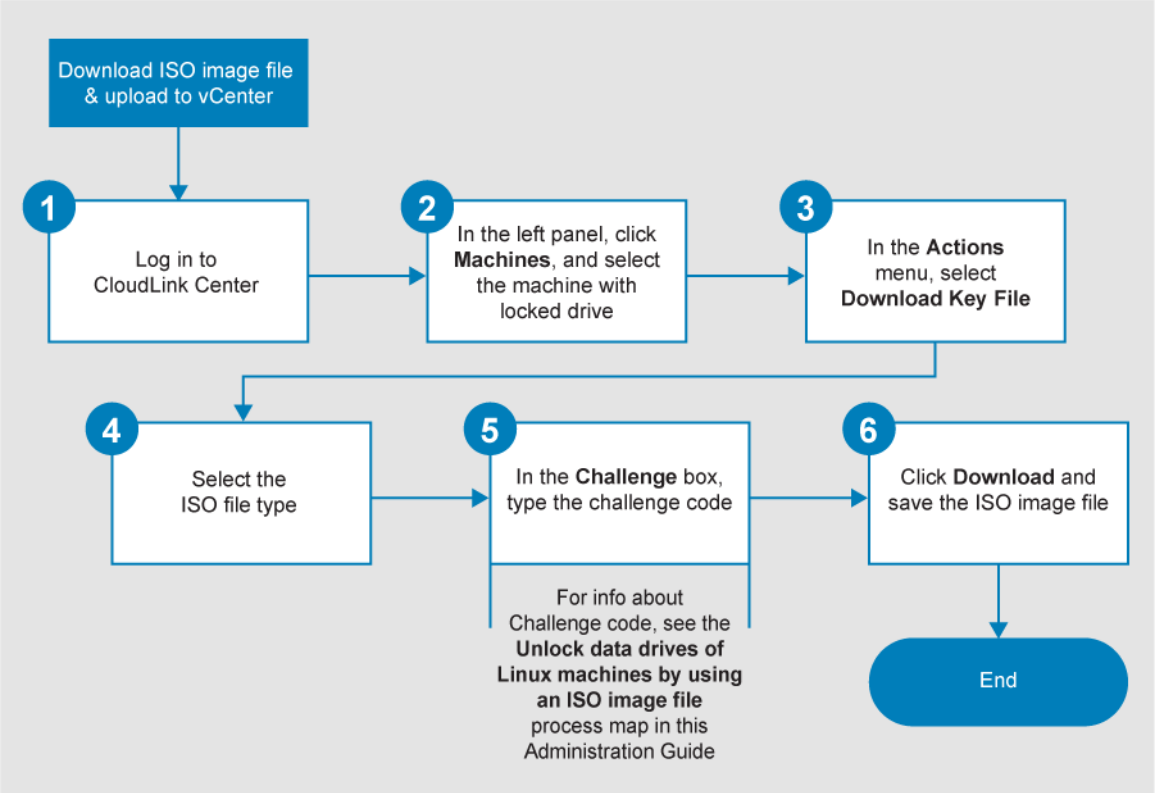
Unlock data disks of Linux machines using an ISO image file

Use this procedure when a Linux machine cannot access CloudLink Center due to networking issues, and the data disk or disks are locked.

Before you begin:

- Root Shell access on the machine that has the locked disk
- CloudLink Center administrative access
- VMware access that enables mounting an ISO image file on the machine with the locked disk


Unlock data drives of Linux machines by using ISO image file
1
Start the Linux machine
2
Log in using "root" credentials, and run the svm status command
3
Ensure the data drive is locked
4
Run the svm unlock command
A challenge code is displayed
5
Save the code, and use it for downloading the ISO image file
6
Download the ISO file and upload to vCenter
See the Download ISO file and upload to vCenter process map in this Administration Guide
7
Add or connect the virtual CD/DVD & attach the ISO image file
8
In vSphere Client, click Edit Settings
9
Ensure the virtual CD/DVD is connected to the machine with locked drive
10
Return to the PowerShell session and press Enter. Run the svm status command
11
Ensure that the data drive is unlocked
Click Edit Settings in vSphere Client
12
Remove the ISO image file from the virtual CD/DVD
End



What to do next:

The svm service may stop running on the Linux machine after the machine is reconnected to CloudLink Center. Run one of the following commands to start the svm service:

- `systemctl restart svmd`
- `service svmd start`

Unlock data disks of Windows machines using an ISO image file

Use this procedure when a Windows machine cannot access CloudLink Center due to networking issues, and the data disk is locked.

Prerequisites

- Administrative access on the machine that has the locked disk
- CloudLink Center administrative access
- VMware access that enables mounting an ISO image file on the machine with the locked disk

Steps

1. Start the Windows machine before you attempt to unlock the disk.
2. Log in as an administrator to the Windows machine that has the locked disk.
3. In PowerShell, confirm that the data disk is locked using the following command:

   ```
   svm status
   ```

4. Run the following PowerShell command at the CLI:

   ```
   svm unlock
   ```

   A challenge code is displayed. You require the code to download the `ISO` file.

5. Leave the PowerShell session open. Do not press **Enter** to continue.
6. In the **Machines** panel in CloudLink Center, select the machine with the locked disk.
7. From the **Actions** menu, select **Download Key File**.
8. From the **File Type** list, select the **ISO** image file .
9. In the **Challenge** box, enter the challenge code.
10. Type the challenge code into the **Challenge** field.
11. Click **Download**, and then save the `ISO` image file.
12. Upload the `ISO` image file to the vCenter Server datastore.
13. Use the **Edit Settings** option in vSphere Client to add or connect to that virtual CD/DVD drive, and then attach the ISO image file in the datastore.
14. Ensure that the virtual CD/DVD is connected to the machine with the locked disk.
15. On the Windows machine that has the locked disk, return to the PowerShell session and press Enter.
16. Confirm that the data disk is unlocked by running the following PowerShell command:

```
svm status
```

17. Use the **Edit Settings** option in vSphere Client to remove the ISO image file from the virtual CD/DVD.

Unlock out-of-band data disks with a RAW file

You can unlock data disks attached to machines that are disconnected from CloudLink Center using a key file.

CAUTION: Because you can possibly lose data, it is recommended that you submit a support ticket at Dell Technologies Online Support so that a CloudLink support team member can assist you.

Unlock data disks of Linux machines using a RAW file

Use this procedure when a Linux machine cannot access CloudLink Center due to networking issues, and the data disk or disks are locked.

Prerequisites

- Root Shell access on the machine that has the locked disk
- CloudLink Center administrative access

Steps

1. Start the Linux machine before you attempt to unlock the disk.
2. Log in as root to the Linux machine that has the locked disk.
3. Confirm that the data drives are locked using the following command:

```
svm status
```

4. Run the following command:

```
svm unlock
```

   A challenge code is displayed. You require this to download the `RAW` file.
5. Keep the session open. Do not press Enter.
6. On the CloudLink Center user interface, in the left panel, select the machine with the locked disk.
7. From the **Actions** menu, select **Download Key File**.
8. From the **File Type** list, select the **RAW** file.
9. In the **Challenge** box, enter the challenge code.
10. Click **Download** and save the `RAW` file.
11. Copy the `RAW` file to the root (`/`) directory of the Linux machine.

12. On the Linux machine that has the locked disk, return to the PowerShell and press **Enter**.
13. Ensure that the data disk is unlocked by running the following command:

```
svm status
```

14. The svm service may stop running on the Linux machine after the machine is reconnected to CloudLink Center. Run one of the following commands to start the svm service:
    - `systemctl restart svmd`
    - `service svmd start`

Unlock data disks of Windows machines using a RAW file

Use this procedure when a Windows machine cannot access CloudLink Center due to networking issues, and the data disk or disks are locked.

Prerequisites

- Administrative access on the machine that has the locked disk
- CloudLink Center administrative access

Steps

1. Start the Windows machine before you attempt to unlock the disk.
2. Log in as an administrator to the Windows machine that has the locked disk.
3. Confirm that the data disks are locked by running the following command:

```
svm status
```

4. Run the following command:

```
svm unlock
```

   A challenge code is displayed. You require the code to download the `RAW` image file.
5. Keep the session open. Do not press Enter.
6. On the CloudLink Center user interface, in the left panel, select the machine with the locked disk.
7. From the **Actions** menu, select **Download Key File**.
8. From the **File Type** list, select the **RAW** image file.
9. In the **Challenge** box, enter the challenge code.
10. Click **Download** and save the `RAW` image file.
11. Copy the `RAW` file to the C drive of the Windows machine.
12. On the Windows machine that has the locked disk, return to the PowerShell session and press **Enter**.
13. Ensure that the data disk is unlocked by running the following PowerShell command:

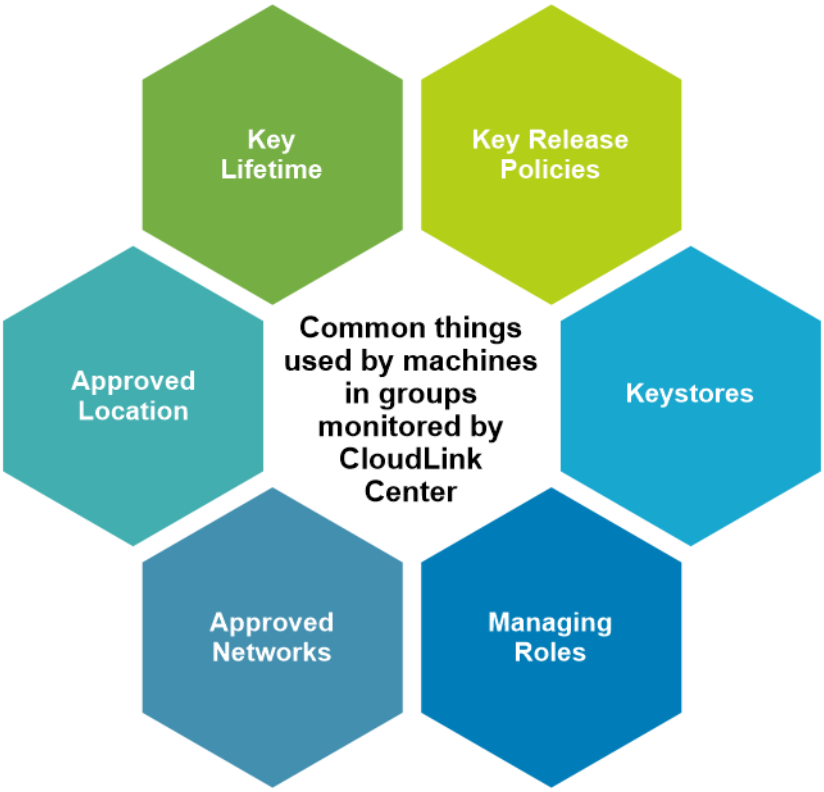
```
svm status
```

9

Manage secure machine groups on CloudLink Center

You can organize machines into groups for administrative or operational purposes. For example, you might group machines for your Finance department where the volume encryption policy requires encryption of all boot and data volumes. You might also group machines for your DevOps department where the volume encryption policy requires encryption of only boot volumes. Each machine group might have a different administrator.

Each machine must belong to a machine group. A machine is assigned to a machine group during deployment. If you do not specify a group during deployment, the machine is assigned to the built-in machine group named **Default**. You can change the machine group that a machine belongs to after deployment.



All machines in a group use the same:

- Key release policies that determine when a machine in the group can start up automatically. For more information, see CloudLink key release policies.
- For Enterprise and Microsoft Azure and Azure Stack—Volume encryption policy that determines the types of volumes that must be encrypted (boot, data, or both boot and data). Volume encryption policy applies to virtual machines (boot and data volumes) or physical machines (data volume only). Volume encryption policy does not apply to a physical machine's boot volume. For more information, see CloudLink Center volume encryption policy.
- Keystore where encryption keys are stored. For more information, see Manage encryption keystores and keys in CloudLink Center.
- Managing roles that determine the roles that administer it. Only users belonging to a managing role for a machine group can view and make changes to it.
- Approved networks from which machines in the machine group can start up automatically. For more information, see Manage approved networks for machine groups.
- Approved location that is used to verify that a machine is in the correct place. For more information, see Manage approved locations for machine groups.

- Key lifetime that determines the frequency that CloudLink Center updates encryption keys for machines in the group. Once a key is updated, the previous key is expired. By default, keys never expire, which is referred to as an infinite lifetime. You can change the key lifetime.

CloudLink key release policies

This topic provides information about the key release policies available in CloudLink.

Before CloudLink Center automatically releases keys, a machine must:

- Fulfill the requirements of a key release policy.
- Use an IP address that belongs to an approved network. For more information, see Manage approved networks for machine groups.
- Belong to an approved location. For more information, see Manage approved locations for machine groups.
- Not have been previously removed. For more information, see Remove a machine from CloudLink Center.

Key release policies may be required to enable:

- A machine to start as part of the prestartup authorization process
- Access to encrypted data volumes

If a machine does not meet the policies, CloudLink Center puts the machine in the pending state. To enable a key release policy, you must manually select a key release policy. For more information about manually enabling key release policies, see Accept or reject pending machines in CloudLink.

Key release policies are set for a machine group. For more information, see Manage secure machine groups on CloudLink Center.

Preboot unlock for PowerFlex devices

To enable preboot unlock of devices that are connected to PowerFlex machines, change all pending policies of a machine group to **Allow Automatically**. This prevents the PowerFlex machine from being placed in the pending state and allows the connected devices to be unlocked.

CloudLink Center Key Release Policies Matrix

Key release policies inform CloudLink Center whether a volume encryption key can be released to an agent when a machine is powered on.

Table 3. Key Release Policies in CloudLink Center

Key Release Policy Name	What is the Policy Checks For?
Pending Machine policy	Determines whether a machine is placed in pending state when it connects to CloudLink Center: • For the first time. • From a network or location not previously associated with the machine. Allows the machine to automatically register to CloudLink Center or remain in pending state and require manual approval. For more information, see CloudLink pending machine policy. Pending Machine policy (New Machine) • Determines if a newly added machine has an approved IP address. • If yes, allows the machine to automatically register to CloudLink Center or remain in pending state and require manual approval.
Volume Encryption policy	Determines the volumes that must be encrypted for VMs. For example, the All Data policy encrypts data on all existing data volumes. When a policy states that a specific volume encryption requirement is optional, then administrators must manually choose which volumes to encrypt using CloudLink Center. By default, this policy is set for a machine group but can be changed. For individual machines in a group, you can enable a selected volume to be exempt from the group policy. For more information, see CloudLink Center volume encryption policy.
	Volume Encryption policy: Boot and Manual Data

Table 3. Key Release Policies in CloudLink Center (continued)

Key Release Policy Name	What is the Policy Checks For?
	Encrypt boot volumes. Data volumes are not encrypted.
	Volume Encryption policy: All Data Encrypt data volumes. Boot volumes are not encrypted. • **Windows machines**: Data volumes are auto encrypted when added. • **Linux machines**: Based on the policy settings, the data volumes are automatically encrypted.
	Volume Encryption policy: Boot and All Data Boot and all data volumes are encrypted. • **Windows machines**: Data volumes are auto encrypted when added. • **Linux machines**: Based on the policy settings, the data volumes are automatically encrypted.
	Volume Encryption policy: Manual The boot and data volumes need not be encrypted. This implies that encryption is optional on all volumes.
Approved networks	Approved networks are definitions within CloudLink Center that specify which IP addresses or address ranges are allowed for CloudLink agents that request for keys. CloudLink Center automatically releases keys if the IP address of a machine is on an approved network. CloudLink Center supports IPv4 and IPv6 addresses for specifying IP addresses for approved networks. Check if the machine IP address belongs to a list of approved networks for its machine group. If the machine belongs and all other key release policies are met, the machine is allowed to start. If not, put the machine in Pending state. However, during initial startup following deployment of CloudLink Agent, CloudLink Center allows the machine to go directly to the connected state. If the machine is restarted, CloudLink Center puts it in the pending state. If a machine starts up and its IP address does not belong to an approved network, the machine is put in the pending state. For more information, see Manage approved networks for machine groups.
Approved locations	Sometimes it is necessary to limit data access by location. Data regulations may mandate that machines only containing specific data reside and run in specific locations or data centers. CloudLink agent can validate that a machine is running in an approved location, within VMware vCenter, Microsoft Azure, or Amazon Web Services environments. For example, you can create an approved location named "US Datacenter" and select it as an approved location for multiple machine groups. An approved location is created using specific locations from a cloud provider. For example, the VMware vCenter provider allows data centers, clusters, ESXi hosts, or vCenter folders to be specified as an approved location. When a machine starts up, its location is checked against a list of approved locations for its machine group. If a machine group has no assigned approved locations, then no check is performed. A machine is allowed to start if it belongs to an approved location for its machine group and all key release policies are met. The location of a machine is periodically checked while it is running to ensure that it has not been moved. Approved locations are also checked when a machine registers with CloudLink Center. CloudLink validates that a machine is running in an approved VMware vCenter, Microsoft Azure subscription, Microsoft Azure Stack subscription, or Amazon Web Services location. For more information, see Manage approved locations for machine groups.

Types of CloudLink key release policies

This topic provides information about the different types of CloudLink key release policies.

CloudLink pending machine policy

This topic provides information about the CloudLink pending machine policy.

The pending machine policy determines whether a machine is placed in the pending state when it connects to CloudLink Center for the first time. Or, when it connects from a network or location not previously associated with the machine. You can select to allow the machine to automatically register with CloudLink Center, or remain in the pending state and require manual approval.

The following pending machine policy is applicable only to Enterprise and Microsoft Azure and Azure Stack:

New Machine—If a new machine that has an approved IP address is added to CloudLink Center, you can choose to allow it to automatically (default) connect to CloudLink Center or require manual approval to connect to CloudLink Center.

This policy applies to VMs.

Change the pending machine policy the same way you change a key release policy. For more information, see Change key release policies of a machine group on CloudLink Center.

CloudLink Center volume encryption policy

This topic provides information about the volume encryption policy in CloudLink Center.

Volume encryption policy determines which volumes must be encrypted for virtual or physical machines. For example, the All Data volume encryption policy requires that all existing data volumes on a machine be encrypted.

Volume encryption policy applies to boot or data volumes for VMs.

Volume encryption policy is set for a machine group and can be changed at any time. For more information, see Manage secure machine groups on CloudLink Center.

For an individual machine in a machine group, you can enable a particular volume to be exempt from the group policy. For more information, see View volume encryption policy compliance.

Types of Volume Encryption Policies in CloudLink Center

This topic provides information about the different types of volume encryption policies available in CloudLink Center.

- **Boot and Manual Data**—The boot volume must be encrypted. Data volumes are not required to be encrypted.
- **All Data**—Data volumes must be encrypted. The boot volume is not required to be encrypted.
  - On Windows machines, data volumes that are added are automatically encrypted.
  - On Linux machines, Based on the policy settings, the data volumes are automatically encrypted.
- **Boot and All Data**—The boot and all data volumes must be encrypted.
  - On Windows machines, data volumes added to the machine are automatically encrypted.
  - On Linux machines, Based on the policy settings, the data volumes are automatically encrypted.
- **Manual**—The boot and data volumes are not required to be encrypted.

NOTE: Key release requires that the boot volume is encrypted. For more information, see CloudLink key release policies.

NOTE: The **Boot and Manual Data** and **Boot and All Data** encryption policies are not supported on the storage node drives installed on a PowerFlex device.

Handle existing encrypted Windows volumes in CloudLink

You can deploy CloudLink Agent to Windows machines with volumes that are already encrypted by BitLocker. During deployment, these volumes remain encrypted, and are put under CloudLink Center management.

CloudLink Center machine group properties

This topic provides information about the group properties of a machine in CloudLink Center.

A machine group is composed of several properties. You define many of these properties when creating a machine group. You can modify these properties later. Other properties are for informational purposes only and cannot be changed.

Machine group properties include:

- **Name** —The unique name of the machine group.
- **Description (optional)**—A brief description of the machine group.
- **Keystore** —The keystore used by all machines in the group. For information, see Set the current keystore.

- **Managed By**—The names of the roles that administer this machine group.
- **Approved Networks (optional)**—The networks to which machines in this group belong. The networks must be defined as an approved network. For more information, see Manage approved networks for machine groups.
- **Approved Locations (optional)**—An approved location is used to verify that a machine is in the correct place when the machine starts. For more information, see Manage approved locations for machine groups.
- **Shutdown on Locations Failure**—Whether to automatically power off a machine that starts up outside of an approved location.
- **Registration Code**—The code used when deploying CloudLink Agent to a machine to assign it to this machine group.

  During deployment of CloudLink Agent to a machine, it is assigned to the Default machine group if no group registration code is provided. As a deployment option, you can assign the machine to another, existing group by specifying the machine group's registration code in the deployment command. For more information, see the *Dell CloudLink Deployment Guide*.
- **Volume Encryption Policy (For Enterprise and Microsoft Azure and Azure Stack)**—The volume encryption policy that applies to all machines in a group. For more information, see CloudLink Center volume encryption policy.
- **Manage SED Drives** —Select **Enabled** to have CloudLink manage SED encryption keys. When a machine with SEDs is registered with this machine group, CloudLink Center controls releasing keys to all SEDs in that machine.

  If you select **Disabled**, a CloudLink administrator must manually select each SED in the machine to control encryption key release.

  This property is only available if you have an SED license and select either the **All Data** or **Boot and All Data** encryption policy.
- **Machine Agent Upgrade** —Whether to automatically upgrade the Agent of a machine CloudLink when you upgrade CloudLink Center.
- **Max Usage Since Last Reset** —The maximum number of encrypted machine instances (instance or socket license) or encrypted capacity used in this group since the last reset. This information might be useful if you are assessing your peak license usage over a specific time frame.
- **Current Usage**—The number of machine instances currently encrypted (instance or socket license) or encrypted capacity used in a group.
- **Key Lifetime**—The frequency that CloudLink Center updates or rotates encryption keys for machines (drives or volumes) in the group. Once a key is updated, the previous key expires. By default, keys never expire, which is referred to as an infinite lifetime. For example, when you set the Key Lifetime property of a machine group to 180 days, CloudLink automatically performs a KEK rotation for all encrypted volumes and managed SEDs defined in that machine group.

  You can trigger automatic encryption key changes based on a time interval of days. For example, if you specify an interval of one day, new encryption keys are generated every day.

  When modifying a key lifetime, you can change the following values:

  - Infinite—The encryption key never expires
  - <Number> days—A list of preset values for the number of days before expiry
  - Custom—A number of days before expiry that you specify
- **Policies** —The current setting of the key release policies that control automatic startup of machines in the group. For more information, see CloudLink key release policies.

View machine groups on CloudLink Center

You can view a list of machine groups for which you have permission to view.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
   The machine groups are displayed.

Create a machine group to CloudLink Center

A machine group must exist before you can assign a machine to it.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
3. Click **Add**.
4. In the **Add New Group** dialog box, provide the following information:
   - Name
   - Description
   - Volume Encryption Policy
   - Manage SED Drives

     NOTE: This property is only available if you have an SED license and select either the All Data or Boot and All Data encryption policy.

   - Keystore
   - Managed By
   - Approved Networks (optional)
   - Key Lifetime
   - Machine Agent Upgrade
5. Click **Add**.

Modify a machine property on a CloudLink Center machine group

You can modify all properties for a machine group, except the name.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
3. Select the check box next to the machine group you want to modify.
4. Click **Actions** > **Modify**.
5. In the **Modify Group** dialog box, modify or provide the following machine properties:
   - Description
   - Volume Encryption Policy
   - Manage SED Drives

     NOTE: This property is only available if you have an SED license and select either the All Data or Boot and All Data encryption policy.

   - Keystore
   - Managed By (not available for the Default machine group)
   - Approved Networks (optional)
   - Key Lifetime
   - Machine Agent Upgrade
6. Click **Modify**.

Change the volume encryption policy

Use this procedure to change the volume encryption policy for a machine group.

About this task

For example, a particular machine group may use the Boot and Manual Data policy. This policy requires that only the boot volume is encrypted for machines in this group. No data volumes must be encrypted. You may want to change to the All Data policy so that data volumes added to Windows machine are automatically encrypted.

Changing the volume encryption policy does not affect the boot volume or any existing data volumes. The new policy is applied when data volumes are added to the machine.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
3. Select the check box next to the machine group whose volume encryption policy you want to change.
4. Click **Action** > **Modify**.

Change the location of a machine group on CloudLink Center

Use this procedure to change the location of a machine group on CloudLink Center.

About this task

You can add an approved location to a machine group. Create an approved location before you add it to a machine group. For more information, see Manage approved locations for machine groups.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
3. Select the checkbox next to the machine group whose group location you want to change.
4. Click **Actions** > **Change Locations**.
5. In the **Change Group Approved Locations** dialog box, change or specify the following machine properties:
   - **Name**
   - **Approved Locations**
   - **Shutdown on Locations Failure**

   NOTE: If you set **Shutdown on Locations Failure** to **Yes**, you must specify the duration for which CloudLink must wait before automatically powering off the machine.
6. Click **Change**.

Change key release policies of a machine group on CloudLink Center

For a machine group, you can modify the key release policies that must be met for a machine in that group to start automatically.

Here are two examples where default policy settings cause CloudLink Center to prevent a machine from starting up automatically:

- A machine attempts to start up with a different IP address than the IP address recorded for that machine in the CloudLink Center database. CloudLink Center puts this machine in the pending state.
- A machine that is registered with CloudLink Center cannot boot, so you move its encrypted data volume or PowerFlex storage node device to another registered machine. When this machine attempts to start, CloudLink Center detects that the data volume or PowerFlex storage node device is associated with a different machine than the machine recorded in the CloudLink Center database. CloudLink Center allows the machine to start up, but puts it in the pending state and locks the moved data volume or PowerFlex storage node device to make it inaccessible.

When machines are in the pending state, you can manually accept or reject the startup. For more information, see Accept or reject pending machines in CloudLink.

To avoid having to manually accept each startup, you can change the default key release policy to allow CloudLink Center to release keys for volumes or devices of a machine when machines start up. For example, you can allow CloudLink Center to release the key for moved data volumes on startup. CloudLink Center also updates its database to associate the moved data volume with the current machine and automatically unlocks data volumes.

- For instructions about unlocking a moved volume or a device, see Unlock a moved volume and Unlock a moved device.
- For instructions about moving a data volume or a PowerFlex storage node device, see Move an encrypted drive to another machine.

Change pending policies of a machine group on CloudLink Center

For a machine group, you can modify the key release policies that must be met for a machine in that group to start automatically.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
3. Select the check box next to the machine group whose key release policies you want to change.
4. Click **Actions** > **Change Pending Policies**.
5. In the **Change Group Policies** dialog box, change or specify the following policy setting values:
   - **Require Manual Approval (default)**—If the key release policy is not met, you must manually accept the startup.
   - **Allow Automatically**—If the key release policy is not met, the startup is allowed to continue. .
   - **Allow If Address Is On The Same Subnet** —This value is only available for the **When a new IP address detected** policy.

     If the IP address of the machine is different from the address stored in the CloudLink Center database, but remains in the same subnet (/24 mask), the startup is allowed to continue. The new IP address is recorded for this machine in the database.

     This option is useful when you know that the IP address of a machine may change. For example, in some cloud environments (such as Microsoft Azure), the public IP address of a machine may change when it is powered off and restarted. A new IP address is assigned from the same subnet as the previous address.
6. Click **Change**.

Generate a registration code for a machine group on CloudLink Center

When you create a machine group, CloudLink Center generates a registration code. You may want to generate a new registration code for a machine group. For example, you might suspect that the existing machine code has been compromised.

About this task

The registration code is used to assign a machine to a machine group during deployment. For more information, see CloudLink Machine group properties.

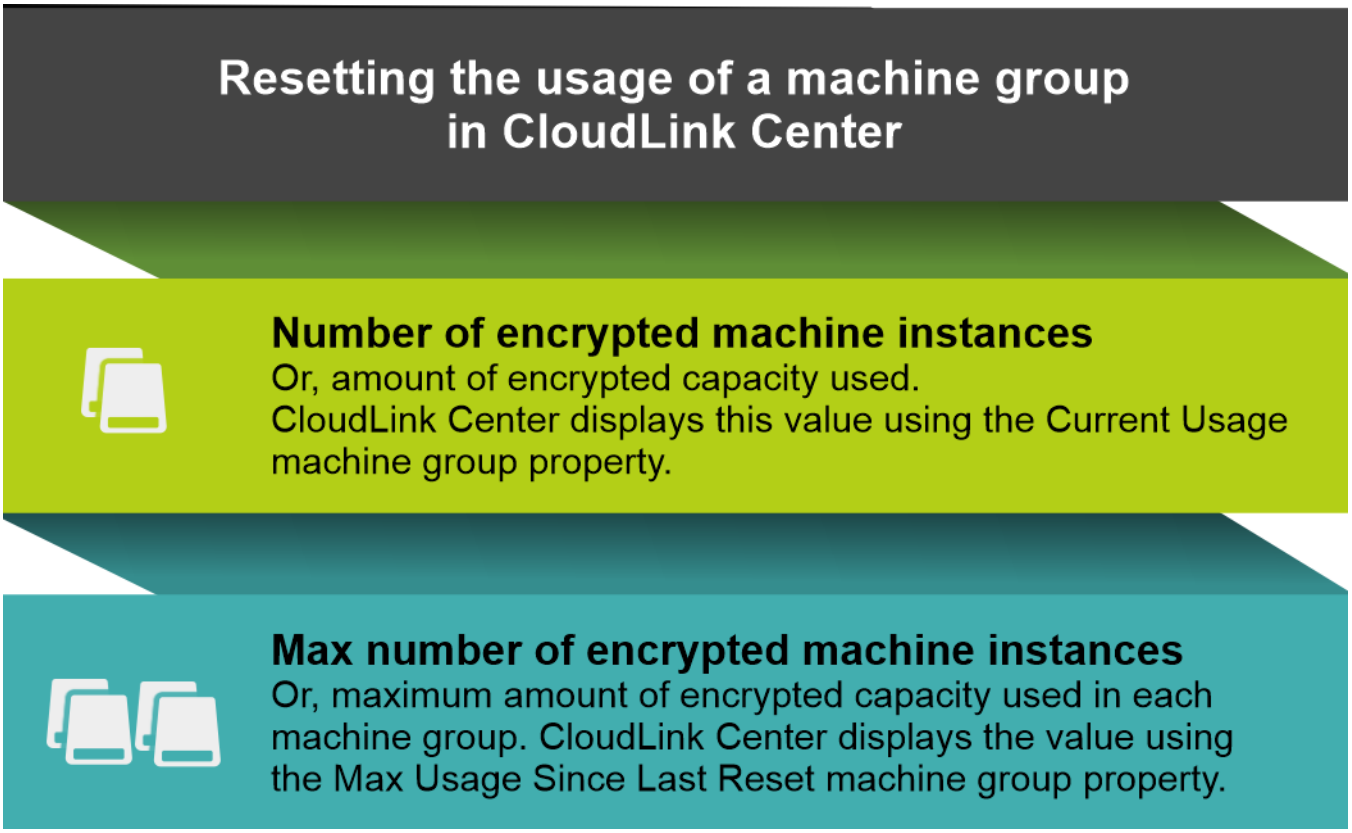
Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
3. Select the checkbox next to the machine group for which you want to generate a new registration code.
4. Click **Actions** > **Generate New Code**.
5. In the **Confirm Code Generation** dialog box, click **Generate**.

Scenarios for using maximum usage of CloudLink licenses

This topic provides information about the scenarios for using maximum usage of CloudLink licenses.

CloudLink Center keeps track of the:



These values work in combination to help you understand your license usage, as described in the following example.

- You encrypt three machines in the Production machine group. Both the Max Usage Since Last Reset and Current Usage values are 3.
- At a later time, you move one machine to another machine group. The Max Usage Since Last Reset value is 3 and the Current Usage value is 2.
- You reset the usage for the Production machine group. The Max Usage Since Last Reset value is 2 and the Current Usage value is 2.

If you reset the usage for a machine group at regular intervals (for example, on the first day of each month), you can monitor your license usage over time.

Reset the license usage for a machine group

About this task

If you are assessing your peak license usage over a specific period, you can reset the license usage for a machine group. For more information, see Scenarios for using maximum usage of CloudLink licenses.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
3. Select the checkbox next to the machine group whose license usage you want to reset.
4. Click **Actions** > **Reset Usage**.
5. In the **Confirm Usage Reset** dialog box, click **Reset**.

Delete a machine group from CloudLink Center

You can delete any machine group except the integrated Default machine group.

Prerequisites

- You must be a user who is assigned to a role that manages the machine group you want to delete.
- The machine group must not have any machines assigned to it.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machine Groups**.
3. Select the checkbox next to the machine group that you want to delete.
4. Click **Actions** > **Delete**.

Manage approved networks for machine groups

When a machine starts up, its IP address is checked against a list of approved networks (or approved IP addresses) for its machine group. If the IP address of the machine belongs to an approved network for the group and all key release policies are met, the machine is allowed to start. For more information, see CloudLink key release policies.

Each approved network is assigned a unique name, which allows approved networks to be reused across machine groups. For example, you might create an approved network named CloudLink Lab and select that name to easily specify it as an approved network for machine groups.

If a machine starts up and its IP address does not belong to an approved network for the group, the machine is put in the pending state.

There is one circumstance in which a machine can automatically start if its IP address does not belong to an approved network. On initial startup following deployment of CloudLink Agent, CloudLink Center allows the machine to go directly to the connected state. However, if the machine is restarted, CloudLink Center puts it in the pending state.

CloudLink Center supports IPv4 and IPv6 addresses for specifying IP addresses for approved networks.

View approved networks on CloudLink Center

You can view the list of approved networks that have been defined in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Approved Networks**.
   The list of approved networks is displayed.

Add an approved network to CloudLink Center

About this task

You add approved networks that you want to assign to machine groups. Adding an approved network involves defining its name and, optionally, description. After adding an approved network, you specify its IP addresses. For more information, see Add IP addresses to an approved network in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Approved Networks** > **Add**.
3. In the **Add Network** dialog box, provide the following:
   - **Name**—The unique name of the approved network. This name makes it easier to assign to a machine group.
   - **Description** (optional)—A brief description of the approved network.
4. Click **Add**.

Add IP addresses to an approved network in CloudLink Center

You add the IP address that you want automatically approved to each approved network. CloudLink Center supports IPv4 and IPv6 addresses for specifying IP addresses for approved networks.

Prerequisites

Before you can add IP addresses, the approved network must exist. For information, see Add an approved network to CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Approved Networks** > **Add**.
3. Select the checkbox next to the network to which you want to add an IP address.
4. Click **Actions** > **Add IP to Network**.
5. In the **Add IP to Network** dialog box, select the required IP address type and enter the IP address.
   The IP address format listed depends on the IP address type that you select from the **Type** list.
6. From the **Type** list, select one of the following IP address type:
   a. If you select IP Address, then enter the IP address in the **IP** box.
   b. If you select the IP Addresses Range, then specify the Start IP and End IP range in the **Start IP** and **End IP** box.
   c. If you select CIDR, then specify a network of IP addresses using CIDR in the **CIDR** box.
7. Click **Add**.

Edit IP addresses of an approved network in CloudLink Center

You can edit or an IP address for an approved network.

About this task

You can edit the approved network type and its associated IP addresses. For more information, see Add IP addresses to an approved network in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Approved Networks**.
3. Select the checkbox next to the network whose IP address you want to edit.
   The IP addresses are displayed below the table entries.
4. To edit an IP address, click **Edit**.

5. In the **Modify IP** dialog box, select the required ACL Type and enter the IP address.
   The IP address format listed depends on the ACL type that you select from the **ACL Type** list.
6. From the **ACL Type** list, select one of the following ACL type:
   a. If you select IP Address, then enter the IP address in the **IP** box.
   b. If you select the IP Addresses Range, then specify the Start IP and End IP range in the **Start IP** and **End IP** box.
   c. If you select CIDR, then specify a network of IP addresses using CIDR in the **CIDR** box.
7. Click **Modify**.

Delete IP addresses of an approved network in CloudLink Center

You can delete an IP address for an approved network.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Approved Networks**.
3. Select the check box next to the network whose IP address you want to delete.
   The IP addresses are displayed below the table entries.
4. To delete an IP address, click **Delete**.
5. In the **Confirm IP Deletion** dialog box, when prompted to confirm the request, click **Delete**.

Modify an approved network

You can modify an approved network by changing its description.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Approved Networks**.
3. Select the check box next to the network whose description you want to change.
4. Click **Actions** > **Modify**.
5. In the **Modify** dialog box, enter the description.
6. Click **Modify**.

Delete an approved network

You can delete an approved network, as long as it has not been assigned to a machine group.

Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Approved Networks**.
3. Select the check box next to the network you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm Network Deletion** dialog box, when prompted to confirm the deletion request, click **Delete**.

Manage approved locations for machine groups

This topic provides information about the approved locations for machine groups in CloudLink Center.

It is sometimes necessary to limit data access by location. Data sovereignty regulations might mandate that machines containing specific data only reside and run in specific locations or data centers.

CloudLink can validate that a machine is running in an approved VMware vCenter, Microsoft Azure subscription, Microsoft Azure Stack subscription, or Amazon Web Services location. Each approved location is given a unique name, which allows

approved locations to be reused across machine groups. For example, you can create an approved location named “US Datacenter” and select it as an approved location for multiple machine groups. An approved location is created using specific locations from a cloud provider. For example, the VMware vCenter provider allows data centers, clusters, ESXi hosts, or vCenter folders to be specified as an approved location.

When a machine starts up, its location is checked against a list of approved locations for its machine group. If a machine group has no assigned approved locations, then no check is performed. A machine is allowed to start if it belongs to an approved location for its machine group and all key release policies are met. For more information, see CloudLink key release policies.

If a machine starts up and it is not located in an approved location for the machine group, CloudLink Center can automatically power off the machine after a specified duration, or leave the machine powered on.

The location of a machine is periodically checked while it is running to ensure it has not been moved. A machine is powered off if its location has changed and it is no longer running in an approved location. Machines may also power off if you change the approved locations in CloudLink Center. This shutdown limits data exposure in an unapproved location. All power off requests is recorded as a security event that includes information about how long the machine may have been running in the unapproved location.

Approved locations are also checked when a machine registers with CloudLink Center. Registration is unsuccessful if the location of a machine is not approved for use within the machine group.

View approved locations

Use this procedure to view the list of approved locations that have been defined in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Location** > **Approved Locations**.
   The list of approved locations is displayed.

Add approved cloud providers to approved locations

A cloud provider is a server or host that provides information about the location of a machine. CloudLink Center supports VMware vCenter, Microsoft Azure, and Amazon Web Services as cloud providers. The VMware vCenter provider enables an entire vCenter, a data center, a cluster, a folder, or an ESXi host to be specified as a location.

Prerequisites

A Cloud Provider host, such as VMware vCenter.

Steps

1. Log in to CloudLink Center.
2. Click **Location** > **Cloud Providers** > **Add**.
3. In the **Add New Provider** dialog box, enter the provider name, description, and select the type of the cloud provider.
   The cloud provider values listed depends on the type of the cloud provider that you select from the **Type** list.
4. From the **Type** list, select the required cloud provider.
   a. If you select VMware vCenter, then enter the following information.
      - **Address**—Cloud provider address, in FQDN format or an IPv4 or IPv6 address
      - **Username**—Domain and username for the cloud provider account
      - **Password**—Password for the cloud provider account
   b. If you select Microsoft Azure Subscription, then enter the following information:
      - **Subscription ID**—Azure subscription ID
      - **Client ID**—Azure client ID
      - **Client Key**—Azure client key

      NOTE: Use Microsoft Azure Subscription only with machines deployed in the new portal at portal.azure.com.

   c. If you select Microsoft Azure Stack Subscription, then enter the following information:
      - **Server**—The server hostname or IP address

- **Subscription ID**—Azure subscription ID
- **Client ID**—Azure client ID
- **Client Key**—Azure client key

d. If you select Amazon Web Services, then enter the following information:

- **Region**—Physical location of the AWS cloud provider
- **Access Key ID**—Access key for your AWS account. This access key must have permission to check for running instances associated with your AWS account. At a minimum, the AWS account must have the Identity and Access Management (IAM) policy Amazon EC2 Read-Only Access.
- **Secret Access Key**—Secret access key for your AWS account

5. Click **Add**.

Add an approved location

You can add approved locations and assign them to machine groups.

Steps

1. Log in to CloudLink Center.
2. Click **Location** > **Approved Locations** > **Add**.
3. In the **Add Location** dialog box, enter the name and description of the approved location.
4. Click **Add**.

Add a cloud provider instance to an approved location

A cloud provider instance is a location in a cloud provider, such as a data center or cluster. The instance can be defined as a cloud provider or a subset of a cloud provider.

Prerequisites

A cloud provider host, such as VMware vCenter. For more information see, Add approved cloud providers to approved locations.

Steps

1. Log in to CloudLink Center.
2. Click **Location** > **Approved Locations**.
3. Select the checkbox next to the approved location to which you want to add a cloud provider instance.
4. Click **Actions** > **Add Instance to Location**.
5. In the **Add Instance to Location** dialog box, select a previously added provider and type of the provider.
6. Click **Add**.

Modify cloud providers

Use this procedure to modify a cloud provider.

Steps

1. Log in to CloudLink Center.
2. Click **Location** > **Cloud Providers**.
3. Select the checkbox next to the cloud provider that you want to modify.
4. Click **Actions** > **Modify**.
5. In the **Modify Provider** dialog box, modify the cloud provider description, username, or password. For more information, see Add approved cloud providers to approved locations.

Delete cloud providers

Use this procedure to delete a cloud provider.

Steps

1. Log in to CloudLink Center.
2. Click **Location** > **Cloud Providers**.
3. Select the check box next to the cloud provider you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm Provider Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

Modify approved locations

Use this procedure to modify an approved location.

Steps

1. Log in to CloudLink Center.
2. Click **Location** > **Approved Locations**.
3. Select the check box next to the approved location you want to modify.
4. Click **Actions** > **Modify**.
5. In the **Modify Location** dialog box, modify the description of the location.

Delete approved locations

Use this procedure to delete an approved location that is not assigned to a machine group.

Steps

1. Log in to CloudLink Center.
2. Click **Location** > **Approved Locations**.
3. Select the check box next to the approved location you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm Location Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

10

Secure CloudLink Center agents using mutual authentication

In CloudLink versions earlier than 7.1.3, agents are typically installed by using certificates that are internally generated by agents. In CloudLink 7.1.3 and later versions, you could configure third-party signed CA certificate on a CloudLink Center server which was used to generate unique individual certificates for each agent. In addition to this, starting from CloudLink 8.0 and later versions, you can enhance your data security by using mutual authentication. Mutual authentication is a security process where the CloudLink agent and the server exchange, verify, and trust certificates of each other. This occurs while the CloudLink agent is connecting to the CloudLink server. The mutual authentication process involves the following certificates:

- securevm.auth—The CA certificate. Used to identify CloudLink Center server to the CloudLink Agent.
- securevm.cer—The third-party signed agent certificate. Used to identify CloudLink Agent to CloudLink Center server.
- securevm.key—Read-only file which contains details of the private key.

Key aspects of this feature:

- You can use mutual authentication on either Linux or Windows agent machines.
- You cannot assign one third-party signed agent certificate to more than one agent machine. To assign certificates to agents, you must separately download certificate for each machine from CloudLink Center.
- You can opt to not use third-party signed certificate for connecting agent machines to CloudLink.

Generate a CSR using CloudLink to get a third-party certificate for an agent machine

By using CloudLink, you can generate a CSR so it can be signed by an external CA. You can also get a certificate that is independently issued by a third party. To generate a CSR by using CloudLink, do the following:

Steps

1. On the CloudLink Center UI, click **AGENTS** > **Agent Download**.
2. Click **Actions** > **Generate CSR**.
3. In the **GENERATE CERTIFICATE SIGNING REQUEST** dialog box, enter or select data.

   NOTE: By default, the **Common Name** field is populated with the hostname of CloudLink Center.

4. Click **Generate**.
   A CSR file is generated.
5. Send it to a third-party to get a CA-approved certificate, and then upload it by selecting the **CLOUDLINK GENERATED** option when uploading to CloudLink. See Upload a third-party signed CA certificate to CloudLink.
6. After you receive the CA certificate, upload it to CloudLink so you can generate individual certificates for each agent on CloudLink, and then assign certificates to connect agents to CloudLink. See Upload a third-party signed CA certificate to CloudLink.

Change CA certificate lifetime to authenticate an agent

The Change CA Certificate option is provided to help generate a new self-signed Client Authentication with a default lifetime of 398 days.

Steps

1. On the CloudLink GUI, in the left pane, click **AGENTS** > **Agent Download**.
2. In the working pane, from the **Actions** list, select **Change CA certificate** .
3. In the **Confirm CA Certificate Change** dialog box, when prompted to confirm the request, click **Change**.

   NOTE: The Change CA certificate operation fails when one or more agents are encrypted. Decrypt all devices before configuring the CA certificate.

Upload a third-party signed CA certificate to CloudLink

To assign a third-party signed certificate to an agent, you must upload the certificate to CloudLink so you can generate, download, and assign a CA-approved certificate for each agent. To upload a certificate, do the following:

Steps

1. On the CloudLink Center user interface, click **AGENTS** > **Agent Download**.
2. Click **Actions** > **Upload CA**.
3. From the **Certificate Format** drop-down menu, select one of the following:
   - **THIRD PARTY GENERATED PEM**—Select to upload a PEM certificate that is issued by an independent CA, and then upload both the certificate and key files.

     NOTE: Only the PEM format is supported.

   - **CLOUDLINK GENERATED**—Select to upload a certificate issued by a CA whose CSR was generated by CloudLink. See Generate a CSR using CloudLink to get a third-party certificate for an agent machine.
4. To review the summary of a certificate, click **Preview**.
   Information about the certificate name, expiry date, and fingerprint is displayed.
5. Click **Upload**. The third-party signed certificate is uploaded to CloudLink. A message is displayed indicating that an SVM CA certificate is successfully uploaded. You can view more information about this in the **Actions** list.

   NOTE: Each time you upload a new third-party signed certificate, the agent machines already assigned—with third-party signed certificates and connected to CloudLink—are automatically disconnected from CloudLink.

   NOTE: You can choose to not use third-party signed certificate for connecting agent machines to CloudLink. However, if a new third-party signed certificate is uploaded to CloudLink, then the already connected agents are disconnected after restarting. To reinitiate the connection, run the following command at PowerShell: `svm /S <CloudLink IP>`.

6. Go to the **Agent Download** page, again download agent certificate for each agent machine, and then assign it to individual agent machine.
   For information about assigning a third-party signed certificate to CloudLink agents, see Assign a CA certificate and a third-party signed certificate to a CloudLink Center Windows agent or Assign a CA certificate and a third-party signed certificate to a CloudLink Center Linux agent.

Download a CA certificate and third-party signed certificate for CloudLink agent

By using the CloudLink GUI, you can generate and download a CA certificate and a certificate for each agent machine, and then assign it to individual agent machines. A third-party signed agent certificate can be used both on a Linux and Windows agent. However, any third-party signed certificate cannot be used on more than one agent. To download a certificate, do the following:

Steps

1. On the CloudLink Center GUI, click **AGENTS** > **Agent Download**.
2. Click **Download Agent Certificate**.
   The agent certificate is downloaded in a compressed file that has the `securevm.auth`, `securevm.cer` and `securevm.key` files. The .AUTH file is a CA certificate file, .CER file is a certificate file and the .KEY file is a private key file.
   NOTE: You can download the third-party certificate to any folder on a Windows system, but you must correctly specify the file folder path when running the `msiexec` or `clagent.bat` command.
3. Expand the compressed file folder.
4. Save all the three expanded files to a folder on the agent machine.
   The folder name must not have any white spaces. For example, `My Folder` is incorrect. However, `My_Folder` is accepted.

Assign a CA certificate and a third-party signed certificate to a CloudLink Center Linux agent

You can enable Linux agents to use mutual authentication during two scenarios:

- A fresh deployment of CloudLink 8.0 or later versions. See Assign a CA certificate and a third-party signed certificate during new installation of CloudLink 8.0 or later versions on Linux agent.
- Upgrade or migration of CloudLink from 7.1.5 or later versions to CloudLink 8.0 and later versions. See Assign a CA certificate and a third-party signed certificate to a Linux agent when upgrading CloudLink from 7.1.5 version.

Assign a CA certificate and a third-party signed certificate during new installation of CloudLink 8.0 or later versions on Linux agent

To download a CA certificate and a third-party signed certificate and assign it to a Linux agent associated with a newly deployed CloudLink 8.0 or later versions, do the following:

Steps

1. Install and connect an agent to CloudLink Center. See the "Install CloudLink Agent using Standard mode" section in the *Dell CloudLink Center Deployment Guide*.
2. Download a CA certificate and a third-party signed certificate for CloudLink agent.
3. Install a CA certificate, a third-party certificate and key on a Linux agent or Install and connect Linux agent to CloudLink by using Linux Installer script.
4. Enable CloudLink to authenticate an agent assigned with a third-party signed certificate.

Enable client authentication

If you are using CloudLink versions earlier than 7.1.3, then agents are installed by using certificates self-signed by the agents. However, in CloudLink 7.1.3 and later versions, the Client Authentication feature enables you to use the third-party signed certificate to connect to CloudLink.

Even after you upgrade CloudLink to 7.1.3 or later versions, the machines are still in a connected state. However, if you enable the Client Authentication mode, then the machines go to disconnected state because the agent machines are not connected to CloudLink by using the third-party signed certificate. If you want to continue your agent machines in the self-signed certificate mode, disable the Client Authentication mode.

NOTE: If you are using a CloudLink version earlier than 7.1.3 where an agent uses self-signed certificate, and you want to assign the third-party signed certificate to that agent, then you must first upgrade to CloudLink 7.1.3 or later versions. For information about upgrading to CloudLink 7.1.3 or later versions, see the *Dell CloudLink Upgrade Guide* available on the support site.

Enable CloudLink to authenticate an agent assigned with a third-party signed certificate

By default, the Client Authentication mode is disabled on CloudLink Center. To enable the Client Authentication feature, do the following:

Steps

1. On the CloudLink GUI, in the left pane, click **AGENTS** > **Agent Download**.
2. In the working pane, from the **Actions** list, select **Change Client Authentication**.
3. In the **CHANGE CLIENT AUTHENTICATION** dialog box, from the drop-down menu, select **Enabled**.
   - **Enabled**—Agent is connected by using only the third party certificate.
   - **Disabled**—Agent is connected by using either self-signed internal certificate or third-party certificate.
4. Click **Change**.
   All the connected agents are verified to detect whether they are connected to CloudLink by using a valid third-party signed certificate. If yes, status of agents is displayed as **Connected**. Else, it is indicated as **Disconnected**.
   NOTE: After uploading the third-part CA certificate to CloudLink, enable client authentication on CloudLink so all the exiting machines will get disconnected. After you individually assign the new third-party signed certificate to individual agents, they will connect to CloudLink by using the new certificates.

   To connect an agent by using a third-party signed certificate, do one of the following:
   - Assign a CA certificate and a third-party signed certificate during new installation of CloudLink 8.0 or later versions on Linux agent.
   - Assign a CA certificate and a third-party signed certificate to a CloudLink Center Windows agent.

Install a CA certificate third-party certificate and key on a Linux agent

To install a Linux agent and connect it to CloudLink by using the Linux Installer package when newly deploying CloudLink 8.0 or later versions, do the following:

Steps

1. Download a CA certificate and a third-party signed certificate for CloudLink agent.
2. Ensure that all the three expanded files are stored to the `/tmp` directory on the agent.
   NOTE: Unlike for Windows agents, you must save the CA certificate and the third-party signed certificate for Linux agents only in the `/tmp` directory.
3. Use a CLI application such as wget or curl.
4. Enter one of the following commands:
   - `wget -O clagent.sh https://clc_address/cloudlink/agent`
   - `curl -o clagent.sh https://clc_address/cloudlink/agent`

   where, *clc_address* is the CloudLink Center server address. The *clc_address* must be in either the FQDN, IPv4, or IPv6 format.
5. Run the following command at the Command Line Interface (CLI): `rpm -ivh <rpm package> svm -v -S <CloudLink IP> -G <Group code> -C /tmp/ -M /tmp/`
   The agent is installed and successfully connected to CloudLink by using the thirty-party signed certificate. After successful connection, the state of agent will be either **Pending** or **Connected** based on the Approved Network key release policy you have set up.
6. Delete the certificates and key files from the folder every time after installing an agent by using a CA certificate and a third-party signed certificate.

Install and connect Linux agent to CloudLink by using Linux Installer script

To install a Linux agent and connect it to CloudLink by using the Linux Installer script when newly deploying CloudLink 7.1.3 or later versions, do the following:

Steps

1. Download a CA certificate and a third-party signed certificate for CloudLink agent.
2. Ensure that all the three expanded files are saved to the `/tmp` directory on the agent.

   NOTE: Unlike for Windows agents, you must save the CA certificate and the third-party signed certificate for Linux agents only in the `/tmp` directory.

3. Use a CLI application such as wget or curl.
4. Enter one of the following commands:
   - `wget -O clagent.sh https://clc_address/cloudlink/agent`
   - `curl -o clagent.sh https://clc_address/cloudlink/agent`

   where, *clc_address* is the CloudLink Center server address. The *clc_address* must be in either the FQDN, IPv4, or IPv6 format.

5. Run the following command at the Command Line Interface (CLI): `sh clagent.sh -S <CloudLink IP> -G <Group code> -C /tmp/ -M /tmp/`
   The agent is installed and successfully connected to CloudLink by using the thirty-party signed certificate. After successful connection, the state of agent will be either **Pending** or **Connected** based on the Approved Network key release policy you have set up.
6. Delete the certificates and key files from the folder every time after installing an agent by using a CA certificate and a third-party signed certificate.

Assign a CA certificate and a third-party signed certificate to a Linux agent when upgrading CloudLink from 7.1.5

If you have a Linux agent that is installed by using a third-party signed certificate then you can reassign a CA certificate and a third-party signed certificate to the agent. This connects CloudLink to the agent in a secure manner. To replace a third-party certificate with a mutual authentication connection, re-register a Linux agent machine to CloudLink, and then do the following:

Steps

1. Upgrade CloudLink Center from 7.1.5 version to 8.0 or later versions. See the latest version of *Dell CloudLink Center Upgrade Guide* available on the support site.
2. Download a CA certificate and a third-party signed certificate for CloudLink agent.
3. Run the following commands at the Command Line Interface (CLI):
   - `svm -M <directory where the securevm.auth file is saved>`. For example, `svm -M /home/my_folder`.
   - `svm -C <directory where the securevm.cer & securevm.key are saved>`. For example, `svm -C /home/my_folder`.

   The following message is displayed: **Configuring third party certificate**. The agent is installed with a CA certificate, third-party certificate and key, and then successfully configured.
4. Restart the Agent service on the agent VM by running the following command: `service svmd restart`.
5. Re-register the agent to CloudLink by running the command: `svm -S <CloudLink IP address>`.
   The agent is connected to CloudLink. The state of agent is either **Pending** or **Connected** based on the Approved Network key release policy that you have set up.

Assign a CA certificate and a third-party signed certificate to a CloudLink Center Windows agent

You can assign a CA certificate and a third-party signed certificate to Windows agents in two scenarios:

- A new deployment of CloudLink 8.0 or later versions. See Assign a CA certificate and a third-party signed certificate during new installation of CloudLink 8.0 or later versions on Windows agent.
- Upgrade of CloudLink from earlier versions. See Assign a CA certificate and a third-party signed certificate to Windows agents during upgradation of CloudLink from earlier versions.

Assign a CA certificate and a third-party signed certificate during new installation of CloudLink 8.0 or later versions on Windows agent

To download a CA certificate and a third-party signed certificate and assign it to a Windows agent associated with a newly deployed CloudLink 8.0 or later versions, do the following:

Steps

1. Download a CA certificate and a third-party signed certificate for CloudLink agent.
2. Install and connect Windows agent to CloudLink by using Windows Installer package or Install and connect Windows agent to CloudLink by using Windows Installer script.
3. Enable CloudLink to authenticate an agent assigned with a third-party signed certificate.

Install and connect Windows agent to CloudLink by using Windows Installer package

To install a Windows agent and connect it to CloudLink by using the Windows Installer package when newly deploying CloudLink 8.0 or later versions, do the following:

Steps

1. On the CloudLink user interface, click **AGENTS** > **Agent Download**.
2. In the working pane, select the checkbox corresponding to **64-bit Installer Package**.
3. Click **Download Selected**.
   The `securevm-windows-x64.msi` file is downloaded.
4. Run the following command at the Command Line Interface (CLI): `msiexec /i securevm-windows-x64.msi CLOUDLINKCENTER=<CloudLink_IP> CAPATH=<folder where the securevm.auth is saved> CERTPATH=<folder where the securevm.cer and securevm.key are saved>`
   The agent is installed and successfully connected to CloudLink by using the thirty-party signed certificate. After successful connection, the state of agent will be either **Pending** or **Connected** based on the Approved Network key release policy you have set up.
5. If you want the agent machine to operate by using the third-party signed certificate mode, Enable CloudLink to authenticate an agent assigned with a third-party signed certificate.
6. Delete the certificate and key files from the folder every time after installing an agent by using a CA certificate and a third-party signed certificate.

Install and connect Windows agent to CloudLink by using Windows Installer script

To install a Windows agent and connect it to CloudLink by using the Windows Installer script when newly deploying CloudLink 8.0 or later versions, do the following:

Steps

1. On the CloudLink user interface, in the left pane, click **AGENTS** > **Agent Download**.
2. In the working pane, select the checkbox corresponding to **Windows Installer Script**.
3. Click **Download Selected**.
   The `Windows clagent.bat agent` file is downloaded.
   NOTE: You can download an agent file by using the PowerShell interface also.
4. Run the following command at the Command Line Interface (CLI): `clagent.bat /S <CloudLink IP> /P <folder where the securevm.cer & securevm.key are saved> /M`
   The agent is installed and successfully connected to CloudLink by using the CA-signed certificate. After successful connection, the state of agent will be either **Pending** or **Connected** based on the Approved Network key release policy you have set up.
5. If you want the agent machine to operate by using the CA-signed certificate mode, Install and connect Windows agent to CloudLink by using Windows Installer script.
6. Delete the certificate and key files from the folder every time after installing an agent by using a CA certificate and a third-party signed certificate.

Assign a CA certificate and a third-party signed certificate to Windows agents during upgradation of CloudLink from earlier versions

If you have a Windows agent that is already installed by using a third-party signed certificate, then you can reassign a CA certificate and a third-party signed certificate to the agent. This connects CloudLink to the agent in a secure manner. To replace a third-party signed certificate with a mutual authentication connection, connect a Windows agent machine to CloudLink, and then do the following:

Steps

1. Run the following commands at the PowerShell interface:
   - `Svm CAPATH=<folder where the securevm.auth file is saved>`. For example, `svm capath c:\my_folder`

     The following message is displayed: **Successfully assigned the CA certificate**.
   - `Svm CERTPATH=<folder where the securevm.cer & securevm.key are saved>`. For example, `svm certpath c:\my_folder`

   The following message is displayed: **Successfully assigned the agent certificate**. The agent is installed and successfully connected to CloudLink by using the CA-signed certificate.
2. Restart the Agent service either by using Windows Service Control Manager or by running the following command at the PowerShell interface: `Restart-Service SecureVMSvc`
   After restarting, the machine status is displayed as **Connected** on the **Machines** page of CloudLink.
3. Delete the certificate and key files from the folder every time after installing an agent by using a third-party signed certificate.

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Manage certificates in CloudLink Center

Import certificate

You can import certificates that you obtained outside of CloudLink Center from a third-party certificate authority (CA).

Prerequisites

Ensure that you provide a certificate and its private key.

Steps

1. Log in to CloudLink Center.
2. On the CloudLink Center UI, click **CERTIFICATE** > **Certificates**.
3. Click **Import Certificate**.
4. In the **Import Certificate** dialog box, do the following:
   a. In the **Certificate** box, click  to select the certificate.
   b. In the **Private Key** box, click  to select the private key.
   c. From the **Group** list, select the required certificate group.
   d. Click **Import**.

Generate CSR

Use this procedure to generate a certificate signing request (CSR).

Steps

1. Log in to CloudLink Center.
2. On the CloudLink Center UI, click **CERTIFICATE** > **Certificates**.
3. Click **Generate CSR**.
4. In the **Generate Certificate Signing Request** dialog box, enter the common name, subject alternative names, organization name, organization unit, city or locality, state or province, country, and the key size and then click **Generate**.

Delete a certificate

By using CloudLink Center UI, you can delete the certificate that you imported.

Steps

1. Log in to CloudLink Center.
2. On the CloudLink Center UI, click **CERTIFICATE** > **Certificates**.
3. Select the check box next to the serial number of the certificate you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm Certificate Deletion** dialog box, when prompted to confirm the delete certificate request, click **Delete**.

Download a certificate and its private key

After a certificate is imported to the CloudLink Center, it can be downloaded at anytime.

Steps

1. Log in to CloudLink Center.
2. On the CloudLink Center UI, click **CERTIFICATE** > **Certificates**.
3. Select the check box next to the serial number of the certificate you want to download.
4. Click **Actions** > **Download Certificate/Private Key**.

Results

The certificate is downloaded in a compressed file that has the certificate and its private key files. The .cer file is the certificate file and the .pem file is the private key file.

Create a certificate group

Create a certificate group to group similar certificates under the same group name.

Steps

1. Log in to CloudLink Center.
2. On the CloudLink Center UI, click **CERTIFICATE** > **Certificate Groups**.
3. Click **Add**.
4. In the **Add New Certificate Group** dialog box, enter the following:
   - **Name**—Name of the group.
   - **Description**—Description for the group.
   - **Keystore**—The keystore used by all the certificates in the group..
   - **Managed By**—The names of the roles that administer this certificate group.
   - **Alarm**—Enabled or Disabled. If the alarm is enabled, CloudLink Center generates an alarm when a certificate is nearing its expiry or if a certificate has already expired.
   - **Expiration Alert**— Specifies the scheduled time that the CloudLink Center checks for certificates nearing its expiry or expired certificates. Specify the duration for when the alert policy is applicable by providing a value.
5. Click **Add**.

Modify a certificate group

Modify a certificate group that you created.

Steps

1. Log in to CloudLink Center.
2. On the CloudLink Center UI, click **CERTIFICATE** > **Certificate Groups**.
3. Select the check box next to the certificate group name you want to modify.
4. Click **Actions** > **Modify**.
5. In the **Change Certificate Group** dialog box, modify the required values.
6. Click **Change**.

Delete a certificate group

Delete a certificate group that you do not require.

Steps

1. Log in to CloudLink Center.
2. On the CloudLink Center UI, click **CERTIFICATE** > **Certificate Groups**.
3. Select the check box next to the certificate group name you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm Certificate Group Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

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Manage CloudLink Encryption for containers

CloudLink encryption for containers enables you to encrypt shared volumes in a Kubernetes cluster. One CloudLink Center instance can support multiple Kubernetes clusters. Each Kubernetes cluster node can have multiple Encryption for Containers agents running on it, which includes one Encryption for Containers agent for each driver.

Using the CloudLink Center web interface, you can add Kubernetes clusters on which you can deploy containerized applications.

CloudLink supports the following:

- Kubernetes version 1.20 and 1.21 (For CloudLink Center 7.1.3-8.0)
- Kubernetes version 1.18 to 1.19 (For CloudLink Center 7.1)
- Kubernetes version 1.14 to 1.17 (For CloudLink Center 7.0)
- VMware Tanzu Kubernetes version 1.1 or later
- OpenShift Cluster version 4.3 or later
- Storage types that support Container Storage Interface (CSI):
  - Generic NFS (Supports File Systems)
  - PowerScale 1.5.0 (Supports File Systems). The CSI specification versions supported by CloudLink when deployed on PowerScale: CloudLink 7.1–7.1.2 (CSI v1.2.0) and CloudLink 7.1.3–8.0 (CSI v1.5.0).
  - PowerFlex (Supports File Systems and Raw Block Volume provisioning). The CSI specification versions supported by CloudLink when deployed on PowerFlex: CloudLink 7.0 (CSI v1.1.3), CloudLink 7.1–7.1.2 (CSI v1.3.0), and CloudLink 7.1.3–8.0 (CSI v1.5.0).
  - PowerStore (Supports File Systems). The CSI specification versions supported by CloudLink when deployed on PowerStore: CloudLink 7.1.3–8.0 (CSI v2.0).
- FIPS validated dm-crypt crypto module for container block volume encryption

Change Kubernetes server certificate lifetime

Use this procedure to change the maximum lifetime for Kubernetes server certificates.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Information**.
3. Click **Change Lifetime**.
   The **Change Cetificate Lifetime** dialog box is displayed.
4. In the **Years** box, enter the required number of years, and then click **Apply**.

Change Kubernetes server certificate

Use this procedure to change the Kubernetes server certificate if required. Change Kubernetes certificates if the hostnames in the certificate are no longer valid.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Information**.
3. Click **Actions** > **Change Server Certificate**.
4. In the **Confirm Certificate Change** dialog box, when prompted to confirm the request, click **Change**.

Download Kubernetes server certificate

Use this procedure to download the current Kubernetes server certificate.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Information**.
3. Click **Actions** > **Download Server Certificate**.
   The Kubernetes server certificate is downloaded to your Downloads folder.

Generate a CSR for Kubernetes

Use this procedure to generate a certificate signing request (CSR), which involves CloudLink Center generating a private key and signing the request. The request is then fulfilled by a certificate authority (CA) and the final certificate is uploaded to CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Kubernestes** > **Information**.
3. Click **Actions** > **Generate CA CSR**.
4. In the **Generate Certificate Signing Request** dialog box, enter the name, organization, organization unit, city or locality, state or province, and select the country, and then click **Generate**.

Upload Kubernetes server CA-signed certificate

Use this procedure to upload a new certificate and an optional private key, the web server restarts and the connection is terminated. After uploading a certificate signed for CloudLink Center, verify the subject, end date, and fingerprint to ensure that it is the correct certificate.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Information**.
3. Click **Actions** > **Upload CA Signed PEM**.
   The **Upload Kubernetes CA Certificate** dialog box is displayed.
4. In the **Certificate** box, click  to select the certificate, and then click **Upload**.

Download the Kubernetes Helm package

Use this procedure to download the Kubernetes Helm package.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Download**.
3. Select the check box next to the CloudLink Kubernetes Helm package, and then click **Download Selected**.
   The Kubernetes Helm package is downloaded to your Downloads folder.

Download the Kubernetes node plugin and dockerfile

Use this procedure to download the Kubernetes node plugin and dockerfile.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Download**.
3. Select the check box next to the CloudLink Kubernetes Node Plugin and Dockerfile, and then click **Download Selected**. The Kubernetes node plugin and dockerfile is downloaded to your Downloads folder.

Add a Kubernetes cluster

Use this procedure to add a Kubernetes cluster.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Clusters** > **Add**
3. In the **Add New Cluster** dialog box, enter a cluster name, description, and select a keystore, key release policy mode, and the managed by role that administers the Kubernetes cluster.
4. Click **Add**.

Modify a Kubernetes cluster

Use this procedure to modify a Kubernetes cluster.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Clusters**.
3. Select the check box next to the Kubernetes cluster you want to modify.
4. Click **Actions** > **Modify**.
5. In the **Modiify Cluster** dialog box, enter the description and select a keystore, key release policy mode, and managed by role, and then click **Modify**.

Generate a new Kubernetes cluster certificate

Use this procedure to change the existing Kubernetes cluster certificate by generating a new certificate.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Clusters**.
3. Select the check box next to the Kubernetes cluster whose certificate you want to change.
4. Click **Actions** > **Generate New Certificate**.
5. In the **Confirm Kubernetes Cluster Certificate Change** dialog box, when prompted to confirm the request, click **Generate**.

Delete a Kubernetes cluster

Use this procedure to delete a Kubernetes cluster.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Clusters**
3. Select the check box next to the Kubernetes cluster you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm Cluster Deletion** dialog box, when prompted to confirm the cluster delete request, click **Delete**.

View the event history of a Kubernetes cluster

Use this procedure to view the event history of a Kubernetes cluster.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Clusters**.
3. Select the check box next to the Kubernetes cluster whose event history you want to view.
4. Click **Actions** > **Show Event History**.

View Kubernetes nodes

Use this procedure to view the Kubernetes nodes.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Nodes**.

View Kubernetes volumes from Kubernetes clusters or Kubernetes nodes

Use this procedure to view the Kubernetes volumes.

Steps

1. Log in to CloudLink Center.
   You can view the Kubernetes volumes either from **Kubernetes Clusters** or from **Kubernetes Nodes**.
2. To view the Kubernetes volumes from Kubernetes Clusters:
   a. Click **Containers** > **Kubernetes Clusters**.
   b. Select the check box next to the Kubernetes cluster whose volumes you want to view.
   c. Click **Actions** > **Show Volumes**.
   The Kubernetes volumes are displayed.
3. To view the Kubernetes volumes from Kubernetes Nodes:
   a. Click **Containers** > **Kubernetes Nodes**.
   b. Select the check box next to the Kubernetes node whose volumes you want to view, and then click **Show Volumes**.
   The Kubernetes volumes are displayed.

Accept Kubernetes volumes

Use this procedure to accept a pending Kubernetes volume.

Prerequisites

You can accept a Kubernetes volume only if the volume is in the pending state.

Steps

1. Log in to CloudLink Center.
2. Click **Containers** > **Kubernetes Nodes**.
3. Select the check box next to the Kubernetes node whose volume you want to accept, and then click **Show Volumes**. The pending volumes are displayed.
4. Select the check box next to the Kubernetes volume you want to accept, and then click **Accept**.
5. In the **Confirm Accept Volume** dialog box, when prompted to confirm the request, click **Accept**.

Supported volume access modes for updating keys for Kubernetes volumes

Supported volume access modes for updating keys for Kubernetes volumes.

CloudLink Center supports three kinds of access modes for PVs or PVCs:

- ReadOnlyMany—the volume can be mounted as read-only by multiple nodes.
- ReadWriteOnce—the volume can be mounted as read/write by a single node.
- ReadWriteMany—the volume can be mounted as read/write by multiple nodes.

Table 4. Supported Volume Access Modes for Updating Key for Kubernetes volumes

Access mode	Update key for Kubernetes volume	
-	**PowerFlex CSI v1.1.5**	**NFS CSI v2.0**
ReadOnlyMany	not supported	not supported
ReadWriteOnce	supported	supported
ReadWriteMany	not supported	supported

Generate an update key for Kubernetes volumes from Kubernetes clusters or Kubernetes nodes

Update key generates a new key for the key encryption key. The following procedure explains how to update the key encryption key for Kubernetes volumes.

Steps

1. Log in to CloudLink Center.
   You can update the key encryption key for the Kubernetes volumes either from **Kubernetes Clusters** or from **Kubernetes Nodes**.
2. To generate an update key for the Kubernetes volumes from Kubernetes Clusters:
   a. Click **Containers** > **Kubernetes Clusters**.
   b. Select the checkbox next to the Kubernetes clusters for which you want to update the key.
   c. Click **Actions** > **Show Volumes**.
   d. Select the checkbox next to the Kubernetes volumes for which you want to update the key.
   e. Click **Update Key**.

f. In the **Confirm Update Key for Volume** dialog box, when prompted to confirm the request, click **Update Key**.

3. To generate an update key for the Kubernetes volumes from Kubernetes Nodes:
   a. Click **Containers** > **Kubernetes Nodes**.
   b. Select the check box next to the Kubernetes node for which you want to update the key, and then click **Show Volumes**.
   c. Select the check box next to the Kubernetes volumes for which you want to update the key, and then click **Update Key**.
   d. In the **Confirm Update Key for Volume** dialog box, when prompted to confirm the request, click **Update Key**.

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Manage CloudLink Center user roles

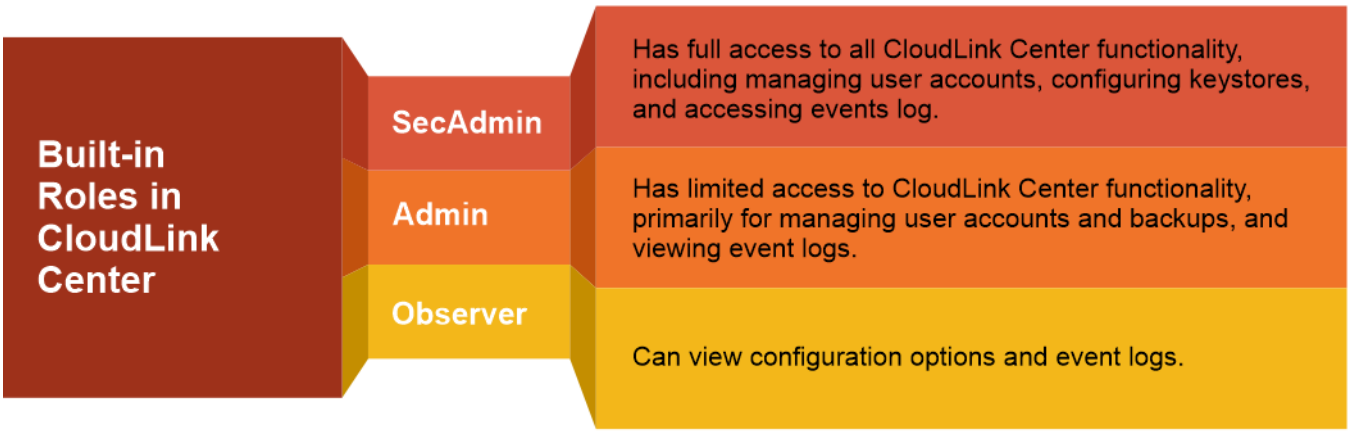
This chapter provides information about CloudLink Center built-in roles, custom roles, and managing these roles.

A role determines the permissions in CloudLink Center of users who are assigned that role. For example, you may assign one or more users the Admin role, which lets them perform many administrative functions in CloudLink Center. However, this role does not permit its users to perform functions that require the higher level of security available in the SecAdmin role, such as keystore configuration.

Built-in CloudLink Center user roles

This topic provides information about built-in roles in CloudLink Center.

CloudLink Center provides three built-in roles: SecAdmin, Admin, and Observer. You can define additional roles to customize the permissions allowed for users assigned to that role, as required.



The SecAdmin role has permission to view and manage all objects (such as a role, user, keystore, or machine group). For example, a user assigned the SecAdmin role can see all machine groups, regardless of the managing role assigned to individual machine groups.

CloudLink Center provides a built-in user account that is assigned the SecAdmin role. For more information, see Secadmin user (built-in) role. CloudLink Center does not provide built-in user accounts for Admin or Observer roles. If built-in roles do not meet your requirements, you can create custom roles. For more information, see Add CloudLink Center user roles.

Implicit user role permissions for using CloudLink Center

This topic provides information about the implicit user role permissions for using CloudLink Center.

For a given object (such as a role, user, keystore, or machine group), a role with permissions that allow changes to that object automatically includes any view permissions. For example, if a role includes the Add User or Delete User permission, the role automatically includes the View Users permission, even if it is not explicitly assigned.

Manage custom roles

This topic provides information about managing custom roles in CloudLink Center.

Every custom role has a role that administers it (referred to as the managing role). Only users belonging to a managing role for a role can administer it and administer users belonging to that role.

For example, to delete a role named TestRole, you must be a user assigned to a role that is identified as a managing role for TestRole and that managing role must contain the Delete Role permission. To add a user in TestRole, you must be a user assigned to a role that is identified as a managing role for TestRole and that managing role must contain the Add User permission.

You can assign one or more managing roles to a role. For example, an IT – Both the IT – North America role and the IT – US role might manage the East Coast role.

You assign managing roles when creating a custom role. For more information, see Add CloudLink Center user role. Managing roles for a custom role can be changed by users belonging to one of those managing roles. For more information, see Change managing roles. For information about permissions available for custom roles, see Role-Based Access Control for CloudLink .

NOTE: This guide assumes that you belong to the appropriate managing roles to perform tasks.

Role administration example

This topic provides information about a role administration example.

To understand custom role management, let us look at the following examples. For these examples, a user who is named John has three roles:

- Role Creator with only the Add Role permission
- East Coast User Management with the Add User and Change User Password permissions
- Role Deleter with Delete Role permission

To create a new role, a user must have the Add Role permission. They can acquire this permission from any role they belong to. For example, if John belongs to three roles and any of those roles contains Add Role, he is allowed to create a new role and set any role as the managing role. John has the Add Role permission from his membership in Role Creator.

To delete a role, the user must have the Delete Role permission and that permission must come from one of the role's managing roles. For example, if John wants to delete TestRole, then Role Deleter must be one of the managing roles. If only East Coast Management was one of the managing roles, John can manage users with this role but John cannot delete it.

To change the password of a user, the user must have the Change User Password permission for all the roles that a user belongs to. For example, to change John's password, another user requires the Change User Password permission using a managing role for Role Creator, East Coast User Management, and Role Deleter.

View CloudLink Center user roles

Use this procedure to view the user roles in CloudLink Center. You can view existing roles, which include the built-in roles and any custom roles that you have defined.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Roles**.
   The CloudLink Center user roles are displayed. The following information is displayed for each user role:
   - **Name**—The name of the role.
   - **Description**—A description that meaningfully identifies the purpose or scope of the role.
   - **Built In**—A flag that identifies whether the role is provided with CloudLink Center.
   - **Managed By**—The names of the roles that administer this role. For information, see Manage custom roles.

     In addition to the information provided in the Roles table, the following information is displayed for each selected user role:

   - **Permissions**—A list of permissions assigned to this role. For information, see Role-Based Access Control for CloudLink.

Add CloudLink Center user role

Use this procedure to add a user role in CloudLink Center.

About this task

If built-in roles do not meet your requirements, you can create roles to customize the permissions for users assigned to that role, as required. For more information about built-in roles, see Built-in CloudLink Center user roles.

A custom role can include any combination of permissions. For example, you might create a custom role named AdminDelegate that allows only a subset of Admin permissions, such as unlocking accounts. If an Admin user is not available, the AdminDelegate is available to help users who are locked out of their accounts. This AdminDelegate role cannot perform more sensitive activities, such as changing passwords.

A user can be assigned multiple roles, which gives them a combined set of permissions. For more information, see Add CloudLink Center users.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Roles** > **Add**.
3. In the **Add New Role** dialog box, enter the following information:
   - **Name**—The name of the role.
   - **Description**—A description that meaningfully identifies the purpose or scope of the role.
   - **Use Permissions From**—An existing role from which you want to copy permissions.
   - **Permissions**—A list of permissions that are assigned to this role. The initial permissions are the same as those for the role that is selected in Use Permissions From. Customize the permissions for the role you are creating. For information about permissions, see Role-Based Access Control for CloudLink .
   - Managed By—The names of the roles that administer this role. For information, see Manage custom roles.
4. Click **Add**.

Modify CloudLink Center user roles

Use this procedure to modify a user role in CloudLink Center. You can modify only custom user roles. You cannot modify built-in roles.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Roles**.
3. Select the check box next to the custom user role you want to modify.
4. Click **Actions** > **Modify**.
5. In the **Modify Role** dialog box, change the description and the permissions.
6. Click **Modify**.

Change managing roles

Use this procedure to change the managing role for a role.

Prerequisites

You must be a user who is assigned to a role that manages the role you want to change.

About this task

You can change the managing role for a role. For example, a change to your organization's structure may mean that a particular managing role no longer has the authority to manage one of its current roles. For more information, see Manage custom roles.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Roles**.
3. Select the check box next to the custom role you want to change.
4. Click **Actions** > **Change Managing Roles**.
5. In the **Change Managing Roles** dialog box, select the required role from the **Managed By** list.
6. Click **Change**.

Delete CloudLink Center custom user roles

Use this procedure to delete custom user roles in CloudLink Center.

Prerequisites

You must be a user who is assigned to a role that manages the role you want to delete.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Roles**.
3. Select the check box next to the custom user role you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm Role Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

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Manage CloudLink Center users and groups

This topic provides information about the different user account types that you can create, logging in using two-factor authentication and managing users in CloudLink Center.

Each person who must work with CloudLink must be defined as a user in CloudLink Center. Each user is assigned a role that determines their permissions in CloudLink Center.

A user can be assigned multiple roles, giving them a combined set of permissions.

For Enterprise and Microsoft Azure and Azure Stack—You can also use existing user accounts in the Microsoft Windows domain of your organization and assign those accounts to the appropriate CloudLink Center role.

Secadmin user (built-in) role

This topic provides information about the secadmin user role which is a built-in role in CloudLink Center.

CloudLink Center provides a built-in user with the username secadmin, and assigns the SecAdmin role to that user. For more information, see Built-in CloudLink Center user roles.

First-time access to CloudLink Center after deployment requires logging in as a secadmin user, and providing a default password. The secadmin user has to change the default password immediately after the first-time login to CloudLink Center. For information about first-time login to CloudLink Center, including the default password that is used, see the *Dell CloudLink Deployment Guide*.

After deployment, you can continue using this built-in secadmin user. You cannot delete this user, change the user name (secadmin), or change the role. These restrictions mean that no administrative activity can result in you being locked out from CloudLink Center. It is recommended that you store the built-in secadmin account credentials safely in a location which only trusted security officers can access. You can create other user accounts for daily operations.

CloudLink Center user types

This topic provides information about the user types in CloudLink Center.

You can create the following types of CloudLink Center user accounts:

- **Local**—These user accounts exist only in CloudLink.
- (For Enterprise only) **Domain**—These user accounts represent existing user accounts in the Microsoft Windows domain of your organization.
- (For Enterprise only) **Domain Group**—These user accounts represent existing group accounts in the Microsoft Windows domain of your organization. CloudLink supports only Active Directory universal groups.
- **Client** —These user accounts are intended only for use with the REST APIs. Provide a name and secret for authenticating the REST APIs from a client application. For more information, see the REST API documentation.

  For more information, see Add CloudLink Center users.

CloudLink local accounts

Local user accounts exist only within CloudLink. To log in to CloudLink Center, local users provide a password that you define when setting up the account.

CloudLink domain and domain group accounts

This topic provides information about the domain and domain group accounts in CloudLink Center.

Rather than creating users for your CloudLink needs, you can use existing accounts in the Microsoft Windows domain of your organization and assign those accounts to the appropriate CloudLink Center role.

Domain accounts are convenient because:

- If CloudLink Center has been configured to use Integrated Windows Authentication (IWA), domain users and groups can use their Windows login credentials to log in to CloudLink Center.
- If you use Windows domain groups, you can create a user in CloudLink Center that corresponds to it. You do not have to create a domain account in CloudLink Center for every user in the Windows domain group.

  This practice is useful if you have a Windows domain group for administrators and you want all administrators to have access to CloudLink Center. For security, CloudLink Center uniquely records audits for actions that are taken by each administrator. If an administrator belongs to more than one Windows domain group, and two or more of these groups have corresponding user accounts in CloudLink Center, the administrator has all the privileges of all their user accounts.

Before you can add domain accounts, you must configure a Microsoft Windows domain. For more information, see Configure Microsoft Windows domain users.

NOTE: To ensure that CloudLink can correctly check the group membership of a user, it is recommended to set the scope of the domain group to Universal.

2-Factor Authentication (2FA) in CloudLink Center

This topic provides information about the 2FA in the CloudLink Center.

For Enterprise and Microsoft Azure and Azure Stack—To access CloudLink Center, a user must provide both a username and password. For increased security, you may require a user to log in using 2FA. After providing the CloudLink Center credentials, the user must provide additional credentials generated by a third-party provider.

CloudLink supports two-factor authentication using these third-party providers:

- Google Authenticator—This application is typically installed on the user's mobile device and generates a token that the user provides as the additional credentials when logging in to CloudLink Center.
- RSA SecurID—A hardware or software token generates a tokencode that the user provides as the additional credentials when logging in to CloudLink Center.

For PowerFlex—To access CloudLink Center, a user must provide both a username and password. For increased security, you may require a user to log in using 2FA. After providing the CloudLink Center credentials, the user must provide additional credentials generated by a third-party provider.

CloudLink supports two-factor authentication using these third-party providers:

- Google Authenticator—This application is typically installed on the user's mobile device and generates a token that the user provides as the additional credentials when logging in to CloudLink Center.
- RSA SecurID—A hardware or software token generates a token code that the user provides as the additional credentials when logging in to CloudLink Center.

Two-factor authentication is available for individual local and domain users, but not for domain group users. For more information, see CloudLink Center user types.

NOTE: If you configure CloudLink Center to use Integrated Windows Authentication (IWA) and 2FA, CloudLink Center uses only the IWA credentials for domain users.

Before you can set up users for two-factor authentication using RSA SecurID, you must configure the RSA Authentication Manager. For more information, see RSA Authentication Manager.

View CloudLink Center users

Use this procedure to view the existing users in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Users & Groups**.

   The following user information is displayed about each user account:
   - **User Name**—Identifies the user in the system. If the user type is domain, this value is the exact username from the domain user account.
   - **Access Roles**—Identifies the role for the user that determines the user's access permissions. For more information, see Manage custom roles.
   - **User Type**—Identifies whether the user is a local, domain, domain group, or client user. For more information, see Add CloudLink Center users.
   - **Built-in**—Displays Yes for built-in users, and No for users you have created.
   - **Locked**—Indicates whether the user has been locked out of CloudLink Center because of too many incorrect login attempts. For more information, see Unlock CloudLink accounts.
   - **2fa**—Indicates whether the user is required to log in to CloudLink Center using two-factor authentication. For more information, see CloudLink local accounts.

     Values include:

     - None
     - RSA SecurID
     - Google

Add CloudLink Center Users

Use this procedure to add users in CloudLink Center.

Prerequisites

If you are adding a user that represents a Windows domain group, the user must exist in the Windows domain. CloudLink does not validate user presence in a Windows domain.

About this task

Every CloudLink Center user must have an account. The account information uniquely identifies the user and determines their access permissions in CloudLink Center.

NOTE: An administrator can add a new user only if the access role is a custom role, and the custom role must belong to a managing role.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Users & Groups** > **Add**.
3. In the **Add New User** dialog box, enter the following information:
   - **User Name**—Identifies the user in the system. If the user type is Domain or Domain Group, type the exact account name as defined in the Windows domain.
   - **Access Roles**—Identifies the role for the user that determines the user's access permissions. For more information, see Manage custom roles.
   - **User Type**—Identifies whether the user is a local, domain user domain group or a client.
   - **2fA Type**—Indicates whether the user is required to log in to CloudLink Center using 2FA. For more information, see 2-Factor Authentication (2FA) in CloudLink
   - **Password** (local users only)—Defines the initial password of the user.
   - **Confirm Password** (local users only)—Retype the initial password of the user.
   - **Change Password** (local users only)—Determines whether the user must change the initial password on first login (On the first login) or when they choose to (On demand).

4. Click **Add**.

Additional Account Set Up for Google two-factor Authentication

This topic gives information about the additional account set up for Google two-factor authentication

For users with 2fa Type set to Google, a dialog box similar to the following is displayed after you create a user:



You must recommend a user to scan the QR code using the Google Authenticator application on the user's mobile device. Alternatively, you can provide the user with the Account Name and Secret Key value, which the user can enter manually into the Google Authenticator application. For manual entry, ensure the time-based option is selected in Google Authenticator.

NOTE: You may also want to provide the user with the Scratch Codes. Each of these codes can be used once instead of the randomly generated token to log in to CloudLink Center. Use these codes only for exceptional circumstances, where the user may not have access to Google Authenticator and must log in to CloudLink Center.

Change user roles in CloudLink Center

Use this procedure to change the role of users to another existing role.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Users & Groups**.
3. Select the check box next to the user account whose role you want to change.
4. Click **Actions** > **Change Roles**.
5. In the **Change User Roles** dialog box, select the required role.

   You can assign more than one role to a user.
6. Click **Change**.

Change user password in CloudLink Center

Use this procedure to change the CloudLink Center password of a user.

About this task

Users can change their password, regardless of whether they have the Change User Password permission. Users can change their passwords using the Change Password option in the **User Name** menu.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Users & Groups**.
3. Select the check box next to the user account whose password you want to change.
4. Click **Actions** > **Change Password**.
5. In the **Change User Password** dialog box, type the password and retype to confirm it.
6. Click **Change**.

Change 2-Factor Authentication (2FA) for accessing CloudLink Center

Use this procedure to change whether a user is required to log in to CloudLink Center using 2FA.

About this task

If you change a user to require Google two-factor authentication, a new account must be created in Google Authenticator using the credentials that CloudLink Center displays after you change the two-factor authentication type. For information, see Additional account set up for Google two-factor authentication.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Users & Groups**.
3. Select the check box next to the user account whose 2FA you want to change.
4. Click **Actions** > **Change 2fa**.
5. In the **Change 2FA** dialog box, select the required 2FA type from the **2fa Type** list.
6. Click **Change**.

Unlock CloudLink accounts

This topic provides information about unlocking user accounts in CloudLink.

For security, local users, including the built-in secadmin user, have a limited number of attempts to provide the correct password when logging in to CloudLink Center. After the specified number of login attempts, the user is locked out of the account.

When a user is locked out of their account, CloudLink Center displays a message below the Password box indicating that the account is locked. The account is automatically unlocked after fifteen minutes. If the user requires immediate access, the account must be manually unlocked.

For more information about setting the number of login attempts, see Change the number of login attempts before lockout.

You can identify users who have been locked out on the Users page. For these users, the table displays **Yes** in the **Locked** column.

Manually unlock local CloudLink Center users

Use this procedure to manually unlock local users in CloudLink Center.

About this task

With the exception of the built-in secadmin user, users are manually unlocked from CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Users & Groups**.
3. Select the check box next to the user account whose account you want to unlock.

4. Click **Actions** > **Unlock**.
5. In the **Unlock** dialog box, select
6. Click **Change**.

Manually unlock built-in secadmin CloudLink Center users

Use this procedure to manually unlock the secadmin (built-in) user in CloudLink Center.

Prerequisites

You must have access to the CloudLink Center console interface, which is used to configure underlying network- or system-level connections. You used the console interface for the first time when deploying CloudLink Center. For more information, see the *Dell CloudLink Deployment Guide*.

About this task

The built-in secadmin user is unlocked from the CloudLink Center console interface. This interface is the only way to unlock this user if there are no other users assigned the SecAdmin role or they have also been locked out. For more information, see Built-in CloudLink Center user roles.

Steps

1. Access the CloudLink Center console interface.
2. From the **Update** menu, select **Unlock User**.
3. Click **OK**.

Delete CloudLink Center users

Use this procedure to delete the user account in CloudLink Center. You can delete any user except the built-in secadmin user.

Prerequisites

You must have the Delete User permission in a managing role for all the roles of the users.

Steps

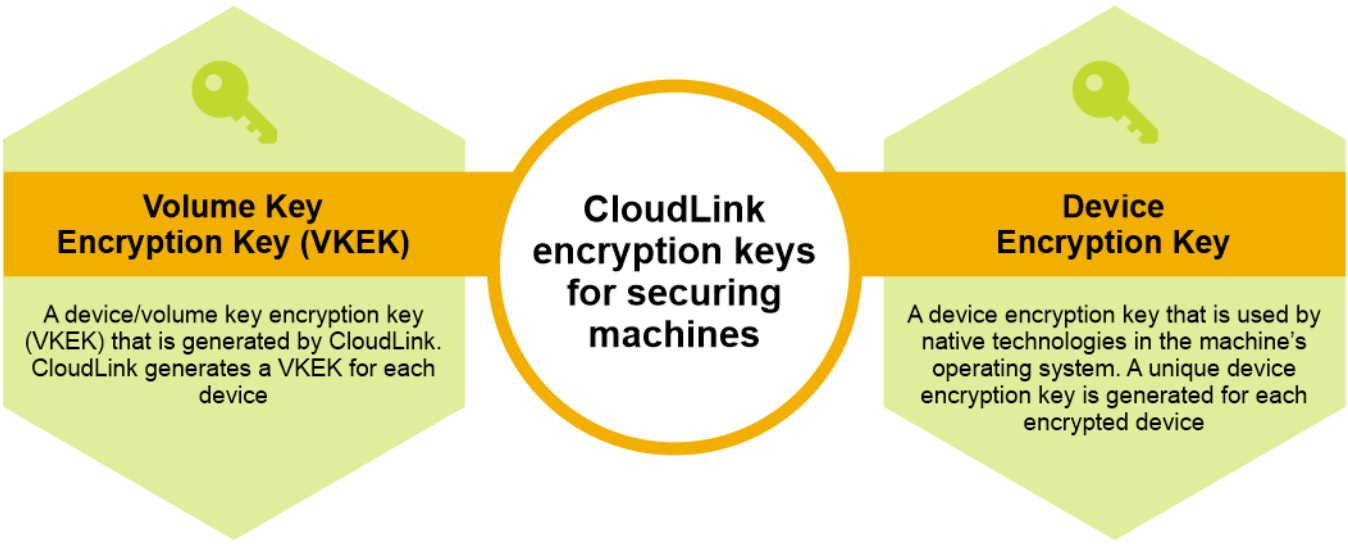
1. Log in to CloudLink Center.
2. Click **System** > **Users & Groups**.
3. Select the check box next to the user account you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm User Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

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Manage encryption keystores and keys in CloudLink Center

This chapter provides information about the encryption keystores, keys, and managing them in CloudLink Center.

CloudLink uses the following types of encryption keys to secure machines:



These keys can be stored in the CloudLink Center keystore or an external keystore. A keystore is a combination of a key location and a key protector. For more information, see Best practices for key location access control and backup.

- For a machine, volume encryption keys secure the boot or data volumes, as determined by the key release policy. For more information, see CloudLink key release policies.
- For a device, device encryption keys secure the encrypted devices. For more information, see CloudLink key release policies.
- The VKEK protects the volume or device encryption keys:
  - When CloudLink Center receives a request from CloudLink Agent to encrypt a volume on its machine, CloudLink Center generates a new VKEK in a keystore and uses it to encrypt the volume encryption key.
  - When a volume requires decryption, CloudLink Center decrypts the volume encryption key using the VKEK and sends it to CloudLink Agent.

Know the difference between the types of encryption keys that are used to secure machines. However, because volume or device encryption keys are created and managed by native technologies in the operating system of the machines, they are not discussed in detail in CloudLink documentation. Unless specified otherwise, the terms encryption keys and keys in this guide, see the *VKEK*.

During deployment, CloudLink Center creates an initial keystore for encryption keys called CloudLink Vault. For more information about using CloudLink Vault, see Manage CloudLink Vault. If you do not want to use the initial, or default, keystore to store encryption keys, external options are available, including Microsoft Active Directory, Amazon S3, S3-compatible bucket, or External KMIP Server.

Encryption keys are also encrypted, or protected, by one or more key protectors, including CloudLink Vault, SafeNet LunaSA, Microsoft Azure Key Vault, a KMIP key manager, or a password.

If you add keystores, only one keystore can be active for each machine group, but multiple keystores can be used in each CloudLink Center or CloudLink Center cluster deployment. Keys that the CloudLink Center generates are stored in a keystore. You can modify and delete keystores.

If you have more than one keystore, you can move keys from a source keystore to a destination keystore. However, you cannot move keys from a keystore that is assigned to a machine group. This approach is useful for keeping as many keys as possible in a keystore. If you prefer, you can leave keys in the keystores where CloudLink Center created them. When CloudLink Center requires a key, it checks each accessible keystore.

You can change the frequency that CloudLink Center automatically updates keys, which are known as the key lifetime. For more information, see the section "Key Lifetime" in the topic CloudLink Machine group properties. You can also manually update keys.

You can view keys in a keystore and the key history for a machine.

CloudLink Center encryption key location and protector options

This topic provides information about the encryption key location and protector options in CloudLink Center.

Keystores

The term keystore implies the combination of a key location and a key protector. Encryption keys are stored in a key location and are encrypted, or protected, by a key protector.

Key locations

CloudLink Center supports several options for the key location that is used to store encryption keys:

- **Local Database**—An internal key location
- **Microsoft Active Directory**—An external key location
- **Amazon S3**—An external key location

  You must have an Amazon Web Services (AWS) account to use this location.

- **S3-compatible bucket**—An external S3-compatible key location
- **External KMIP Server**—In CloudLink Center 7.1.3 later versions, you can configure an external KMIP server so it can generate and store keys.

Key protectors

CloudLink Center supports several options for encryption key protectors.

NOTE: The type of available key protector depends on the selected key location.

- **CloudLink Vault**—An internal key protector
- **SafeNet LunaSA**—An external key protector using a hardware security module (HSM) for protection
- **Microsoft Azure or Azure Stack Key Vault**—An external key protector using an Azure or Azure Stack Key Vault for protection
- **KMIP server**—An external key protector using a Key Management Interoperability Protocol (KMIP) server for protection
- **Password**—The encryption key is protected with a password.

Best practices for key location access control and backup

This topic provides information about the best practices for saving, backing up, and restoring CloudLink Center machine encryption keys.

You are responsible for your encryption keys and for ensuring that the appropriate access control and backup policies and procedures are in place to protect the keys against loss or theft. If your keys become unavailable, you cannot access any data that was encrypted using those keys.

CloudLink Center backups are critical for restoring CloudLink Center. Have a backup of CloudLink Center so that you can deploy a new server and restore CloudLink Center. If you are using the local database, volume encryption keys or device encryption keys are stored in CloudLink Center. Backups are the only method of restoring keys so that you can access encrypted data.

NOTE: Ensure that you meet all prerequisites for restoring CloudLink Center from backup, otherwise you cannot access encrypted data after restoring from a backup file.

For more information about CloudLink Center backups and restoring from a backup file, see Back up and restore CloudLink Center.

The following identifies the key protectors that are available for each type of key location.

Key Protector—CloudLink Vault

- Local database key location—Yes
- Microsoft Active Directory key location—No
- Amazon S3 key location—No
- S3-compatible bucket key location—No
- External KMIP Server—No

Key Protector—SafeNet LunaSA

- Local database key location—Yes
- Microsoft Active Directory key location—No
- Amazon S3 key location—No
- S3-compatible bucket key location—No
- External KMIP Server—No

Key Protector—Microsoft Azure or Azure Stack Key Vault

- Local database key location—Yes
- Microsoft Active Directory key location—No
- Amazon S3 key location—No
- S3-compatible bucket key location—No
- External KMIP Server—No

Key Protector—KMIP key manager

- Local database key location—Yes
- Microsoft Active Directory key location—Yes
- Amazon S3 key location—Yes
- S3-compatible bucket key location—Yes
- External KMIP Server—Yes

Key Protector—Password

- Local database key location—Yes
- Microsoft Active Directory key location—Yes
- Amazon S3 key location—Yes
- S3-compatible bucket key location—YesExternal KMIP Server—Yes

CloudLink Center key location

This topic provides information about the key locations in CloudLink Center.

Local Database

CloudLink Center includes a secure local database that is protected by CloudLink Vault. This initial key location encrypts credentials that are used to access remote resources. For example, CloudLink Vault stores credentials required to access the Microsoft Windows domain, FTP or SFTP servers, and external key locations.

You can continue to use this initial key location or configure a different key location. When used as the key location, the CloudLink Vault local database encrypts and stores VKEKs. If you delete a CloudLink Vault key location, all keys are destroyed.

For CloudLink Center clusters, CloudLink Vault is replicated to each server in a cluster. The key location is automatically available on each cluster server.

CloudLink Center backups must be configured when using CloudLink Vault as the key location. For more information, see Best practices for key location access control and backup.

NOTE: If you do not use the initial CloudLink Vault as the key location, CloudLink Center still requires it to store credentials that are used to access remote resources. For more information about working with CloudLink Vault, see Manage CloudLink Vault.

Microsoft Active Directory organizational unit

The authentication credentials for Active Directory are stored in the CloudLink Vault. Ensure key safety by backing up the Active Directory server.

If you delete an Active Directory keystore, the keys remain in the Active Directory base container. You can add this container to CloudLink Center with the same key protector to regain access.

Amazon S3 bucket

The authentication credentials for Amazon S3 are stored in the CloudLink Vault. Ensure key safety by protecting your S3 bucket from accidental deletion.

If you delete an Amazon S3 keystore, the keys remain in the S3 bucket. You can add this bucket to CloudLink Center with the same key encryption password to regain access.

S3-compatible bucket

The authentication credentials for an S3-compatible bucket are stored in the CloudLink Vault. Ensure key safety by protecting your S3-compatible bucket from accidental deletion.

If you delete an S3-compatible keystore, the keys remain in the S3-compatible bucket. You can add this bucket to CloudLink Center with the same key encryption password to regain access.

External KMIP Server

In CloudLink Center 7.1.3 and later versions, you can configure an external KMIP server so it can generate and store encryption keys that are required for agent or machine encryption. However, you cannot transfer or move keys if the source or destination keystore is linked to a KMIP Proxy Key Protector. If you want to use an external KMIP server as the key location, it must be done only at the initial stage that is, the key must be generated directly on the external KMIP server during the initial configuration.

The key release policies applicable for an agent are applicable for external KMIP servers too. It can be used for encrypting machines, PowerFlex devices, containers, and SED key management. However, the aforesaid features are not applicable for Key Management over KMIP, because in this case, CloudLink acts as a KMIP server.

CloudLink key protectors

A key protector is the protection mechanism that is used to encrypt and protect the volume or device encryption keys. Key protectors include:

CloudLink Vault

For more information, see Manage CloudLink Vault.

SafeNet LunaSA

A SafeNet LunaSA key protector uses a hardware security module (HSM) to protect encryption keys.

The main authorization credentials to LunaSA HSM are configured on each server in a cluster. These credentials are not stored in the CloudLink Vault.

For PowerFlex—If you delete a LunaSA keystore, the keys remain in the HSM. You can add this HSM to CloudLink Center with the same password to regain access.

Microsoft Azure Key Vault

A Microsoft Azure Key Vault is a service that Microsoft Azure provides to protect your encryption keys.

KMIP server

A KMIP server can be used to protect your encryption keys. A KMIP server key protector protects all encryption keys that are stored in a location.

Password

Encryption keys can be password protected.

View keystores

Use this procedure to view the keystores added. A keystore is a combination of a key location and a key protector.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Keystores**.
   The list of keystores is displayed.

Configure a keystore

This topic provides information about configuring a keystore.

The initial CloudLink Vault keystore is configured during server setup. You can configure one or more additional keystores.

NOTE: You can have only one current, or active, keystore for each machine group, but multiple keystores can be used in each CloudLink Center or CloudLink Center cluster deployment.

Add a keystore

Use this procedure to add a keystore, in addition to the initial CloudLink Vault keystore.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Keystores** > **Add**.
3. In the **Add New Keystore** dialog box, enter the name and description, and then click **Next**.
4. Select the required keystore location from the **Key Location Type** list, and then click **Next**.
   For instructions on adding a key location, see Add an encryption key location.
5. Select the required key protector from the **Protector Type** list.
6. Click **Add**.

Add an encryption key location

You can use the Local Database, Active Directory, Amazon S3 bucket, S3-compatible bucket, or an External KMIP Server to create and store keys.

Local Database

You can use the local CloudLink Vault database to store encryption keys.

Active Directory

To use Microsoft Active Directory for the CloudLink keystore, you must have a Microsoft Windows domain controller that is reachable by CloudLink Center. For information about configuring an Active Directory on a Windows server, see Configure Active Directory (AD) for the CloudLink encryption keystore.

Verify that CloudLink Center has a DNS configured. For more information, see Configure CloudLink Center DNS properties.

When creating an Active Directory keystore, provide the following values:

- **Domain**—The domain name configured on the Active Directory host, such as example.com
- **Base DN**—The name of the container configured on the Active Directory host

  For example: CN=MyKeys,OU=MyOU,DC=cloudlink,DC=com

- **Hosts (comma separated)**—The Active Directory hostname

  For example: clc.example.com.

  The Active Directory host is a Windows Server where Active Directory is configured. You can add additional Active Directory hostnames for redundancy. Do not use an IP address.

- **Username**—The login name for the bind user
- **Password**—The password configured for the bind user

Amazon S3 Bucket

When creating an Amazon S3 Bucket keystore, provide the following values:

- **Bucket Name**—The name of the bucket resource for the keystore

  If the bucket does not exist, CloudLink Center creates it. If the bucket exists, CloudLink Center tries to connect to it. If CloudLink Center cannot connect to the bucket, CloudLink Center displays an Access-Denied error.

- **Region**—The region where your bucket is expected to be located.
- **Access Key ID**—The Access Key ID associated with your AWS account
- **Access Key**—The Secret Access Key associated with your AWS account

S3 Compatible Bucket

When creating an S3-compatible bucket keystore, provide the following values:

- **S3 Endpoint**—The endpoint for your S3-compatible bucket
- **Bucket Name**—The name of the bucket resource for the keystore

  If the bucket does not exist, CloudLink Center creates it. If the bucket exists, CloudLink Center tries to connect to it. If CloudLink Center cannot connect to the bucket, CloudLink Center displays an **Access Denied** error.

- **URL Style**—Either a virtual-hosted-style URL where the bucket name is a subdomain, or path-style URL where the bucket name is appended to the domain name and is a part of URL path.
- **Access Key**—The access key associated with your S3-compatible storage account.
- **Secret Key**—The secret key associated with your S3-compatible storage account.

NOTE: You can test the configuration values using the **Test** button.

Set the current keystore

This topic provides information about setting the current keystore.

CloudLink Center stores new encryption keys in the current keystore for the machine group. Only one keystore can be current at a time for a machine group. You might want to change the current keystore after adding a new keystore in which you want CloudLink Center to store new encryption keys.

When you switch the current keystore for a machine group, encryption keys are not automatically moved to the new keystore. You can move keys to the new keystore, except for keys stored in SafeNet LunaSA.

You set the current keystore for a machine group by modifying the machine group. For more information, see Modify a machine property on a CloudLink Center machine group .

Modify key location of a keystore

Use this procedure to modify key location and key protector properties after adding a keystore.

About this task

NOTE: Only one keystore can be active at a time for each Machine Group.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Keystores**.
3. Select the check box next to the keystore whose key location you want to modify.
4. Click **Actions** > **Modify Key Location**.

Modify key protector of a keystore

Use this procedure to modify key protector properties after adding a keystore.

About this task

NOTE: Only one keystore can be active at a time for each Machine Group.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Keystores**.
3. Select the check box next to the keystore whose key protector you want to modify.
4. Click **Actions** > **Modify Protector**.

Delete a keystore

Use this procedure to delete a keystore.

About this task

You can delete any keystore that is not used in a machine group. A keystore in use by a machine group must be removed from the machine group before it can be deleted. The results and behavior following deletion depend on the keystore type. For information, see CloudLink encryption key location and protector options.

If you delete a keystore and later determine that it contained keys that are required to access encrypted volumes or devices, you may be able to restore the keystore from a backup. For more information, see Restore keystores from a backup file.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Keystores**.
3. Select the checkbox next to the keystore that you want to delete.
4. Click **Actions** > **Delete**.
5. In the **Confirm Keystore Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

Resolve missing CloudLink Center key alarm

This topic provides information about resolving the missing CloudLink Center key alarm.

Each day, CloudLink Center checks if it can access volume or device keys for connected and registered machines. If CloudLink Center detects that it cannot access one or more volume or device keys, it generates the Missing Key alarm for each key. After raising this alarm, CloudLink Center checks hourly to see whether the key has become available. CloudLink Center lowers the alarm only when the key is available during the check. The alarm is also lowered when a machine is disconnected.

This alarm occurs if a keystore containing the required key becomes inaccessible because CloudLink Center cannot access the volume or device keys that are stored in the keystore. This alarm also occurs if an administrator deletes a keystore that contains required volume or device keys, or deletes a key from a keystore during a shred operation or by using external tools.

This alarm is intended to notify you in a timely manner that volume or device keys are missing. Actions that you must take to resolve this alarm may include checking network connectivity for the keystore or restoring a keystore from a backup file. For more information, see Restore keystores from a backup file.

Show keys in a keystore

Use this procedure to view the keys stored in a selected keystore.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Keystores**.
3. Select the check box next to the keystore whose keys you want to view.
4. Click **Actions** > **Show Keys** .
   The keys are displayed.

Move keys to another keystore

Use this procedure to move keys from one keystore to another. For example, you may want to use an external keystore such as Microsoft Active Directory instead of the initial keystore. After configuring the external keystore, move keys to it from the initial keystore.

Prerequisites

The source and destination keystores must both be accessible by CloudLink Center.

About this task

NOTE: Keys cannot be moved from a keystore that is assigned to a machine group.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Keystores**.
3. Select the check box next to the keystore whose keys you want to move.
4. Click **Actions** > **Move Keys**.
5. In the **Move Keys** dialog box, select the required keystore from the **Move To** list.
6. Click **Move**.

View event history of a keystore

Use this procedure to view the history of the encryption keys for a machine.

Steps

1. Log in to CloudLink Center.

2. Click **System** > **Keystores**.
3. Select the check box next to the keystore whose event history you want to view.
4. Click **Actions** > **Show Key History**.

Update keys

Use this procedure to update keys to reencrypt all the volume encryption keys of a machine with new volume key encryption keys (VKEK).

Prerequisites

The machine must be in the connected state.

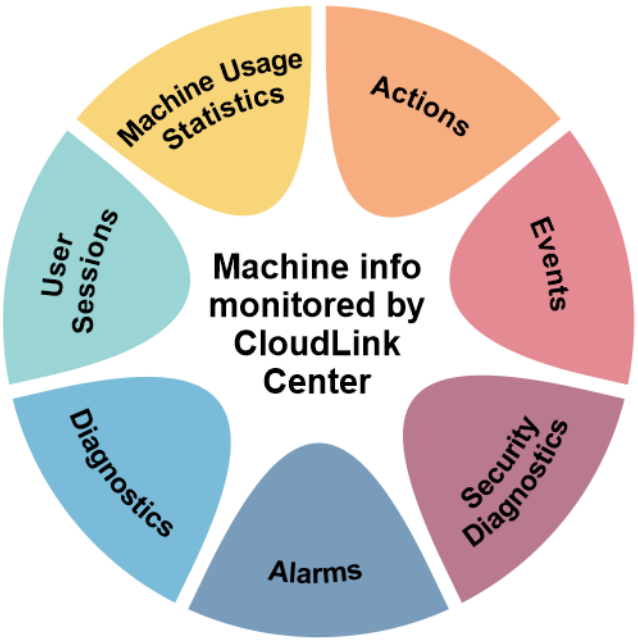
Steps

1. Log in to CloudLink Center.
2. Click **Agents** > **Machines**.
3. Select the check box next to the machine whose keys you want to update.
4. Click **Actions** > **Update Keys**.

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Monitor CloudLink Center

This chapter provides information about comprehensive monitoring of the CloudLink environment and its machines in CloudLink Center.



Actions, events, security events, and alarms in CloudLink

When an action, event, or security event occurs, CloudLink Center displays a notification header and all notifications in the bottom-right corner of the window. You can show or hide notifications using the icons in the notification header. The notification header and notifications automatically disappear after a few seconds.

Each hour, CloudLink Center checks to determine if more than 10,000 actions, events, and security events exist. If so, CloudLink Center deletes entries older than four days, starting with the oldest entry, until the total number of entries is less than 10,000.

In addition to action and event notifications, CloudLink Center raises alarms to make you aware of critical states or conditions. For more information, see View and manage alarms.

View CloudLink Center actions

Use this procedure to view the actions initiated by users, such as uploading or assigning licenses, accepting a pending machine, or setting the CloudLink Vault mode.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Actions**.
   You can view the actions that happened in the last 10, 30 minutes, hour and all.

CloudLink events and corresponding syslog severity numbers

CloudLink assigns one of six levels to each event to indicate its importance. These levels map to syslog levels 1 to 6. For example, Alert is level 1, Critical is level 2 and so on.

Following are the event levels and their corresponding syslog number:

- Alert=1
- Critical=2
- Error=3
- Warning=4
- Notice=5
- Information=6

For alarms, CloudLink Center creates an event when an alarm is raised, assigning it a level of Error if the severity is high or Warn if the severity is low. CloudLink Center creates another event if the condition or state that caused the alarm is resolved, assigning this event the Information level.

For actions, CloudLink Center creates an event when the user initiates the action, assigning it to the Notice level. On completion of the action, CloudLink Center may create additional events based on the internal processing of an action (if needed). CloudLink Center always creates an event that indicates when an action was completed.

CloudLink writes events to the configured syslog server. Details about when events were initiated and completed are available only in the syslog server. These details are not provided in the Events table.

View CloudLink Center events

Use this procedure to view the internal activity in the system, as well as activity related to actions and alarms.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Events**.
   You can view the events that happened in the last 10, 30 minutes, one hour and also all the events .

Security events in CloudLink

The Security Events page displays all the configuration changes in CloudLink that result in security events, such as:

- Logins by users
- Failed attempts to unlock the CloudLink Vault using a passcode
- Registrations for machines
- Changes to the CloudLink Vault mode
- Successful or failed attempts to start a secure user action
- Key activities such as requests, updates, or moves

CloudLink Center assigns each security event a level to indicate its importance. The level types are the same for events. For information, see View CloudLink Center security events.

CloudLink writes security events to the configured syslog server.

View CloudLink Center security events

Use this procedure to view the security events in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Security Events**.
   You can view the security events that happened in the last 10, 30 minutes, one hour and also all the security events.

View CloudLink Center alarms

For more information about viewing CloudLink Center alarms, see View and manage CloudLink Center alarms.

Change the CloudLink Center alarm state

CloudLink Center provides two alarm states: Watched and Ignored.

About this task

To change the alarm notification states and email notification status in CloudLink Center:

Steps

1. Log in to CloudLink Center.
2. Select **System** > **Alarms Configuration**.
3. Select the alarm you want to change.
4. Select **Actions** > **Change**.
5. In the **Change**, dialog box, select the following:
   - From the **State** list, select one of the following values:
     - **Watched**—Future notifications for the selected alarm are reported.
     - **Ignored**—Future notifications for the selected alarm are ignored.
   - From the **Email Notification** list, select one of the following:
     - **Send**
     - **Do not Send**
6. Click **Change**.

Manage email notifications in CloudLink Center

This topic provides information about the email notifications that can be sent when a CloudLink Center alarm is raised or updated. You can configure the CloudLink Center server that generates an alarm to send an email notification whenever an alarm is generated or updated.

Send test email in CloudLink Center

Use this procedure to send a test email in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Email Notifications**.
3. Click **Actions** > **Send Test Email**.

4. In the **Confirm Sending Test Email** dialog box, when prompted to confirm the request, click **Send**.

Change email subject format in CloudLink Center

Use this procedure to change the email subject in CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Email Notifications**.
3. Click **Actions** > **Change Email Subject Format**.
   The **Change Email Subject Format** dialog box is displayed.
4. In the **Email Subject Format** box, enter an email subject format, and then click **Change**.

Add recipient in CloudLink Center

Use this procedure to add a recipient.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Email Notifications**.
3. Click **Actions** > **Add Recipient**.
   The **Add Recipient** dialog box is displayed.
4. In the **Email Subject** box, enter an email address, and then click **Add**.

Delete recipient from CloudLink Center

Use this procedure to delete a recipient.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Email Notifications**.
3. Select the check box next to the recipient you want to delete.
4. Click **Actions** > **Delete Recipient**.
5. In the **Confirm Recipient** dialog box, when prompted to confirm the delete request, click **Delete**.

Change email server configuration in CloudLink Center

Use this procedure to add an email server and account to use to send notifications.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Email Notifications**.
3. Click **Change Configuration** .
4. In the **Change Email Server Configuration** dialog box, select SMTP from the **Server Type** list, and then enter the following information:
   - **Server Address**—The SMTP server address
   - **Port**—The SMTP server port
   - **Sender Address**—The originating email address
   - **User Name**—The email account username
   - **Password**—The email account password

5. Click **Change**.

View individual log files

Use this procedure to view individual CloudLink Center log files. You can view the tail of an individual log file and specify the frequency that it automatically refreshes.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Diagnostics**.
3. Click **View Logs**.
4. In the **Select Logs To View** dialog box, do the following:
   a. From the **Log File** list, select a log file.
   b. In the **Lines To Show** box, enter the number of lines from the tail of the log file to view.
   c. In the **Update Interval (sec)** box, enter the frequency (in seconds) to update the view with the current file content.
5. Click **Show Logs**.

Download log files

Use this procedure to download all the CloudLink Center log files in a compressed ZIP file. For example, when reporting an issue to Dell Technologies Customer Support, this file is required.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Diagnostics**.
3. Click **View Logs**.
4. In the **Select Logs To View** dialog box, select a log file, and enter the required values for which you want to download the log files.
5. Click **Show Logs**.
   The required log file is displayed.
6. Click **Actions** > **Download Logs**.
   The log file is downloaded to your Downloads folder.

Generate diagnostic log files

The Generate Diagnostics command creates a diagnostics log. Dell Technologies Customer Support uses this file if there is a support issue. It is downloaded with the log files to your Downloads folder.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Diagnostics**.
3. Click **View Logs**.
4. In the **Select Logs To View** dialog box, select a log file, and enter the required values for which you want to create a diagnostic file.
5. Click **Show Logs**.
   The required log file is displayed.
6. Click **Actions** > **Generate Diagnostics**.
   The **Generate Diagnostics** dialog box is displayed and diagnostic log file is generated.

Enable the debug mode in CloudLink Center

Only a Dell authorized support representative should perform this procedure. The authorized support representative uses the debug mode when investigating support issues.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Diagnostics**.
3. Click **View Logs**.
4. In the **Select Logs To View** dialog box, select a log file, and enter the required values for which you want to enable the debug mode.
5. Click **Show Logs**.
   The required log file is displayed.
6. Click **Actions** > **Enable Debug**.
   A challenge code and a key code are generated.
7. In the **Enable Debug** dialog box, enter the response in the **Response** box.
8. Click **Enable**.

View user sessions in CloudLink Center

Use this procedure to view information about the current session of each user. For users who are not currently logged in, you can view information about their last session.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Users Sessions**.
   The current user sessions are displayed.

End user sessions in CloudLink Center

Use this procedure to terminate a user session to immediately log out a user. The next time that the user performs a task in the web browser, CloudLink Center displays its login screen.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Users Sessions**.
3. Select the check box next to the user whose user session you want to terminate.
4. Click **Terminate**.
5. In the **Confirm User Session Termination** dialog box, when prompted to confirm the termination request, click **Terminate**.

View usage in CloudLink Center

Use this procedure to view usage statistics for machine instance, physical machines with SEDs, encrypted capacity, or KMIP client. There is no usage statistics for VMware ESXi socket licenses.

About this task

You can view a chart of machine instance, physical machines with SEDs, encrypted storage capacity, or KMIP clients and summary information about registered instances, registered physical machines with SEDs, SED encrypted storage capacity, and KMIP clients.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Usage**.
3. In the **License Usage** page, select instance usage, capacity usage, SED instance usage, or KMIP client license usage.

   The following usage statistics information is displayed corresponding to the usage type that you select:

   - **Last Usage Reset**—The date and time that license usage statistics were reset. This statistic is provided for machine instance, physical machines with SEDs, or encrypted storage capacity. For more information, see Scenarios for using maximum usage of CloudLink licenses.
   - **Max Usage Since Last Reset**—The maximum number of machine instances registered, physical machines with SEDs registered, or encrypted capacity used since the last reset. This information might be useful if you are assessing your peak license usage over a specific time.
   - **Current Usage**—The number of machine instances registered, physical machines with SEDs registered, or encrypted capacity used.

     The vertical axis of the graph represents the number of instances or amount of encrypted capacity that is registered or used. The horizontal line represents dates.

     The graph displays a horizontal blue or green line that shows the total assigned licenses. Hover over the blue line to view information about that license.

     The graph shows a blue or green line representing the number of registered machine instances, registered physical machines with SEDs, or used encrypted capacity. The number of machine instances, physical machines with SEDs, or encrypted capacity that can be registered with CloudLink Center on any given date equals the sum of the machine instances, physical machines with SEDs, or encrypted capacity for all licenses valid on that date.

Reset license usage in CloudLink Center

Use this procedure to reset the value for the maximum number of registered machine instances, registered physical machines with SEDs, or used encrypted capacity. For example, if you are monitoring the maximum number of machine instances registered, physical machines with SEDs registered, or encrypted capacity used each month, reset the usage on the first day of each month.

Steps

1. Log in to CloudLink Center.
2. Click **Monitoring** > **Usage**.
3. Select instance usage, capacity usage, SED instance usage, or KMIP client license usage from the list.
4. Click **Reset Usage**.
5. In the **Confirm Usage Reset** dialog box, when prompted to confirm the reset licenses usage request, click **Reset**.

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Back up and restore CloudLink Center

System issues such as power interruptions or hardware failures may cause problems in CloudLink Center, such as data loss or database corruption. If problems occur, it is important to have a backup of CloudLink Center so that you can deploy a new server and restore CloudLink Center from the backup.

CloudLink Center backup

This topic provides information about CloudLink Center Backup.

A backup file includes all critical information to get CloudLink Center up and running, including keystore configuration, user accounts, machine registrations and policies, and events. You can create backup files manually or automatically. You can create a backup store where CloudLink Center stores automatic backups.

NOTE: A backup file does not include CloudLink Center cluster information or licenses.

CloudLink backup key pairs and backup files

This topic provides information about the backup key pairs and backup files in CloudLink Center.

A backup is stored in a file that is encrypted using an AES-256 key that is protected by an RSA-2048 key pair.

You generate the RSA-2048 key pair. The public key is stored in CloudLink Center. Download and save the private key when the key pair is generated. To restore CloudLink Center from a backup file, both the backup file and its private key must be available to you.

The key pair is assigned a sixteen-digit ID. This ID is in the filenames for the private key that is required to use the backup and for the backup file. The filename for the private key uses the prefix cckey. The filename for the backup file uses the prefix `ccbackup`, by default. For more information about changing the prefix for the filename for the backup file, see Change the filename prefix for the backup file.

The following are example file names for a private key and a backup file that requires the private key:

- **Private Key File Name**—`cckey-189a6361dc9772060730b654d9422b5f.pem`
- **Backup File Name**—`ccbackup-189a6361dc9772060730b654d9422b5f-2015-03-23_09-15-01.bak`

Change the filename prefix for the backup file

Use this procedure to change the filename prefix for the backup file.

About this task

By default, the filename for the backup file uses the prefix `ccbackup`. You can change this prefix. For example, if you have multiple CloudLink Centers, you can use a prefix that uniquely identifies their backups. Using a different prefix for each CloudLink Center can help you to better identify the required backup if you must restore a CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Actions** > **Change Backup File Prefix**.
   The **Change Backup Prefix** dialog box is displayed.
4. In the **Change Backup File Prefix** box, enter the prefix for the filename.
5. Click **Change**.

View CloudLink Center backup information

Use this procedure to view the CloudLink Center backup information.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Backup**.

   The **Backup** page includes the following additional information if you use an FTP, SFTP, FTPS, or S3-compatible backup store.

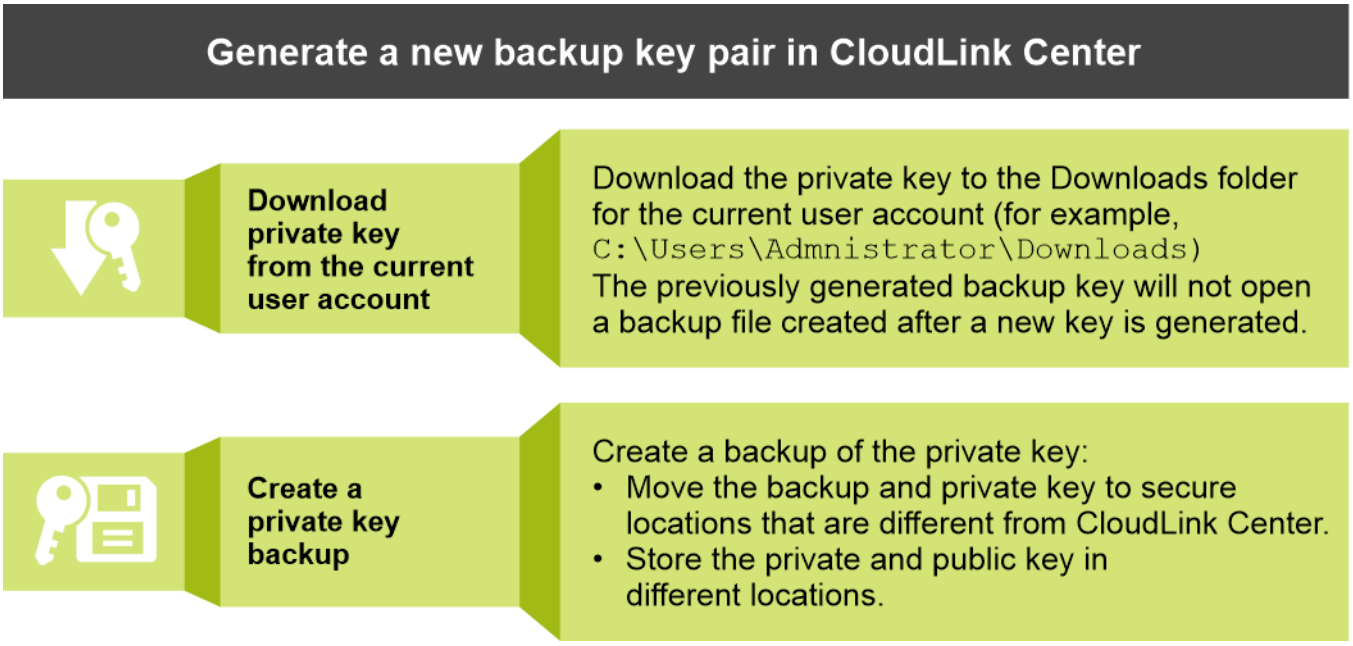
   - **Host**—The remote FTP, SFTP, or FTPS host where a user is intended to save the CloudLink Center backups. You can set this value to the host IP address or hostname (if DNS is configured). This field also lists the endpoint if an S3-compatible backup store is used.
   - **Port**—The port used to access the backup store.
   - **User**—The user with permission to access the backup store.
   - **Directory**—The directory in the backup store where backup files are available.

Generate a backup key pair

Use this procedure to generate a new backup key pair. For example, if the private key for the backup key pair is lost, you can generate a new key pair. You cannot access your backup files without the associated private key. When you generate a new key pair, CloudLink Center automatically generates a new backup file to ensure that the current backup can be opened with the private key of the current key pair.

About this task

It is recommended that you follow the following practices when you generate a new backup key pair to ensure that you have both a private key and backup file that it can open:



Steps

1. Log in to CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Actions** > **Generate and Download New Key**.
   The back up key pair is generated and downloaded to your Downloads folder.

Change the backup store for automatic backups

Use this procedure to change the backup store for automatic backups.

About this task

CloudLink Center automatically generates a backup file. You can use local storage or configure a remote backup store where CloudLink Center stores the backup file.

Regardless of whether you have configured a backup store, you can manually download the backup file. For more information, Download the current backup file.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Actions** > **Change Backup Store**.
   The **Change Backup Store** dialog box is displayed.
4. From the **Store Type** list, select a backup store, and enter the following information depending on the backup store you selected:
   - **Host**—The remote FTP, SFTP, or FTPS host where a user is intended to save the CloudLink Center backups. You can set this value to the host IP address or hostname (if DNS is configured).
   - **Port**—The port used to access the backup store
   - **User**—The user with permission to access the backup store
   - **Password**—The password used to access the backup store
   - **Directory**—The directory in the backup store where backup files are available
   - **Access Key ID**—The Access Key ID associated with your Amazon Web Services (AWS) account or S3-compatible storage
   - **Access Key**—The Secret Access Key that is associated with your AWS account or S3-compatible storage
   - **Bucket Name**—The name of the bucket resource for the AWS or S3-compatible backup store
   - **Region**—The region where your AWS bucket is located.
   - **EndPoint**—The endpoint for an S3-compatible backup store
   - **URL Style**—Either a virtual-hosted-style URL where the bucket name is a subdomain, or path-style URL where the bucket name is appended to the domain name and is a part of URL path
   - **Certificate**—You can upload a self-signed certificate for an S3-compatible backup store.
5. Click **Change**.

Next steps

NOTE: Click **Test** to test the backup store values.

Change the schedule for automatic backups

Use this procedure to change the schedule for generating automatic backups. CloudLink Center automatically generates a backup file each day at midnight (UTC time).

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Actions** > **Change Backup Schedule**.
4. In the **Change Backup Schedule** dialog box, provide the following values:
   a. From the **Days** list, select the days on which you want to automatically generate a backup file.
   b. In the **Hour, UTC** list, select the hour at which you want to automatically generate a backup file.
5. Click **Change**.

Generate a backup file manually

Use this procedure to generate a backup manually if you want to preserve CloudLink Center before the next automatic back up. When you generate a backup file manually, it is recommended that you download the backup file.

About this task

An alarm occurs under these conditions:

- A backup has never been created (either manually or automatically)
- A backup has not been generated (manually or automatically) in the last seven days

For example, this alarm may be triggered if CloudLink Center cannot write the automatically created backup file to disk. The alarm may also be triggered if you are not using a backup store and have not manually downloaded the backup file in the last seven days.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Generate New Backup**.
   A backup file is generated.

Download the current backup file

Use this procedure to download the current backup file at any time.

Prerequisites

You must have access to the last backup file that CloudLink Center generated automatically or you must manually generate a backup file.

About this task

The current backup file is either:

- The last backup file that CloudLink Center automatically created
- The last backup file that you manually generated after the last automatic backup

When you download the current backup file, CloudLink Center displays the age of the backup file. For example, the message might indicate that the current backup was generated 14 hours and 23 minutes ago. CloudLink Center also displays the identifier of the private key required to access the backup file.

The backup file is saved to your Downloads folder (for example, `C:\Users\Administrator\Downloads`). For information about backup file names, see CloudLink backup key pairs and backup files.

After downloading the file, it is recommended that you move it to a location that is different from the CloudLink Center server. If CloudLink Center fails, you can access the backup file and its private key.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Actions** > **Download Backup**.
4. In the **Download Current Backup** dialog box, click **Download**.

Restore CloudLink Center from a backup file

Use this procedure to restore CloudLink Center from a backup file.

Prerequisites

- You must be a secadmin user to restore CloudLink Center from a backup file.
- You need a CloudLink license. Licenses are not included in backups. You must upload a license when restoring from the backup.
- Backup file representing CloudLink Center at the time you want to restore
- Private key for that backup file
- Passcodes for unlocking CloudLink Vault
- If using an external keystore, access to the keystore that contains the keys used at the time the backup file was generated

About this task

If problems occur with CloudLink Center, you can deploy a new server and restore CloudLink Center from the backup file.

Steps

1. Deploy a new CloudLink Center server. For more information, see the *Dell CloudLink Deployment Guide*.
2. In the **Initial Configuration** dialog box, select **Restore from Backup** from the **Deployment Type** list, and then click **Next**.
3. Select the backup file that you want to restore from and its backup key, and click **Restore**.
4. If the CloudLink Center server deployed in Step 1 has a different IP address than the original server, for each machine that was under the control of the previous CloudLink Center, configure the server address of the new CloudLink Center.

Restore a CloudLink Center cluster

Do not restore a CloudLink Center server in a cluster from a backup file. It is not restored as part of the cluster. It is restored as a standalone CloudLink Center.

CloudLink Center backups do not contain CloudLink Center cluster information. To restore a CloudLink Center that was part of a cluster, you must delete the CloudLink Center you want to restore from the cluster. For more information, see Remove a CloudLink Center cluster server. Next, deploy a new CloudLink Center and join it to the cluster. For more information, see Associate a server to a CloudLink Center cluster.

Restore keystores from a backup file

If you delete a keystore and later determine that it contained keys required to access encrypted volumes, you might be able to restore the keystore from a backup.

Prerequisites

NOTE: The Restore Keystore feature is unavailable from CloudLink Center version 7.1.3 or later versions. However, you can restore the complete CloudLink Center information by using a backup file. For information about restoring, see Restore CloudLink Center from a backup file.

- You must be a secadmin user
- Backup file that contains the keystore before it was deleted
- Private key for that backup file
- If manual unlock was used for CloudLink Vault when the backup was created, passcodes for unlocking CloudLink Vault then

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Backup**.
3. Click **Actions** > **Restore Keystores**.

4. In the **Restore Keystores** dialog box, do the following:
   a. In the **Key** box, click  to select your private key.
   b. In the **Backup** box, click  to select your backup file.
   c. In the **Unlock Passcode** box, type the passcode for CloudLink Vault.

Next steps

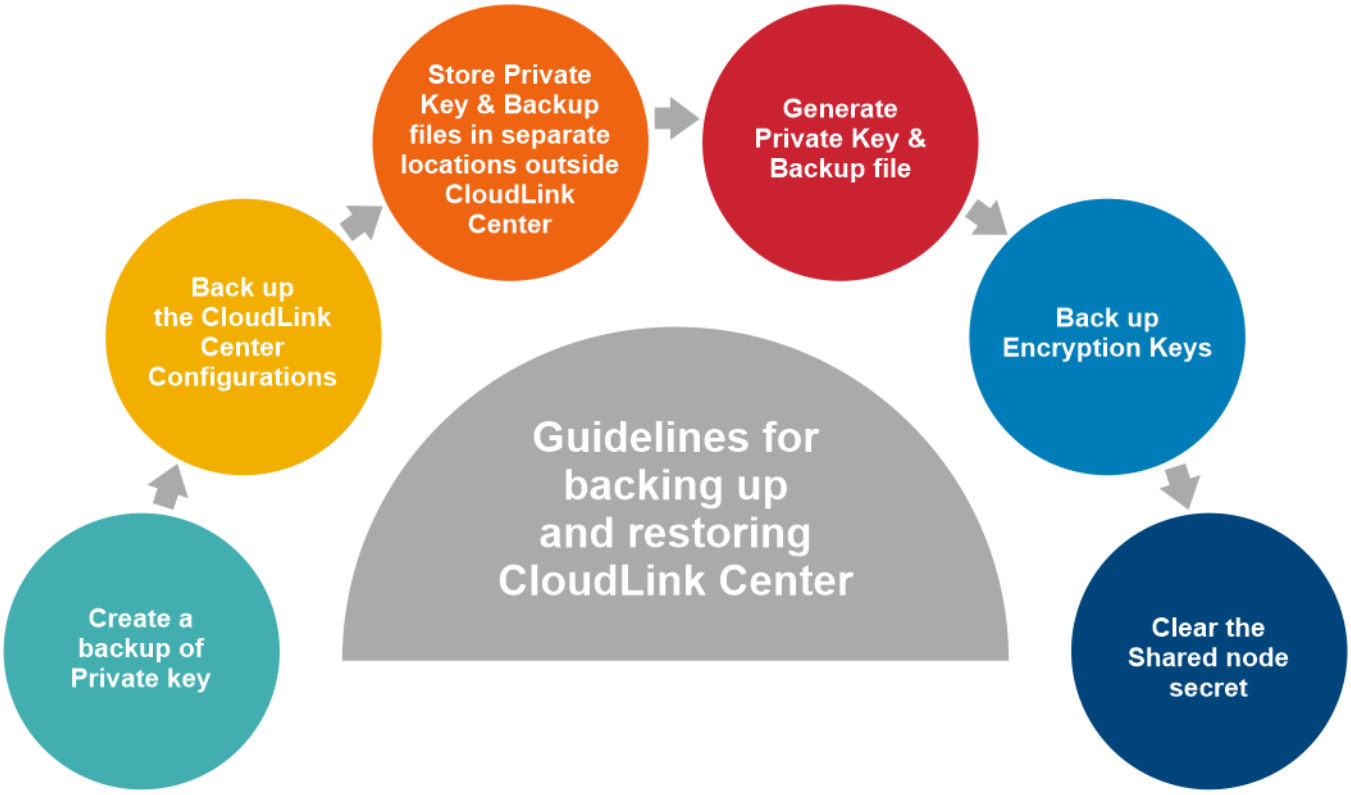
When you restore keystores from a backup file, all keystores in the backup are restored. On the Keystores page, CloudLink Center lists both existing and restored keystores. The names of restored keystore begin with the prefix restored.

You can show the keys for each restored keystore to find the ones that are missing. You can then move all keys from a restored keystore to the keystore for the appropriate machine group.

It is recommended that you delete restored keystores after obtaining the keys from a restored keystore.

Best practices for restoring and backing up keys and files in CloudLink Center

This topic provides the best practices for restoring and backing up keys and files in CloudLink Center.



Keep the following information in mind when working with backup keys and files.

- During initial server configuration, you downloaded the private key for the RSA-2048 key pair. Ensure that you create a backup of this key.
- After you have completed configuration following deployment (such as assigning licenses, creating user accounts, and setting machine boot policy), create a backup.
- Ensure that you store the private key and backup files in separate, secure locations that are different from CloudLink Center. To restore CloudLink Center from a backup file, both the backup file and its private key must be available to you.
- If you have to regenerate the private key (for example, the original key was lost), generate both the private key and a backup file. Previous backup files are not accessible using the new private key.
- A backup includes the CloudLink Vault, Microsoft Azure Key Vault, and SafeNet LunaSA keystores. If you are using a different keystore type, you must back up your encryption keys. For information, see Manage encryption keystores and keys in CloudLink Center.

- If you are using RSA SecurID for two-factor authentication, you must clear the shared node secret after restoring a keystore and before the first authentication attempt. For more information, see Clear an RSA node shared secret.
- A CloudLink Center cluster server that is restored from backup is not restored as part of a cluster. It is restored as a standalone server. Cluster configuration is not part of the backup file. You must delete the server from the cluster and redeploy it as a new CloudLink Center server, then associate it to the cluster.

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Create and manage CloudLink Center cluster

This chapter provides information about creating and managing a CloudLink Center cluster.

A CloudLink Center cluster provides high availability if one CloudLink Center server in the cluster becomes unavailable. For example, a server may become unavailable unexpectedly due to a connection issue. A server may also become unavailable during periods of planned maintenance, when a server is taken offline.

A CloudLink Center cluster consists of up to four CloudLink Center servers, where each is active always. There is no primary server. The agents can be connected to any server in the cluster.

CloudLink Center replicates configuration information between all servers in a cluster. This replication means that all servers contain the same critical configuration information: CloudLink licenses, volume encryption policy, user accounts, manual passcodes for unlocking CloudLink Vault, actions, alarms, and security events.



A CloudLink Center cluster only replicates CloudLink Center server data. Data from external resources, such as key locations, key protectors, and key management servers, are not replicated.

You can remove a server from a CloudLink Center cluster at any time. See Remove a CloudLink Center cluster server for more information.

For information about upgrading a CloudLink Center cluster, see the *Dell CloudLink Upgrade Guide*.

Create a CloudLink Center cluster

Use this procedure to create a CloudLink Center cluster.

Prerequisites

CloudLink Center server should be configured and deployed which acts as the initial server in the cluster.

Steps

1. Deploy and configure one or more CloudLink Center servers.

   Each server must be deployed as a clean installation of CloudLink Center. For more information about deploying and configuring CloudLink Center servers, see the *Dell CloudLink Deployment Guide*.
2. Associate each additional server to the existing server using the **Initial Configuration** wizard. For more information, see Associate a server to a CloudLink Center cluster

CloudLink Center server addresses in CloudLink clusters

This topic provides information about the CloudLink Center server addresses in CloudLink clusters.

In a CloudLink Center cluster, servers and CloudLink Agents use the CloudLink Center server address for communication. You can specify this server address as a static IP address or a hostname in fully qualified domain name (FQDN) format, such as clc.example.com.

Specify the server address in the format you prefer before creating the cluster. You can use static IPv4 addresses, static IPv6 addresses, and FQDNs to create a CloudLink Center cluster. You cannot change the format after creating the cluster. For more information about prerequisites for server addresses in clusters, see the *Dell CloudLink Deployment Guide*.

Associate a server to a CloudLink Center cluster

Use this procedure to associate a server to a CloudLink Center cluster.

Prerequisites

- Deployed and configured CloudLink Center server
- Server address for the CloudLink Center server is the initial server. If you have configured the DNS server first, then use the hostname.
- For the initial server, CloudLink Center login credentials (username and password) for a user with Join To Cluster permissions.
- Public server address for the new server, referred to as the Cluster Server Name or Address.

About this task

To create a CloudLink Center cluster, you associate servers to the existing server. You can associate one server at a time.

During the association of servers process, a server uses the same CloudLink Vault mode (automatic or manual) as the existing server. You can change the CloudLink Vault mode for a new server after the association is complete. For more information, see Domain Name System servers in CloudLink.

NOTE: Passcodes for unlocking CloudLink Vault operating in manual mode are global across the cluster.

Steps

1. On the **Initial Configuration** wizard, select **Add as a Cluster Member** from the **Deployment Type** list.
2. Click **Next**.
3. In the **Server Name or IP Address** box, enter the DNS name or IP address that is used to connect to this server.
4. Click **Next**.
5. Enter the following information:
   - In the **Server** box, enter the IP address for the existing server.
   - In the **User Name** and **Password** boxes, type the login credentials for a user that has the *Join to Cluster* permission.

6. Click **Join Cluster**.

   After associating the server to the cluster, verify the status of both servers. For information, see View CloudLink Center cluster servers.

Change CA certificate lifetime to authenticate a cluster

Using the CA Certificate generation, users can generate the CA certificate. This action will automatically trigger the generation of server certificates. This process only needs to be performed on one node of the cluster, and the CA certificate will propagate to other nodes seamlessly. After generating the certificates, users must restart the application service on each node using the command sh /opt/vcg/cloudlink-installer.sh -it

Steps

1. On the CloudLink GUI, in the left pane, click **Cluster** > **Actions**.
2. In the working pane, from the **Actions** list, select **Change CA certificate** .
3. In the **Confirm CA Certificate Change** dialog box, when prompted to confirm the request, click **Change**.

Upload a third-party signed certificate to communicate among cluster nodes

To enhance the security between nodes in a cluster, you can upload a third-party signed certificate. The certificate you upload is used for inter-node communication.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Cluster**.
3. In the **UPLOAD CLUSTER CA CERTIFICATE** dialog box, enter or select the following information:
   a. From the **CSR Generated By** drop-down menu, select the third-party certificate format. Currently, PEM and PKCS12 are supported.
   b. In the **Certificate** box, click  to select and upload the certificate file.
   c. In the **Key** box, click  to select and upload the key file.
   To view the summary of files you uploaded, click **Preview**.
4. Click **Upload**. The CA-signed certificate file is uploaded to the cluster.

Administer a cluster

When accessing CloudLink Center to perform administration tasks, you can use any server in the cluster. The web application shows the server that the agent is connected to, but operations can be initiated from any server.

Guidelines for working with CloudLink Center clusters

This topic provides the guidelines for working with CloudLink Center clusters.

When working with CloudLink Center clusters, keep the following in mind:

- Each server in a cluster has its own CloudLink Vault. For a CloudLink Vault in automatic mode, unlocking the CloudLink Vault unlocks it only for the current server. CloudLink Vault for other servers in the cluster remains locked.
- If you want to use a keystore other than CloudLink Vault, for all servers in the cluster, configure DNS as described in Domain Name System servers in CloudLink.

- For user accounts, configure the Microsoft Windows domain as described in Domain Name System servers in CloudLink.
- If a CloudLink Agent is disconnected from a cluster when waiting for a pending key release, it reconnects to a cluster server if it is accepted.
- If you use a vSphere key management server (KMS), separately join each CloudLink Center in the cluster to the KMS.

View CloudLink Center cluster servers

Use this procedure to view the servers belonging to a CloudLink Center cluster. The Cluster Server Name or the Address identifies the server on which you are using CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Cluster**.
   The following information is displayed:
   - **Host**—The common name of the server
   - **Server Name/Address**—The hostname or IP address of the server
   - **Incoming Sync State**—See "Outgoing Sync State" for sync state values
   - **Outgoing Sync State**—Indicates whether information is being replicated across cluster servers. Following are the sync states:
     - **Initializing**—status check initialization, show after restart or during the join process while sync is establishing.
     - **Ok**—The server has recorded a heartbeat within the last 15 minutes.
     - **Error**—Database synchronization error
     - **Off**—sync is not working at all, most likely due to network connectivity.
     - **Time_Difference**—time difference between two nodes is more than one minute.
   - **Awaiting Outgoing Monitoring Batches**—The number of monitoring batches waiting to be sent to the selected cluster server.
   - **Awaiting Outgoing Config Batches**—The number of configuration batches waiting to be sent to the selected cluster server.
   - **Awaiting Outgoing Heartbeat Batches**—The number of heartbeat batches waiting to be sent to the selected cluster server.

Change a CloudLink Center cluster server name or address

Use this procedure to change a cluster server name or address after it is associated to a cluster.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Cluster**.
3. Select the check box next to the CloudLink Cluster whose server name or address you want to change.
4. Click **Change Server Name/Address**.
   The **Change Server Name/Address** dialog box is displayed.
5. In the **Server Name/Address**, enter the cluster server name or address.
6. Click **Change**.

Remove a CloudLink Center cluster server

Use this procedure to remove a server from a CloudLink Center cluster.

Prerequisites

Before removing a server, power-off the server.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Cluster**.
3. Select the check box next to the cluster server you want to remove.
4. Click **Delete**.
5. In the **Confirm Cluster Member Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

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Manage CloudLink Vault

This chapter provides information about CloudLink Vault and managing it.

CloudLink Center includes an encrypted container, referred to as the CloudLink Vault that encrypts and protects:

- Credentials used to access remote resources

  For example, CloudLink Vault stores credentials that are required to access the Microsoft Windows domain, FTP or SFTP servers, and external keystores.

- Device or volume key encryption key (VKEK), if CloudLink Vault is used as the keystore.

  For more information about using CloudLink Vault as the keystore, see CloudLink encryption key location and protector options.

When a CloudLink Center server restarts, it must unlock CloudLink Vault before CloudLink Center can authorize machine operations, ensuring that a stolen copy of CloudLink Vault or the disk on which it is stored does not contain any unprotected secrets or encryption keys.

CloudLink Vault was configured during initial server setup. For more information, see *Dell CloudLink Deployment Guide*. You can view and change the configuration at any time to:

- Change the mode for opening the CloudLink Vault (automatic or manual).
- Change passcodes used to unlock the CloudLink Vault in manual mode.

For information about CloudLink Vault and CloudLink Center clusters, see Create and manage CloudLink Center cluster.

View the CloudLink Vault settings

Use this procedure to view the CloudLink Vault settings.

Steps

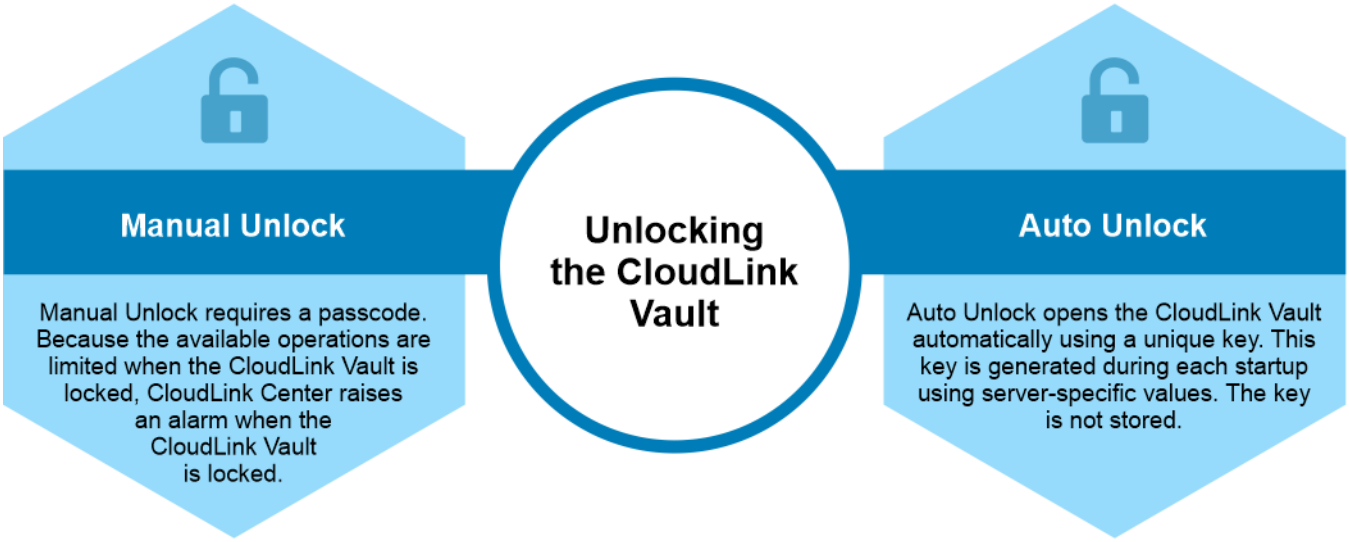
1. Log in to CloudLink Center.
2. Click **System** > **Vault**.
   The following information is displayed:
   - **State**—Identifies whether the CloudLink Vault is locked or unlocked. When locked, CloudLink Center displays a Manual lock on the home page.
   - **Unlock Mode**—Indicates whether the CloudLink Vault is set to automatic or manual mode.

Change the CloudLink Vault mode

Use this procedure to change the CloudLink Vault mode.

About this task

Two modes are available for unlocking the CloudLink Vault when CloudLink Center starts:



Steps

1. Log in to CloudLink Center.
2. Click **System** > **Vault**.
3. Click **Actions** > **Change Mode**.
4. In the **Confirm Mode Unlock Change** dialog box, when prompted to confirm the unlock mode change request, click **Change**.

Set the CloudLink Vault Passcodes

Use this procedure to set the CloudLink Vault passcodes.

About this task

In manual mode, when CloudLink Center starts, an administrator must provide a passcode to unlock the CloudLink Vault.

One to three passcodes were defined during the initial server configuration. You can change passcodes at any time.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Vault**.
3. Click **Actions** > **Set Passcodes**.
4. In the **Set Passcodes** dialog box, type the passcode three times and retype to confirm them, and then click **Set Passcodes**.

Unlock the CloudLink Vault

Use this procedure to unlock the CloudLink Vault.

About this task

If CloudLink Center cannot unlock the CloudLink Vault on startup, CloudLink Center triggers an alarm. You must manually unlock the CloudLink Vault by providing a CloudLink Vault passcode. For more information, see Set the CloudLink Vault passcodes.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Vault**.
3. Click **Unlock**.

Guidelines for working with CloudLink Vaults

This topic provides information about the guidelines for working with CloudLink Vaults.

- When you restore a standalone CloudLink Center from a CloudLink backup file (see Restore CloudLink Center from a backup file), and the CloudLink Vault is configured in Manual Unlock mode, you must use one of the defined passcodes to unlock the CloudLink Vault.
- When you restore a standalone CloudLink Center from a CloudLink backup file (see Restore CloudLink Center from a backup file), and the CloudLink Vault is configured in Auto unlock mode, the vault is automatically unlocked.
- When you restore a standalone CloudLink Center from a VMware backup or a clone, you must use one of the defined passcodes to unlock the CloudLink vault for either Manual or Auto Unlock modes.
- When you upgrade to a new version of CloudLink, CloudLink Vault must be set to Auto Unlock mode. If CloudLink Vault is set to Manual Unlock mode before the upgrade, it remains in Manual Unlock mode after the upgrade and the vault is locked. For more information, see the *Dell CloudLink Upgrade Guide*.

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Assign Microsoft Windows user account for CloudLink Center user roles

This topic provides information about assigning Microsoft Windows user account for CloudLink Center user roles and configuring them.

Rather than creating users for your CloudLink needs, you can use existing user accounts in the Microsoft Windows domain of your organization and assign those accounts to the appropriate CloudLink Center roles.

Before you can add Domain accounts, you must configure a Microsoft Windows domain. For information, see CloudLink Center user types.

After joining the domain, domain users can log in to CloudLink Center using their Windows credentials.

View Microsoft Windows domain configuration

Use this procedure to view the Microsoft Windows domain configuration.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Windows Domain**.
   The following information is displayed:
   - **Host Name**—The Windows domain controller hostname
   - **Address**—The domain controller host address
   - **Global Catalog Port Status**
     - **Accessible**—CloudLink Center can reach the Global Catalog port.
     - **Inaccessible**—The Global Catalog port cannot be reached. While the Global Catalog port is inaccessible, all domain login attempts fail.
   - **LDAP Port Status**
     - **Accessible**—CloudLink Center can reach the LDAP port.
     - **Inaccessible**—The LDAP port cannot be reached. While the LDAP port is inaccessible, all domain login attempts fail.

Configure Microsoft Windows domain

The following procedure explains how to configure a Microsoft Windows domain.

Prerequisites

Ensure that you meet the following prerequisites before associating CloudLink Center to a Microsoft Windows domain:

- Configure a Microsoft Windows domain. For more information, see CloudLink Center user types.
- Verify that CloudLink Center has a DNS server configured. For more information, see Configure CloudLink Center DNS properties.
- On the DNS server, create a reverse lookup zone for the Microsoft Windows domain controller subnet.
- On the DNS reverse lookup zone, configure a pointer to the Microsoft Windows domain controller.

About this task

CloudLink Center dynamically locates the Microsoft Windows domain controller. When specifying the Windows domain parameters, omit the **Primary Host** and **Secondary Host** values.

The username and the password that is provided are used to:

- Associate this server to an Active Directory domain and associate any future servers added to a CloudLink Center cluster.
- Validate the domain group membership of a domain user and determine the roles of the users.

NOTE: Ensure that the username and password that is used to associate the Windows domain is a dedicated user whose password does not expire, because these credentials are saved and used.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Windows Domain**.
3. Click **Configure**.
4. In the **Configure Windows Domain** dialog box, enter the following information:
   - **Domain**—The domain name configured on the Microsoft Windows domain host

     For example: example.com
   - **Protocol**—Select one of the following protocols:
     - LDAP using SASL or GSSAPI
     - LDAP over SSL
     - LDAP over TLS or StartTLS

     If you choose LDAP over SSL, you cannot create a system account and you must upload a certificate.

     NOTE: It is recommended that you integrate to AD/LDAP with TLS, and integrating AD/LDAP with SSL is not recommended.
   - **Create Computer Account**—If you choose to create a system account, CloudLink creates the system account and its principal in the Active Directory (AD) and Key Distribution Center (KDC) of the domain. This option is required only for Integrated Windows Authentication (IWA).

     If you choose not to create a system account, no changes are made in AD. All domain user and domain group features (except IWA) work without creating a system account.
   - **Port Priority**—Select Global Catalog or LDAP
   - **Allow Port Fallback**—Select yes or no
   - **Primary Host**—The Microsoft Windows domain controller hostname, which is a Windows server where the Microsoft Windows domain is configured.

     For example: clc.example.com
   - **Secondary Host**—The Microsoft Windows domain hostname available for redundancy
   - **User**—A user in a Microsoft Windows domain who has permission to add a server to the Microsoft Windows domain.
   - **Password**—The password configured for the user

Modify Microsoft Windows domain

Use this procedure to modify Microsoft Windows domain.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Windows Domain**.
3. Select the check box next to the Windows Domain you want to modify.
4. Click **Modify**.
5. In the **Modify Domain Configuration** dialog box, change the required values.
6. Click **Modify**.

Leave Microsoft Windows domain

Use this procedure to leave the Microsoft Windows domain.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **Windows Domain**.
3. Select the check box next to the Windows Domain you want to leave.
4. Click **Leave**.
5. In the **Confirm Domain Leaving** dialog box, when prompted to confirm the request, click **Leave**.

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RSA Authentication Manager

For increased security, you may require a user to log in using RSA SecurID two-factor authentication, where the user provides a passcode, in addition to CloudLink Center credentials. For more information, see 2-Factor Authentication (2FA) in CloudLink.

Before you can set up users for this type of 2FA, you must configure RSA Authentication Manager.

Upload an RSA Authentication Manager server certificate

Use this procedure to upload an RSA Authentication Manager server certificate to CloudLink Center, before you configure users for 2FA using RSA SecurID.

Prerequisites

- Ensure that the server certificate generated from **RSA Security Console**, is available in a location where it is accessible for uploading to CloudLink Center.
- RSA server dns should be pingable from cloudlink.DNS host resolution has to be configured as per the environment.
  - In QE environment, added RSA server entry—`rsa-dev-am.ciam-dev.net` to the `/etc/hosts` file through CloudLink CLI Diagnostics.
  - In the client environment, support team to understand and do the changes needed for DNS name resolution of RSA from CloudLink.
- Register the hostname and IP address of the CloudLink Controller in the RSA Authentication Manager.
  - Go to **Access** > **Authentication Agents** > **Add New**. Click **Resolve** and then click **Save**.
- Configure RSA Authentication Manager settings on the CloudLink interface.

About this task

You generate the server certificate using **RSA Security Console**, which puts the certificate in a compressed file (ZIP). You download the ZIP file and extract the certificate from it. For more information, see the RSA Security Console documentation.

You can replace the current server certificate with a new one. Ensure that you clear the shared node secret using RSA Security Console. For more information, see Clear the shared node secret.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **RSA Authentication Manager** > **Add Configuration**.
3. In the **Add RSA Authentication Manager Configuration** window, enter the server address, port, client ID, and client key details.
4. Go to the location where the server certificate is located or click  to select the certificate and click **Upload**.
5. Click **Add**.

Modify RSA SecurId authentication configuration

You can modify the RSA SecurId authentication configuration, if required.

Prerequisites

You must have the permissions to modify the configuration.

About this task

You can modify the authentication configuration, if there are errors in the authentication configuration, or if you want to change the server certificate.

Steps

1. Log in to **CloudLink Center**.
2. Click **System** > **RSA Authentication Manager** > **Modify Configuration**.
3. In the **Modify RSA Authentication Manager Configuration** window, change the server address, port, client ID, client key details, and the server certificate, if required.
4. Click **Modify**.

Delete an RSA Authentication Manager configuration

If you no longer require two-factor authentication using RSA SecurID, you can delete its configuration.

About this task

If any users are set up to use RSA SecurID two-factor authentication, CloudLink Center displays a message and does not delete the configuration.

Ensure that you clear the shared node secret using RSA Security Console. For more information, see Clear the shared node secret.

Steps

1. Log in to CloudLink Center.
2. Click **System** > **RSA Authentication Manager**.
3. Select the checkbox next to the RSA Authentication Manager configuration that you want to delete.
4. Click **Delete Configuration**.
5. In the **Confirm RSA Authentication Manager Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

Clear the shared node secret

This topic provides information about clearing the shared node secret in CloudLink Center.

RSA Authentication Manager requires a unique node secret for each Authentication Agent. RSA Authentication Manager automatically creates the node secret it shares with CloudLink Center (an Authentication Agent) during its first successful authentication.

If you delete or replace the configuration files (`REC`), the secret is deleted in CloudLink Center. CloudLink Center cannot present this node secret when requesting authentication, and RSA Authentication Manager will not perform the authentication.

You must clear this shared node secret in the RSA Security Console so that a new shared secret can be generated. For more information, see the RSA documentation.

Configure RSA Authentication Manager settings in CloudLink

This section provides instructions and procedure for configuring RSA Authentication Manager settings in the CloudLink interface.

Prerequisites

- Log in to the CloudLink interface and create a username with **2fa Type** as RSA SecurID.
- RSA server DNS should be pingable from CloudLink. Configure the DNS host resolution as per the environment.
- Register the hostname and IP address of the CloudLink Controller using the RSA Authentication Manager interface.

- Configure RSA Authentication Manager settings on the CloudLink interface.

About this task

CloudLink supports multifactor authentication using RSA SecurID to enhance authentication and identity.

To configure RSA Authentication Manager settings, do the following steps:

Steps

1. Log in to the CloudLink interface.
2. Add CloudLink to RSA Authentication Manager. For details, see the RSA Authentication Manager documentation.
3. Configure the RSA Authentication Manager settings by going to the **System** > **RSA Autentication Manager** > **Add Configuration**
   The **ADD RSA AUTHENTICATION MANAGER CONFIGURATION** window is displayed.
4. Enter the server address, port, and client details.
5. Click **Upload** to save the configuration.

Add CloudLink to RSA SecurId Authentication Manager

You must configure the CloudLink hostname and IP address in **RSA SecurID Authentication Manager** to activate the agent.

Prerequisites

You must have the permissions to register the CloudLink hostname and IP details in the RSA SecurId Authentication Manager.

Steps

1. Log in to RSA SecurId Authentication Manager
2. Go to **Access** > **Authentication Agents** > **Add New**.
3. On the **Add New Authentication Agent** page, select the system domain from the dropdown and enter the CloudLink hostname, IP address, and alternate IP addresses.
4. Click **Save**.

   You can enter multiple IP addresses.

   A message is displayed, prompting you to confirm whether you want to save the agent.
5. Click **Yes, Save Agent**.

Authenticate CloudLink users using system generated PIN

This section provides information about authenticating CloudLink users using the RSA multifactor authentication process.

Prerequisites

- Retrieve the client key from the RSA server and export the certificate. For the instructions for retrieving the client key, see the RSA user documentation.

About this task

Steps

1. Configure the token policy to **System Generated PIN** in the RSA Security console.

   For details about configure the token policy, see the RSA user documentation for *Edit a token policy*.
2. Log in to the CloudLink interface.
3. Open the **RSA SecurID** and press [Enter].
4. Copy the **Tokencode** and paste in the **RSA Passcode** field on the CloudLink interface.
   On successful validation, the system-generated PIN is displayed on the next page.
5. Copy and paste the PIN in the **Confirm PIN** field.
6. Click **Set PIN**.

The RSA SecurID generates a new tokencode.

7. Paste the new tokencode in the **RSA Passcode** field on the CloudLink interface.
   The **CloudLink Center Home** page is displayed.

Authenticate CloudLink users with user generated PIN

This section provides information about authenticating CloudLinks users using the RSA multifactor authentication process.

About this task

Steps

1. Log in to the CloudLink interface.
2. Open the **RSA SecurID** and press [Enter].
3. Copy the **Tokencode** and paste in the **RSA Passcode** field on the CloudLink interface.
   On successful validation, RSA prompts you to enter a new PIN.
4. Enter the PIN of your choice and enter it on the CloudLink interface to confirm.
5. Click **Set PIN**.
   The RSA SecurID generates a new tokencode.
6. Paste the new tokencode in **RSA Passcode** field on the CloudLink interface.
   The **CloudLink Center Home** page is displayed.

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Manage CloudLink SysLog data

This chapter provides information about SysLog data, viewing, configuring and changing the syslog message format.

The security events visible in CloudLink Center can be automatically exported to a syslog server. It is recommended that you configure CloudLink to forward this information to a centralized log server or SIEM through syslog. All existing and new information is forwarded through syslog.

Add Multiple Syslog Server Configuration

In CloudLink 8.3.x and later, users can add up to four syslog servers.

About this task

CloudLink 8.3.x extends syslog functionality to include events, security events, audit logs, and authentication logs across multiple servers, enhancing monitoring capabilities beyond previous versions. Global customization of the syslog format and the ability to postpone and resume actions are now available across multiple syslog servers. Syslog servers are deleted individually, one at a time.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog** > **Add Syslog Server**.
3. In the **Add Syslog Server** dialog box, enter the following information:
   - From the **Server Status**—select any one of the available statuses:
     - Postponed—Logs are not sent out until the syslog status is explicitly set to Resume.
     - Active—Messages are sent to the configured syslog server until the Postpone command is selected.
   - In the **Host** box, enter the IP address of the host to store the system logs.
   - In the **Port** box, enter a port number to gather log messages.
   - From the **Protocol** list, select one of the following protocols:
     - **UDP**
     - **TCP**
     - **Secure TCP (TLS 1.2)**
   - From the **Facility** list, select a facility to log the syslog messages.
   - From the **Format** list, select one of the following message formats:
     - **CEF** (default)—The Common Event Format (CEF) uses the CEF:0|Device Vendor|Device Product|Device Version|Signature ID|Name|Severity|Extension message format.
     - **LEEF1**—The Log Event Extended Format (LEEF) version 1.0 uses the LEEF:1.0|Vendor|Product|Version|EventID|Extension message format.
     - **LEEF2**—LEEF version 2.0 uses the LEEF:2.0|Vendor|Product|Version|EventID|^|Extension message format.
     - **Custom**—The custom format is defined by an input string and variables.
4. Click **Add**.

View syslog configuration

Use the following procedure to view the Syslog status and configuration.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog**.
   The following information is displayed:

- **Status**—Following are the available statuses:
  - Postponed—Logs are not sent out until the syslog status is explicitly set to Resume.
  - Active—Messages are sent to the configured syslog server until the Postpone command is selected.
- **Host**—The host where system logs are sent.
- **Facility**—The facility on which you log the syslog messages are logged.

Change syslog server configuration

The following procedure explains how to change the syslog server configuration.

About this task

If your organization requires the long-term retention of system events, it is recommended that you configure CloudLink to forward its security events to a centralized log server or SIEM using syslog. By default, CloudLink Center writes events to syslog using the common event format (CEF).

You can configure a syslog logger to direct all system log messages to the configured remote host. For information about network ports required, see the *Dell CloudLink Deployment Guide*.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog** > **Change Configuration**.
3. In the **Change Syslog Server Configuration** dialog box, enter the following information:
   a. In the **Host** box, enter the IP address of the host to store the system logs.
   b. In the **Port** box, enter a port number to gather log messages.
   c. From the **Facility** list, select a facility to log the syslog messages.
   d. From the **Protocol** list, select one of the following protocols:
      - **UDP**
      - **TCP**
      - **Secure TCP (TLS 1.2)**
   e. If you select **Secure TCP (TLS 1.2)** protocol, do the following:
      i. In the **Server Certificate** box, click  to select the server certificate.
      ii. In the **Certificate Format** list, select PEM certificate, or a password-protected PKCS12 (RC2) certificate.
      iii. (Optional) In the **PKCS12 Password** box, enter a password for the PKCS12 (RC2) certificate.
      iv. In the **Confirm PKCS12 Password** box, confirm the password for the PKCS12 (RC2) certificate.
4. Click **Change**.

Change syslog message format

Use the following procedure to change the syslog message format.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Syslog** > **Change Syslog Format**.
   The **Change Syslog Message Format** dialog box is displayed.
3. From the **Syslog Format** list, select one of the following message format:
   - **CEF** (default)—The Common Event Format (CEF) uses the CEF:0|Device Vendor|Device Product|Device Version|Signature ID|Name|Severity|Extension message format.
   - **LEEF1**—The Log Event Extended Format (LEEF) version 1.0 uses the LEEF:1.0|Vendor|Product|Version|EventID|Extension message format.
   - **LEEF2**—LEEF version 2.0 uses the LEEF:2.0|Vendor|Product|Version|EventID|^|Extension message format.
   - **Custom**—The custom format is defined by an input string and variables.
4. Click **Change**.

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Manage CloudLink Center network settings

This chapter provides information about managing network settings in CloudLink Center.

You can change the CloudLink Center hostname, and enable or disable SSH access to CloudLink Center.

If CloudLink Center is deployed on Microsoft Azure, or Amazon Web Services you cannot reenable SSH access, if you are blocked from the CloudLink Center console.

Change CloudLink Center hostname configuration settings

Use this procedure to change the CloudLink Center server hostname.

About this task

You are locked out of the CloudLink Center server if:

- You disable SSH access.
- You are blocked from the CloudLink Center server console.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Network**.
3. Select the check box next to the interface whose hostname you want to change.
4. Click **Change Hostname**.
   The **Change Hostname** dialog box is displayed.
5. In the **Hostname** box, enter a hostname, and then click **Change**.

Change CloudLink Center SSH configuration settings

Use this procedure to enable or disable SSH access to the CloudLink Center server.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Network**.
3. Select the check box next to the interface whose SSH access you want to enable or disable.
4. Click **Change SSH**.
   The **Change SSH** dialog box is displayed.
5. From the **SSH** list, select either enabled or disabled, and then click **Change**.

24

Configure CloudLink Center DNS properties

This chapter provides information about configuring DNS properties in CloudLink Center.

DNS servers in CloudLink Center

This topic provides information about the Domain Name System (DNS) servers in CloudLink Center.

You can configure CloudLink Center to resolve hostnames using a DNS server.

Configure CloudLink with a DNS server if you want to use hostnames or domain names for other servers that CloudLink interacts with, such as:

- Resolving Network Time Protocol (NTP) server hostnames. For more information, see Manage NTP servers associated with CloudLink Center.
- Creating a CloudLink Center cluster and use hostnames for servers in the cluster. For more information, see Create a CloudLink Center cluster.

Add DNS for accessing CloudLink Center

Use this procedure to add a DNS server. You can add a maximum of three DNS servers, if you are not using DHCP.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **DNS**.
3. Click **Add**.
   The **Add DNS Server** dialog box is displayed.
4. In the **IP Address** box, enter the IP address of the DNS server you want to add, and then click **Add**.

Set DNS server as the primary server for CloudLink Center

Use this procedure to set DNS server as the primary server for CloudLink Center.

About this task

If more than one DNS server has been configured, one must be set as the primary DNS server.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **DNS**.
3. Select the check box next to the DNS server which you want to set as the primary DNS server, and then click **Set Primary**.
4. In the **Confirm Primary DNS Server Change** dialog box, when prompted to confirm the request, click **Set Primary**.

PING a DNS server to test connection

Use this procedure to check whether CloudLink Center has network access to an external resource.

About this task

You can test the network configuration from CloudLink Center. The ping option returns the same information as using the ping network utility from a command prompt.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **DNS** > **Ping**.
3. In the **Ping** dialog box, enter the IP address of the CloudLink Center, and then click **Ping**.

Delete a DNS associated with CloudLink Center

Use this procedure to delete any DNS server you have manually added.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **DNS**.
3. Select the check box next to the DNS server you added manually, and then click **Delete**.
4. In the **Confirm DNS Server Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

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Manage NTP servers associated with CloudLink Center

This chapter provides information about managing NTP servers associated with CloudLink Center.

You can synchronize CloudLink Center with the date and time obtained from Network Time Protocol (NTP) servers. By default, CloudLink Center is configured with four global NTP servers.

Force an NTP server time synchronization with CloudLink Center

Use this procedure to force synchronization with an NTP server if the time of CloudLink Center is incorrect.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Time**.
3. Select the check box next to the NTP server address whose time you want to forcefully sync.
4. Click **Force Sync With NTP Server**.
5. In the **Confirm Force Sync With NTP Server** dialog box, when to confirm the request, click **Sync**.

Add an NTP server for CloudLink Center

Use this procedure to add an NTP server for CloudLink Center.

About this task

When you add an NTP server, you provide its IP address or hostname. To use a hostname, ensure that you have configured at least one DNS server first. For more information, see Domain Name System servers in CloudLink.

If DNS servers have not been configured, the NTP servers do not work.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Time** > **Add**.
   The **Add NTP Server** dialog box is displayed.
3. In the **NTP Server** box, enter the hostname or IP address of the NTP server, and then click **Add**.

Delete NTP server

Use this procedure to delete an NTP server.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **Time**.
3. Select the check box next to the NTP server you want to delete, and then click **Delete NTP Server**.
4. In the **Confirm NTP Server Deletion** dialog box, when prompted to confirm the delete request, click **Delete**.

26

Manage web TLS and FQDN certificates

This chapter provides information about the web TLS and FQDN certificates and managing them.

CloudLink Center Certificates

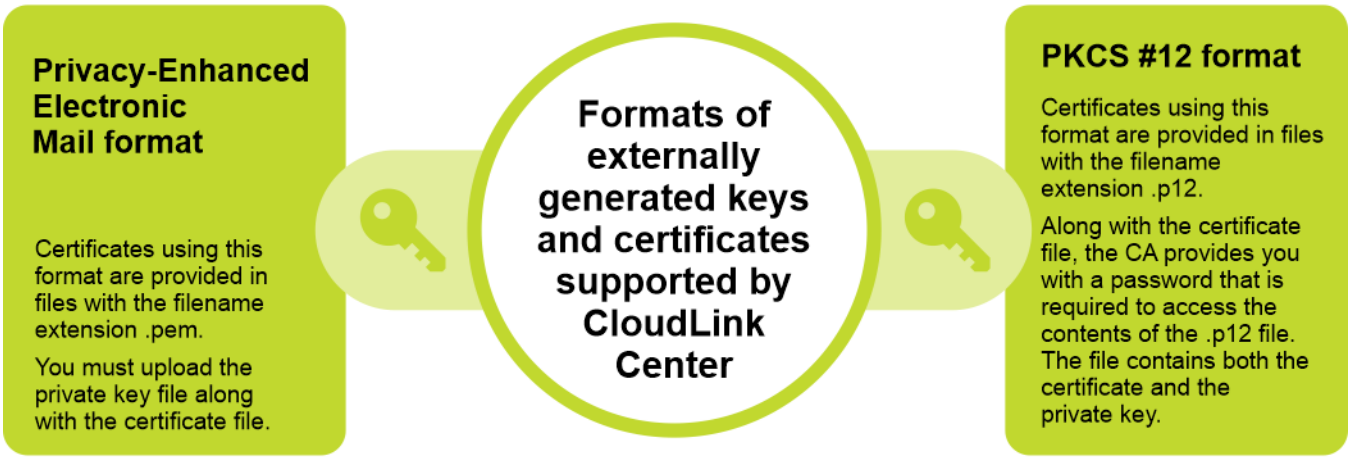
This topic provides information about the web TLS certificates, and the certificate formats for externally generated keys and certificates

By default, the CloudLink Center uses a self-signed certificate. When connecting to CloudLink Center, the web browser may display several security warnings. These warnings are displayed because self-signed certificates do not have the same level of trust as certificates issued and signed by a trusted certification authority (CA).

To stop these warnings from being displayed, you can obtain and upload to CloudLink Center a certificate that has been signed for CloudLink by a trusted CA.

You can upload an externally generated certificate and private key, or generate a certificate signing request for a private key generated by CloudLink Center.

CloudLink supports two formats for externally generated keys and certificates:



Change CA certificate lifetime for TLS

The Change CA Certificate option helps generate a new self-signed CA with a default lifetime of 398 days. Refresh the browser after completing this operation.

Steps

1. On the CloudLink GUI, in the left pane, click **Server** > **TLS** > **Change Certificate** > .
2. In the **Confirm Certificate Change** dialog box, when prompted to confirm the request, click **Change**.

Upload a new TLS certificate

Use this procedure to upload a new TLS certificate.

About this task

When you upload a new certificate and an optional private key, the web server restarts and the connection is terminated.

After uploading a certificate signed for CloudLink Center, verify the subject, fingerprint, and end date to ensure that it is the certificate you want to use.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **TLS** > **Upload**.
   The **Upload Web TLS Certificate** is displayed.
3. In the **Upload Web TLS Certificate** dialog box, enter the following information:
   a. From the **Certificate Format** list, select the required certificate format.
   b. In the **Certificate** box, click  to select the certificate.
   c. In the **Key** box, click  to select the key.
4. Click **Upload**

Generate a CSR certificate

Use this procedure to generate a certificate signing request (CSR) certificate.

About this task

A certificate signing request (CSR) involves CloudLink Center generating a private key and signing the request. The request is then fulfilled by a certificate authority (CA) and the final certificate is uploaded to CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **TLS** > **Generate CSR**.
3. In the **Generate Certificate Signing Request** dialog box, enter the name, organization, organization unit, city or locality, state or province, and country.
4. Click **Generate**.

Configure FQDN in certificates

Certificate authority standards require that Common Name (CN) or Subject Alternative Names (SANs) for the certificate have a Fully Qualified Domain Name (FQDN). Clients require CN or SANs for validating server identity.

CA certificate `'Basic constraints'` field should have the value `" Subject Type=CA"` and `'Key Usage'` field should have the value `"Certificate Signing".`

Generate CloudLink Center certificates with an FQDN in SAN

The Subject Alternative Names (SANs) for newly generated certificates will have FQDN after CloudLink Center is configured with WindowsDomain.

Prerequisites

Ensure that CloudLink Center Windows domain is configured. For example, **`domain name: test.com`**.

Steps

1. Generate a CSR request and upload third party signed certificate for various TLS connections such as agents, KMIP server, cluster, and Kubernetes cluster.
2. Upload third-party CA certificate.

Results

The Subject Alternative Names (SANs) for newly generated certificates have FQDN of the format: `nodehostname.requiredDomain`

Example

A node with the hostname, `cloudlink-4ucui` has FQDN of the format `SAN = cloudlink-4ucui.test.com`.

Generate CloudLink Center certificates with an FQDN in Common Name

Certificates of port 4443, 1194, 5696, and 50000 have FQDN in CN.

Before you begin:

Ensure that you have a third-party CA certificate with Common Name of the format `cloudlink.requiredDomain`. For example, **`CN = cloudlink.test.com`**.

About this task:

When you upload a third-party CA certificate with Common Name of the format `cloudlink.requiredDomain`, the CloudLink certificate that is generated has CN of the format `nodehostname.requiredDomain`.

A node with the hostname, `cloudlink-4ucui` has FQDN of the format `CN = cloudlink-4ucui.test.com`.

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Configure SNMP for CloudLink Center

This chapter provides information about SNMP traps and configuring them in CloudLink Center.

CloudLink Center can send simple network management protocol (SNMP) traps to a single target when CloudLink alarms are raised, updated, or lowered.

Each server in a CloudLink cluster must be configured separately, because cluster members do not synchronize SNMP configuration.

If a CloudLink alarm is set to **Ignored**, then SNMP traps are not sent for that alarm.

Add an SNMP configuration to CloudLink Center

Add an SNMP configuration to the CloudLink Center by selecting SNMPv2 or SNMPv3.

Steps

1. Log in to CloudLink Center.
2. Click **Server** > **SNMP** > **Add**.
3. From **Add New SNMP Configuration**, in the **Target Version** list, select **SNMPv2** or **SNMPv3**.
   - If you select SNMPv2, then enter:
     - **Host**—The IP address or FQDN to where the SNMP traps must be sent.
     - **Port**—The receiving host port number.
     - **Description**—Description for the selected target.
     - **Community**—The SNMP trap community.
   - If you select SNMPv3, then enter:
     - **Host**—The IP address or FQDN to where the SNMP traps must be sent.
     - **Port**—The receiving host port number.
     - **Description**—Description for the selected target.
     - **Engine ID (HEX)**—The remote SNMP protocol engine identifier. This value must be between five and 32 octets in length.
     - **User Name**—The username is required to communicate with the SNMP agent.
     - (Optional) **Authentication Type**—Specifies the type of authentication used.
     - **Authentication Password**—Password used to authenticate a connection with an SNMP agent. Must be of at least eight characters in length.
     - (Optional) **Privacy Type**—Specifies the type of encryption used.
     - **Privacy Password**—Password used to encrypt a connection with an SNMP agent. Must be at least eight characters in length.
4. Click **Add**. The SNMP configuration is added to CloudLink Center.

Modify the SNMP configuration in CloudLink Center

Modify the SNMP configuration in CloudLink Center by updating the required values.

Steps

1. Log in to CloudLink Center.
2. Select **Server** > **SNMP**.
3. Select the SNMP and click **Modify**.
4. In the **Modify SNMP Configuration** dialog box, modify the required values, and then click **Modify**.

Send a test SNMP trap in the CloudLink Center

Test your SNMP configuration to ensure that the target device receives SNMP traps from the CloudLink Center.

About this task

Following traps are sent from CloudLink Center:

- alarmUpTrap when a new alarm is raised on the CloudLink Center server.
- alarmDownTrap when the cause of the alarm is corrected or removed.

The information about the CloudLink alarm is contained in the alarmUpTrap, and the alarm identifier is contained in the alarmDownTrap.

You can use the clAlarmId for unique value for each alarm instance to track the raising and lowering of a specific alarm on the CloudLink Center server.

Steps

1. Log in to CloudLink Center.
2. Select **Server** > **SNMP** > **Send Test Trap**.
3. In the **Confirm Sending Test SNMP Trap** dialog box, when prompted to confirm the request, click **Send**.

Download MIB files

Import or upload the MIB file to the trap receiver or network management system application. CloudLink alarms use the Object Identifier (OID) 1.3.6.1.4.1.1139 for traps.

Steps

1. Log in to CloudLink Center.
2. Select **Server** > **SNMP** > **Download MIB**.

Delete the SNMP configuration in the CloudLink Center

Delete the SNMP configuration in the CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Select **Server** > **SNMP**.
3. Select the SNMP host, and click **Delete**.
4. In the **Confirm SNMP Configuration Deletion** dialog box, when prompted to confirm the request, click **Delete**.

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Linux commands for CloudLink Agent

When performing many tasks related to CloudLink Agent on Linux machines, you use the `svm` command. This command has several subcommands that are used to perform actions including registering, encrypting, reloading machines, and removing mount points.

The following table lists the actions for each `svm` subcommand and identifies the topic in the CloudLink documentation where it is referenced.

Table 5. Command actions

Action	Commands	Procedure
About CloudLink Agent	`svm about`	Displays general information about CloudLink Agent
Decrypt boot volumes	Encrypted using CloudLink 6.7 and later `svm decrypt / [-v ]`	For more information, see Decrypt a volume.
Decrypt mounted volumes	Encrypted using CloudLink 6.7 and later: `svm decrypt [mount_point]`	For more information, see Decrypt a volume.
Encrypt	`svm encrypt [mount_point]`	Encrypt a mount point or block device. For more information, see Encrypt a volume.
	`svm encrypt [device_name]`	
Erase	`svm erase [device_name]`	Erases a device
Generate	`svm gentechsupport [script]`	Creates a tech support file with an optional script file that runs more commands
Manage	`svm manage [device_name]`	Take control of a self-encrypting drive.
Network status	`svm netstat`	Displays the network connection status
Recover	`svm recover [mount_point]`	For more information, see Move an encrypted drive to another machine.
	`svm recover [device_name]`	
Refresh	`svm refresh`	For more information, see "Refresh the CloudLink Agent service on Linux machines" in the *Dell CloudLink Deployment Guide*.
Register	`svm [-v ] [-G group_regcode] -S clc_address`	For more information, see the topic "Download, install, and configure CloudLink Agent using Custom mode on Linux Machines " in the *Dell CloudLink Deployment Guide*.
	`svm [-v ] [-G group_regcode] -S clc_address -t`	Use the -t option to set Trusted Platform Module (TPM) mode on a physical machine that has a TPM version 2.0 chip installed. This option is only available during CloudLink Agent installation.

Table 5. Command actions (continued)

Action	Commands	Procedure
	`svm -S [clc_address] [-v ]`	For more information, see Change the CloudLink Center IP address on a Linux machine.
	`svm -C <directory where the securevm.cer & securevm.key are saved>.`	See Assign a CA certificate and a third-party signed certificate to a Linux agent when upgrading CloudLink from 7.1.5.
Release	`svm release [device_name]`	Release control of a self-encrypting drive.
Reload	`svm reload [-v ]`	Scan for new disks. For more information, see the topic "Refresh the CloudLink Agent service on Linux machines" in the *Dell CloudLink Deployment Guide*.
Remove	`svm removemp [mount_point]`	Remove a mount point without saving any data.
Status	`svm status`	Shows the connection and mount point statuses For more information, see the topic "Verify successful deployment" in the *Dell CloudLink Deployment Guide*.

	`svm status connection`
	or
	`svm status co`
		Shows the CloudLink cluster nodes statuses

Uninstall	`svm uninstall`	For more information, see the chapter "Uninstall CloudLink Agent" in the *Dell CloudLink Deployment Guide*.

Command Variables

This topic gives information about the command variables that are used in CloudLink.

The following table lists the parameters that are used with one or more `svm` subcommands.

Table 6. Command Options

Option	Description
-f	Force encryption or decryption. The machine reboots when this option is used.
-h	Help with the command.
-S	Specify the server address (*clc_address*) for CloudLink Center.
-v	Use verbose mode.

- mount_point—The mount point to be encrypted.
- device_name—The device must be identified when it is not mounted and listed in `/etc/fstab`.
- SED—The self-encrypting drive you want to control with CloudLink Center.

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Command actions for Windows PowerShell

When performing many tasks related to CloudLink Agent on Windows machines, you use the `svm` command. This command has several subcommands that are used to perform actions including encrypting, decrypting, setting dependencies, and showing status.

The following table lists the actions for each `svm` subcommand and identifies the topic in the CloudLink documentation where it is referenced.

Table 7. Command Actions

Action	Commands	Procedure

Clear dependencies	`svm cleardeps <Microsoft SQL Server service name>`
	Example:
	`svm cleardeps MSSQLServer`
		For more information, see the topic "CloudLink Agent for Microsoft SQL Server" in the *Dell CloudLink Deployment Guide*.

Decrypt	`svm decrypt [disk_volume]`	For more information, see Decrypt a volume.
Encrypt	`svm encrypt [disk_volume]`	Encrypt a mount point or block device. For more information, see Encrypt a volume.
Manage	`svm manage [device_name]`	Take control of a self-encrypting drive.
Register	`clagent.bat /S [clc_address] [/g [group_regcode]] [/t]`	Use the /t option to set Trusted Platform Module (TPM) mode on a physical machine that has a TPM version 2.0 chip installed. This option is only available during CloudLink Agent installation.
	`Svm CERTPATH=<folder where the securevm.cer & securevm.key are saved>`	See Assign a CA certificate and a third-party signed certificate to Windows agents during upgradation of CloudLink from earlier versions.
Release	`svm release [device_name]`	Release control of a self-encrypting drive.

Set dependencies	`svm setdeps <Microsoft SQL Server service name>`
	Example:
	`svm setdeps MSSQLServer`
		For more information, see the topic "CloudLink Agent for Microsoft SQL Server" in the *Dell CloudLinkDeployment Guide*.

Show dependencies	`svm showdeps`	For more information, see the topic "CloudLink Agent for Microsoft SQL Server" in the *Dell CloudLink Deployment Guide*.

Table 7. Command Actions (continued)

Action	Commands	Procedure
Status	`svm status`	For more information, see the topic "Verify successful deployment" in the *Dell CloudLink Deployment Guide*.

Command variables

This topic provides information about the command variables that are used in CloudLink.

The following lists the parameters that are used with one or more `svm` subcommands.

- device_name—The device must be identified when it is not mounted and listed in `/etc/fstab`.
- disk_volume—C drive, and D drive
- mount_point—The mount point to be encrypted.
- SED—The self-encrypting drive you want to control with CloudLink Center.

A

Role-based access control for CloudLink

The CloudLink Center 7.1.3 and later versions helps you to enable or disable usage of special characters in the CloudLink Center password for a `secadmin` (UI) and `cloudlink` (console) user.

The following table lists the CloudLink permissions and the default roles to which they are assigned.

NOTE: The policy that you set for using special characters affects both the UI and console. After you enable, it is mandatory to use special characters in a password that you use for the CloudLink UI and console.

Table 8. Permissions and Roles

Name	Role		
-	SecAdmin	Admin	Observer
Users			
View Users	x	x	x
Add User	x	x	-
Delete User	x	x	-
Change User Roles	x	x	-
Change User Password	x	x	-
Change User Second Factor	x	x	-
Unlock User	x	x	-
Roles			
View Roles	x	x	x
Add Role	x	-	-
Delete Role	x	-	-
Modify Role	x	-	-
Change Role Administration	x	-	-
Backup and Restore			
View Backup Configuration	x	x	x
Generate Backup Key	x	-	-
Generate Backup	x	x	-
Download Backup	x	x	-
Change Backup Configuration	x	x	-
Restore Backup	x	-	-
Keystores and Keys			
View Keystores	x	-	x
Add Keystore	x	-	-
Delete Keystore	x	-	-
Modify Keystore	x	-	-
Move Keys	x	-	-

Table 8. Permissions and Roles (continued)

Name	Role		
Machines			
View Machines	x	-	x
Remove Machine	x	-	-
Control Machine Boot	x	-	-
Change VM Encryption Policy	x	-	-
Change Machine Encryption	x	-	-
Change Machine Keys	x	-	-
Shred Machine	x	-	-
Move Machine	x	-	-
View Approved Networks	x	-	x
Add Approved Network	x	-	-
Modify Approved Network	x	-	-
Delete Approved Network	x	-	-
Machine Groups			
View Groups	x	-	x
Add Group	x	-	-
Modify Group	x	-	-
Delete Group	x	-	-
View Machines Usage	x	x	x
Reset Machines Usage	x	-	-
CloudLink Vault			
View Vault Mode	x	x	x
Change Vault Mode	x	-	-
Set Vault Unlock Passcodes	x	-	-
Unlock Vault	x	x	-
Monitoring			
View Actions	x	x	x
View Alarms	x	x	x
View Events	x	x	x
View Security Events	x	x	x
CloudLink Center Clusters			
View Cluster Members	x	x	x
Add Cluster Member	x	-	-
Delete Cluster Member	x	-	-
Join To Cluster	x	-	-
Check cluster concurrent session	x	-	-
CloudLink Licenses			

Table 8. Permissions and Roles (continued)

Name	Role		
View Licenses	x	x	x
Upload License	x	-	-
Assign License	x	-	-
Delete License	x	-	-
Configuration			
View Syslog Configuration	x	x	x
Change Syslog Configuration	x	-	-
Update System	x	-	-
View System Configuration	x	x	x
Change System Configuration	x	x	-
View Server Specific Configuration	x	x	x
Change Server Specific Configuration	x	x	-
View User Sessions	x	x	x
Terminate User Session	x	x	-
View Server Logs	x	-	-
View Server Performance	x	-	-
KMIP			
View KMIP Partitions	x	-	x
Add KMIP Partition	x	-	-
Modify KMIP Partition	x	-	-
Shred KMIP Partition	x	-	-
View KMIP Objects	x	-	x
View KMIP Clients	x	-	x
Add KMIP Client	x	-	-
Modify KMIP Client	x	-	-
Delete KMIP Client	x	-	-
Location			
View Providers	x	-	x
Add Provider	x	-	-
Modify Provider	x	-	-
Delete Provider	x	-	-
View Approved Locations	x	-	x
Add Approved Location	x	-	-
Modify Approved Location	x	-	-
Delete Approved Location	x	-	-
View Licensed Hosts	x	-	-

Table 8. Permissions and Roles (continued)

Name	Role		
Add Licensed Host	x	-	-
Delete Licensed Host	x	-	-

B

Configure Active Directory for the CloudLink encryption keystore

You must deploy a Windows Server that is accessible by CloudLink Center to use Active Directory to store CloudLink encryption keys. This procedure shows you how to configure Active Directory for the CloudLink encryption keystore on a Windows Server that is configured as a domain controller.

About this task

You must provide the hostname of the Windows Server during configuration. You must also add your DNS server to CloudLink Center. For more information, see Domain Name System servers in CloudLink.

Steps

1. Set up the Organization unit on Windows Server:
   a. On the Windows taskbar, click **Start** > **All Programs** > **Administrative Tools**, and then select **Active Directory Users and Computers**.
   b. Create an organization unit by expanding your domain name, and then right-click **New** > **Organizational Unit**.
   c. Enter a **Name** (for example, CloudLink_OU).
   d. Right-click the **Organization Unit** (for example, CloudLink_OU), and select **New** > **Group**.
   e. Enter the **Group Name** (for example, CloudLink_Group).
   f. Select **Global and Security**.
2. Create a bind user:
   a. Right-click the **Organization Unit** (for example, CloudLink_OU), and select **New, User**.
   b. Enter the **First Name** (for example, Cloud), **Last Name** (for example, Link), and login name. Click **Next**.
   c. Type the **Password**, and then click **Finish**.
   d. Right-click the **Organization Unit** (for example, CloudLink_OU), and then select **Delegate Control**.
   e. Click **Next** to follow setup wizard.
   f. Click **Add** and enter the CloudLink group name (for example, CloudLink_Group).
   g. Click **OK**, and then click **Next**.
   h. Select **Create a custom task** to delegate, and then click **Next**.
   i. Select **This folder, existing objects in this folder, and creation of new objects in this folder**, and then click**Next**.
   j. Select **Full Control**, and then click **Next**.
   k. Click **Finish**.
3. Add the bind user to the security group:
   a. Double-click **Security Group**.
   b. Click the **Members** tab.
   c. Click **Add**.
   d. Type the bind username.
   e. Click **OK**.
4. Record the DN of CloudLink:
   a. Click **Start**, and then select **Run**.
   b. Type **cmd**, and then click **OK**.
   c. Enter **dsquery OU** (Support tool is required) and record the DN (for example, OU=CloudLink_OU,DC=company,DC=com).

C

Update the Microsoft Azure Linux agent in a CloudLink Center

Use this procedure to update the Microsoft Azure Linux agent in a CloudLink Center.

Steps

1. Log in to CloudLink Center.
2. Click **About** > **REST Documentation** > **Open Documentation**.
   The **CloudLink Center REST API** page opens in a new window.
3. Click **server : Server Configuration**.
4. Click **PUT /cloudlink/rest/server/azure_agent_install**.
5. You must have the permission to access the REST API. To grant permission to access the REST API, click 

.
6. In the **Select OAuth2.0 Scopes** dialog box, select the checkbox next to **all**, and then click **Authorize**.
   You are logged out of CloudLink Center.
7. Log in to CloudLink Center again.
8. Ensure that you are granted the permission to access the REST API by verifying that this button is in 

state, and then click **Try it out**.
9. Ensure that the REST API call is succeeded.
   The Response Code: 200 is displayed if the REST API call is succeeded.

D

Restore VM agent connection to CloudLink Center

If an encrypted VM uses an incorrect IPv4 address for CloudLink Center, the VM may not start. The boot screen displays the IP address of the encrypted VM along with the hostname or the IP address that the VM is using for CloudLink. After few minutes, the encrypted VM automatically searches all the IPs in the subnet the VM is located, and tries to identify the correct IP address of the CloudLink Center.

If the encrypted VM is unable to locate the CloudLink Center, perform the following workarounds:

- Applies for IPv4 address—Deploy a new CloudLink cluster member on the subnet in which the encrypted VM failing to restart is located.

  If the VM restarts, ensure that the agent has the correct CloudLink Center hostname or IP address. If the correct hostname or IP address is not displayed, decrypt the VM and reinstall the agent with the correct CloudLink Center hostname or IP address.

- Applies for IPv4 and IPv6 addresses—If the VM does not restart using the preceding workaround, then:
  - Power off the VM and attach the encrypted boot drive as a data drive on a Windows VM that has the CloudLink Agent.
  - Use CloudLink Center to decrypt the data drive. The data drive displays the status as Pending. Manually accept the data drive in CloudLink Center. After the drive is decrypted, remove the data drive from the Windows VM and power on the VM.

  NOTE: Do not attach a Windows boot drive to a Windows VM that has the same disk signature. Microsoft operating system changes the disk signature which corrupts the data drive.

NOTE: If the preceding workarounds do not work, then contact Dell Technologies Support to Install the redirect application.

E

Install the redirect application

The following procedure explains how to install the redirect application.

About this task

Contact Dell Technologies Support to install the Redirect Application.

In certain circumstances, a machine that is registered with CloudLink Center may not start because the machine is unable to connect to CloudLink Center. Following are the circumstances:

- The IP address of CloudLink Center changes and DNS is not configured for CloudLink Center.
- The IP address for CloudLink Center might change if a CloudLink Center server is replaced.
- The IP address might change if CloudLink Center is deployed in a cloud environment.
- The public IP address of a machine might change when it is shut down and restarted. A new IP address is typically assigned from the same subnet as the previous address.

When a machine loses its connection to CloudLink Center, CloudLink Agent scans its subnet (/24 mask) to locate CloudLink Center by hostname on port 1194. If CloudLink Center is found, CloudLink Agent reconnects automatically and updates its configuration with the current connection information for CloudLink Center.

If CloudLink Agent cannot find CloudLink Center, it scans the same subnet for an Ubuntu server that is running the Redirect application of the CloudLink. When contacted by a CloudLink Agent, the Redirect application sends CloudLink Agent to the active CloudLink Center server. The configuration of the machine is updated with the connection information of the active CloudLink Center.

Steps

1. On the Ubuntu server, enter the following:

   ```
   wget https://clc_address/cloudlink/securevm/agenttools
   ```

2. Type the following:

   ```
   chmod +x agenttools
   ```

3. Run the Redirect application by typing:

   ```
   ./agenttools
   ```

4. Type **1** to select **Recovery**.

   Type the IP address of the active CloudLink Center server.

Results

After installation, the Redirect application runs as a console application, displaying a message each time it redirects a machine.

Machines are redirected to the new CloudLink Center to restart. However, the machine is assigned the disconnected state in CloudLink Center. Manually change the CloudLink Center server address on the machine. For more information, see Change the CloudLink Center IP address.

F

Move an encrypted drive to a another machine

In some circumstances, you may need to move an encrypted disk from one machine registered with CloudLink Center to another registered machine. For example, if a machine cannot boot, you can move its encrypted disk to another machine.

After moving the disk, you must register the new machine with the same CloudLink Center instance as the original machine. On startup, CloudLink Center determines whether to release keys for the moved disk based on the **Moved Device** or **Moved Volume** key release policy. For information, see Change key release policies of a machine group on CloudLink Center.

If CloudLink Center cannot release keys for the moved volume based on the current **Moved Volume** or **Moved Device** key release policy, it puts the machine in the pending state and locks the volume. You must manually accept the moved volume or device. After it is accepted, CloudLink Center allows the machine to start up and puts it in the connected state. CloudLink Center unlocks the volume or device and displays it as encrypted.

For the purposes of the following procedures, assume that VM-A is the machine with the encrypted disk (/data1/dir1) that you want to move to VM-B.

Move an encrypted disk to Windows machine

Use this procedure to move an encrypted disk to a Windows machine.

Steps

1. Detach the disk from VM-A.
2. Attach the disk to VM-B.

   If the machine is put in the pending state because CloudLink Center cannot release keys for the moved volume, manually accept the moved volume, as follows:

   a. In CloudLink Center, select **Machines**.
   b. Select the machine with the moved volume.
   c. Select **Actions** > **Pending Volumes**.
   d. Select the moved volume and click **Accept**.

Move an encrypted disk to Linux machine

Use this procedure to move an encrypted disk to a Linux machine.

Steps

1. Detach the disk from VM-A.
2. Attach the disk to VM-B.
3. Type the following command or reboot VM-B.

```
svm reload
```

4. Create a mount point on VM-B for the disk from VM-A.
   Example:

```
mkdir -p  /data1/dir
```

5. Type this command: `svm recover /mount_point /device`

Example:

```
svm recover /data1/dir /dev/sdb1
```

6. Restart the CloudLink Agent service. For more information, see Restart the CloudLink Agent service on Linux machines.
7. If the machine is put in the pending state because CloudLink Center cannot release keys for the moved volume, you must manually accept the moved volume, as follows:
   a. In CloudLink Center, click **Machines**.
   b. Select the machine with the moved volume or a moved device.
   c. Click **Actions** > **Pending Volumes**.
   d. Select the moved volume or a moved device, and then click **Accept**.
8. Verify that the encrypted volume or device is accessible.
9. In CloudLink Center, verify that the mount points attached to VM-B are identified as encrypted.

G

Recover an encrypted Linux boot volume

Use this procedure to restore an encrypted Linux boot volume. This procedure can be used to restore Linux boot volumes encrypted with CloudLink version 5.5 or higher.

Steps

1. Shut down the machine (CloudLinkVM-1) with the encrypted Linux boot volume that needs to be recovered.
2. Deploy a machine (CloudLinkVM-2) that uses a Linux distribution supported by CloudLink.

   NOTE: When you create CloudLinkVM-2, use a template different from the one used for CloudLinkVM-1 to prevent volumes from having the same root volume UUID.

3. Install CloudLink Agent on the new machine (CloudLinkVM-2).
4. Connect the new machine (CloudLinkVM-2) to the same CloudLink Center used by the old machine (CloudLinkVM-1).
5. Move the disk containing the encrypted root volume from the old machine (CloudLinkVM-1) and attach it to the new machine (CloudLinkVM-2).

   NOTE: For a root logical volume on LVM: before attaching the disk to CloudLinkVM-2 for recovery, ensure CloudLinkVM-2 does not have an LVM group with the same name.

6. Ensure the root volume is detected by the operating system.
   - Newer Linux distributions are configured to automatically detect block devices when attached. If this does not happen, run the following command:

     ```
     svm reload
     ```

   - If the original machine has the root file system on an LVM volume, you may need to activate the volume so it appears as a block device.

   WARNING: DO NOT restart CloudLinkVM-2 because it can cause a Linux kernel panic.

7. When the attached root volume is available on CloudLinkVM-2, run the following command:

   ```
   svm recover <existing empty directory> <root volume name>
   ```

   Example commands for a root file system on LVM:

   ```
   svm recover /mnt/recovery /dev/system/root
   ```

   ```
   svm recover /mnt/recovery /dev/mapper/centos-root
   ```

   ```
   svm recover /mnt/recovery /dev/ubuntu-vg/root
   ```

   Example commands for a regular root partition:

   ```
   svm recover /mnt/recovery /dev/sdb2
   ```

   ```
   svm recover /mnt/recovery /dev/xvdc1
   ```

8. Run the following command to restart the svmd service to recognize and unlock the volume:

   ```
   service svmd restart
   ```

9. Verify that the volume's status is `locked` when you run the following command:

```
svm status
```

10. Accept the pending volume to release the encryption keys. If the machine is put in the pending state because CloudLink Center cannot release the keys for the recovered volume, you need to manually accept the recovered volume, as follows:
    a. In CloudLink Center, go to **Machines**.
    b. Select the machine with the recovered volume.
    c. Select **Actions** > **Pending Volumes**.
    d. Select the recovered volume and click **Accept**.
11. After the volume is accepted in CloudLink Center, verify that the volume's status is `encrypted` when you run the following command:

```
svm status
```

Results

The data can now be copied from the volume or the volume can be decrypted.